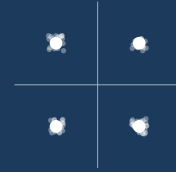
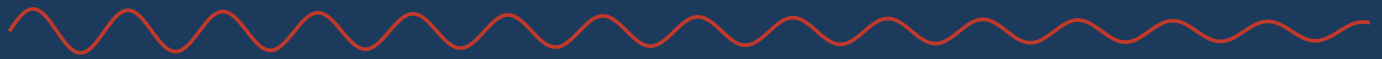


Modern Digital Communications



From the Telegraph to 6G



Theory, history and practice of communication systems,
with Python simulation labs and software-defined radio

- Part I Foundations: history, signals, noise, analog radio, digital telephony
- Part II Digital signal processing for radio
- Part III Digital transmission
- Part IV Information and coding
- Part V Systems: cellular, Wi-Fi, satellite, optical, 6G

Course Edition, 2026

Companion code: `moderncomms-labs`

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Preface

Every time you unlock a phone, stream a video, pay with a card or glance at a map, you rely on a chain of ideas that runs back through the space program, the transistor, radar, the vacuum tube, the telephone and the telegraph. Each link in that chain was forged by engineers solving a concrete problem—how to send more messages over the same wire, how to hear a ship beyond the horizon, how to fit a thousand conversations into a slice of spectrum—and each solution left behind a principle that is still at work inside today’s modems. This book is an attempt to hand those principles to you the way a senior engineer hands them to an apprentice: with the mathematics that makes them exact, the history that explains why they look the way they do, the hardware that makes them real, and a large number of simulations that let you break them and watch what happens.

A communication engineer’s knowledge is peculiar. It is deeply mathematical—the matched filter, the Shannon limit and the Viterbi algorithm are theorems—and yet it is also a craft, full of rules of thumb about loop bandwidths, back-off, guard bands and link margins that you only learn by building things that fail. I have tried to give both halves equal weight. When a result is exact, it is derived. When it is a rule of thumb, the text says so and explains where it comes from, so that you will know when it stops applying.

Who this book is for

The book is written for engineers and advanced students in electrical and computer engineering. It assumes you are comfortable with calculus, linear algebra, basic probability and a programming language; it does not assume a prior course in communications or signal processing. Early chapters build the signal-processing and probability foundations quickly but completely, so that a computer engineer can follow the whole arc, while the boxed discussions and the problem sets push well into material normally reserved for graduate courses.

How the book is organised

Part I, Foundations, opens with the history of telecommunication and then develops signals and spectra, random processes and noise, the classic analog systems—AM, FM, single sideband, the superheterodyne receiver, analog television—and the step into digital telephony through sampling, quantization and PCM.

Part II, Digital Signal Processing for Radio, covers the filters, multirate structures, oscillators and converters that every modern radio is built from, and dissects the transceiver and the software-defined radio.

Part III, Digital Transmission, is the heart of physical-layer engineering: pulse shaping, modulation and detection, synchronization, propagation and equalization.

Part IV, Information and Coding, gives Shannon’s theory and the codes that approach its limits, from Hamming and Reed–Solomon to turbo, LDPC and polar codes, together with the source coding that compresses voice, audio and images.

Part V, Systems, assembles everything into real systems: OFDM, spread spectrum and GPS, multiple antennas, multiple access and the cellular concept, every cellular generation from AMPS to 5G, Wi-Fi and the Internet of Things, satellite links from Telstar to Starlink, wireline and fibre-optic systems, and the research frontier of 6G.

A one-semester introductory course can cover Chapters 1–5 and 8–11. A second course can take Chapters 12–19. Chapters 20–25 read well on their own as a systems course or as professional reference.

Features

Five kinds of boxes appear throughout the text.

HISTORY

The people and events behind an idea, and why the solution took the form it did.

IN PRACTICE

Where the idea lives in deployed equipment and standards, with real numbers.

KEY IDEA

A principle worth remembering long after the derivation is forgotten.

PITFALL

A mistake that experienced engineers still make.

SIMULATION LAB

A Python simulation in the companion labs that lets you explore the topic.

Every data figure in the book was produced by Python code in the companion repository, usually by the same library functions the labs use, so any plot can be regenerated, modified and extended. Worked examples show the arithmetic of real designs, and every chapter ends with problems graded from routine to open-ended and with annotated further reading.

The companion labs

The `moderncomms-labs` repository contains the `commlib` signal-processing library, interactive Jupyter labs for each major topic, the scripts that generate every figure in this book, and GNU Radio flowgraphs for the Ettus USRP B200 software-defined radio. The labs run entirely in simulation; the hardware examples are optional but strongly encouraged. Appendix B explains how to install and use them.

A note on numbers and standards

Standards evolve. Figures for 5G, Wi-Fi 7, satellite constellations and the 6G study reflect publicly available information as of September 2026. Where a number is an engineering approximation rather than a specification value, the text says so.

PART I

Foundations

The Story of Telecommunication

Communication engineering is a young discipline built on old problems. This chapter follows the problem of sending messages at a distance from signal fires to 5G, and at each step extracts the technical idea that survives in today's equipment: the code that compresses text, the cable that smears pulses, the amplifier that made long distance possible, the carrier that let many conversations share one wire, the receiver architecture inside every radio, the theorems that define what is possible, and the engineering choices that shaped satellites, cellular networks and the Internet. Read it as a map of the rest of the book.

1.1 Why an engineer should know the history

It is possible to design a modem without knowing who Harry Nyquist was, just as it is possible to write software without knowing why a byte has eight bits. But communication systems are unusually shaped by their past. The 3.4 kHz bandwidth of a telephone channel, the 8 kHz sampling rate of digital voice, the 1.544 Mb/s of a T1 line, the 10.7 MHz intermediate frequency of an FM radio, the 15 kHz subcarrier spacing of LTE and 5G, and the 30 kHz channels of the first cellular systems are all decisions made by engineers under particular constraints, and they propagate forward because new systems must coexist with old ones. When you understand why a number was chosen, you know when you may safely change it.

History also teaches humility. The transatlantic cable of 1858 failed because its engineers did not understand the physics of a long line. The early theorists of frequency modulation concluded that it had no advantage, and they were right about the question they asked and wrong about the one that mattered. Code-division multiple access was declared impractical by experts shortly before it became the basis of third-generation cellular. Low-density parity-check codes were invented in 1960, forgotten for thirty-five years, and are now in every 5G phone. Engineering is a conversation across generations, and it is worth hearing the earlier voices.

1.2 Before electricity

The oldest long-distance communication systems were visual. Greek historians describe chains of signal fires; the Roman army used towers; and the Great Wall of China had beacon towers that could relay a warning hundreds of kilometres in hours. These systems had enormous speed—the message moved at the speed of light between towers—but almost no *capacity*. A fire is either lit or not, so the message had to be agreed in advance: “the enemy is coming.” In modern language, the alphabet had two symbols and the system sent one bit per event.

1.2.1 The optical telegraph

The first system that could send arbitrary messages over long distances was the optical telegraph of Claude Chappe. Beginning with the Paris–Lille line in 1794, the French government built a network of towers topped by articulated arms whose positions could be read by telescope from the next tower, some 10 to 15 km away. Chappe coined the word *télégraphe*, “far writer.” The arms could take 98 distinguishable positions, of which a subset was used for control and the rest indexed a codebook of words and phrases: a message was a sequence of page and line numbers. At its height the French network had more than 500 stations, and a message could cross the country in a few hours.

The Chappe telegraph already contained ideas that recur throughout this book. It used a finite *alphabet* of signal states chosen to be distinguishable in the presence of “noise”—haze, fatigue and bad telescopes. It used a *codebook* to map frequent messages into short symbol sequences. It had a *protocol*: control signals for “start”, “error” and “repeat”. And it had a *capacity limit*, set by how quickly an operator could move the arms and how reliably the next operator could read them, of roughly one symbol every 20 to 30 seconds.

1.3 The electric telegraph

1.3.1 Cooke, Wheatstone, Morse and Vail

Electricity offered a signal that moved along a wire faster than any horse and did not care about fog. William Cooke and Charles Wheatstone patented a needle telegraph in Britain in 1837; it used several needles that pointed at letters on a board, so an untrained operator could read it. In the United States, Samuel Morse and his collaborator Alfred Vail chose a different path: a single circuit, switched on and off by a key, whose pattern of short and long pulses encoded letters. On 24 May 1844 Morse sent “What hath God wrought” from Washington to Baltimore, and within a decade telegraph wires followed the railways across the continent.

The Morse code is a remarkable piece of engineering. Its designers wanted to minimise the time needed to send typical English text, and they realised that the answer was to give short codes to frequent letters. According to the traditional account, Vail estimated letter frequencies by counting the type in a printer’s type case: the printer kept 12 000 E’s but only 200 Z’s. The code gives E a single dot and T a single dash, while Q, Z and J receive four symbols. This is a *variable-length source code*, a century before David Huffman proved (in 1952) how to construct the optimal one, and Morse code is within about 15% of the efficiency of a Huffman code for English letters. We will return to this in Chapter ??.

The timing rules of International Morse code are strict: a dot lasts one unit, a dash three, the gap between symbols one, between letters three and between words seven. Figure 1.1 shows the keyed envelope of “SOS” and something the early radio operators discovered the hard way: the spectrum of the keyed signal. Switching a carrier on and off instantaneously produces sharp edges, and sharp edges have wide spectra. A radio telegraph with hard keying spattered interference—“key clicks”—across neighbouring channels. The cure was to shape the edges of each pulse, and the lower curve shows that raised-cosine edges lower the spectral sidelobes by tens of decibels. Pulse shaping to control bandwidth is exactly the subject of Chapter ??, where it becomes the root-raised-cosine filter inside every modern

modem.

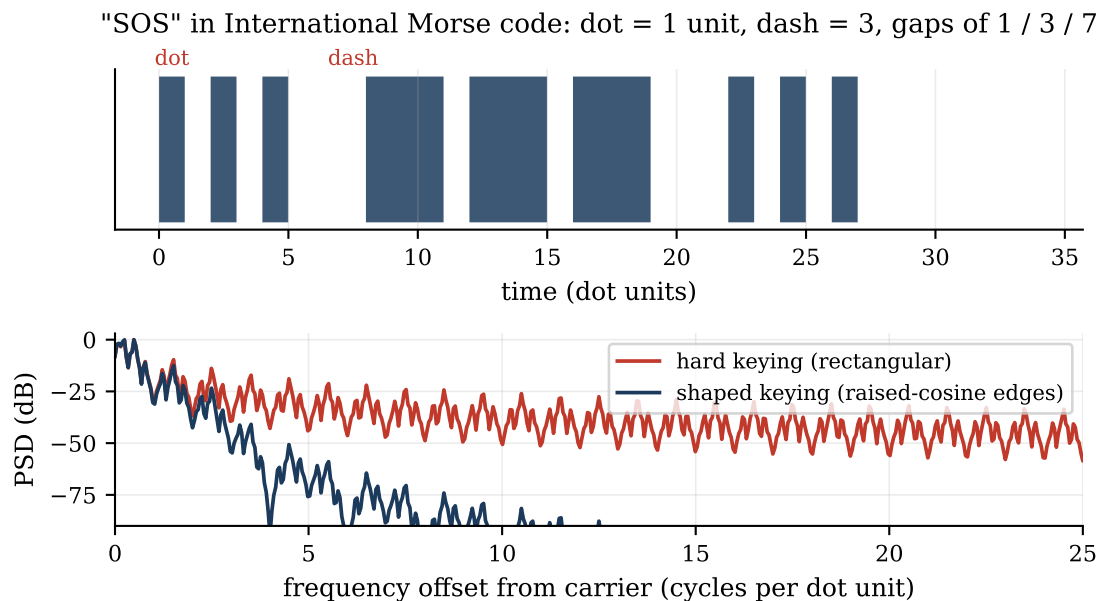


Figure 1.1: Top: the on-off keyed envelope of “SOS”. Bottom: the power spectrum of repeated Morse traffic with instantaneous (hard) keying and with raised-cosine shaped edges. Shaping the pulses reduces the out-of-band spectrum by more than 40 dB, which is why operators added “click filters” to their keys.

1.3.2 Multiplexing: sharing one wire

Telegraph wires were expensive, and a single operator could send only 30 or 40 words per minute. The first attack on this bottleneck was *duplex* and *quadruplex* operation: Thomas Edison’s quadruplex of 1874 sent two messages in each direction at once by using both the polarity and the magnitude of the line current as independent signals—an early form of what we now call multilevel signalling and full-duplex operation.

The second attack was time-division multiplexing. Émile Baudot’s printing telegraph of 1874 used a rotating distributor that connected several operators to the line in turn, and each operator entered characters using five keys. Every character was therefore a fixed-length five-bit word, the Baudot code, the ancestor of ASCII. The unit of signalling rate, the *baud*, honours him: one baud is one symbol per second. Note the distinction, which we will use constantly: the baud rate counts symbols, while the bit rate counts bits, and the two differ whenever a symbol carries more than one bit.

KEY IDEA Three ways to share a channel

Every multiplexing scheme ever built divides the channel’s resources along some dimension so that the signals do not interfere. Baudot divided *time*. Carrier telephony (below) divided *frequency*. Code-division systems, a century later, divided the signal space using nearly orthogonal *codes*, and multi-antenna systems divide *space*. Chapter ?? treats all four as one idea: making the signals of different users approximately orthogonal.

1.3.3 The Atlantic cable and the physics of a long line

The most dramatic engineering story of the nineteenth century is the laying of the transatlantic telegraph cable. The first cable, completed in August 1858, carried Queen Victoria’s 98-word message to President Buchanan—which took about sixteen hours to transmit. Within weeks the cable failed entirely. The company’s electrician, Wildman Whitehouse, had tried to overcome the weak, sluggish signals by applying about 2000 volts, and the insulation broke down.

The man who understood what was really happening was William Thomson, later Lord Kelvin. In 1855 he had analysed a long submarine cable as a distributed resistance R and capacitance C per unit length. Neglecting inductance and leakage, the voltage obeys a diffusion equation,

$$\frac{\partial v}{\partial t} = \frac{1}{RC} \frac{\partial^2 v}{\partial x^2}, \quad (1.1)$$

whose response at distance ℓ to a voltage step applied at $t = 0$ is

$$v(\ell, t) = \operatorname{erfc}\left(\frac{\ell}{2} \sqrt{\frac{RC}{t}}\right). \quad (1.2)$$

The time for the received voltage to reach a given fraction of its final value grows as $RC\ell^2$: double the length of the cable and you must signal four times more slowly. This is Thomson’s “law of squares”. Figure 1.2 shows the consequence. When dots are sent slowly, each has time to rise and fall; when they are sent faster, the slow tails of earlier pulses pile up on later ones and the pattern is lost. The name for this is **intersymbol interference (ISI)**, and it is the central problem of Chapters ?? and ??.

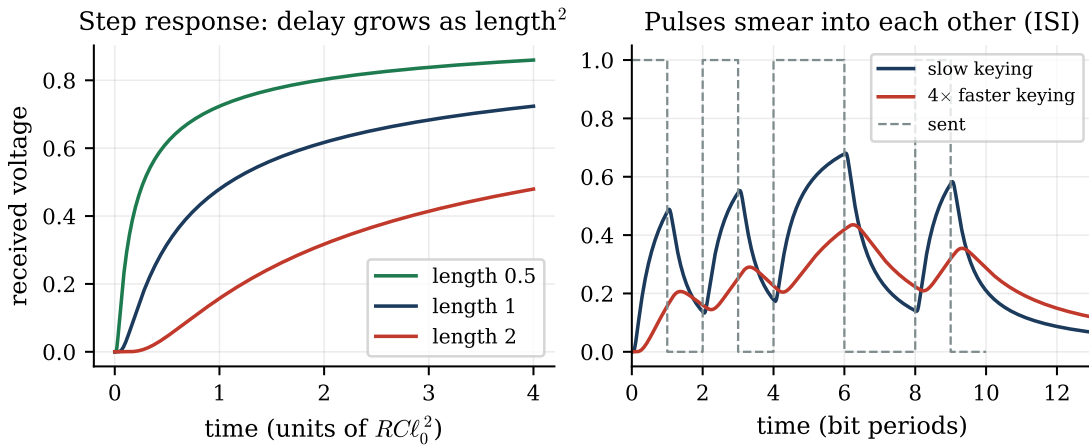


Figure 1.2: Kelvin’s model of a long submarine cable. Left: the response to a step of voltage at three cable lengths; the delay scales with the square of the length. Right: a train of pulses through the same cable at two keying speeds. At the faster rate each pulse is smeared into its neighbours—intersymbol interference.

Thomson’s remedies were engineering, not force. He invented the sensitive mirror galvanometer so that small currents could be read, and he argued for a thicker conductor and better insulation to lower R and C . The 1866 cable, laid by the steamship *Great Eastern*,

embodied his design and worked for years, carrying about eight words per minute. Thomson was knighted for it.

HISTORY Heaviside and the distortionless line

Oliver Heaviside, a self-taught telegraph engineer, extended Thomson's analysis in the 1880s by including the inductance L and leakage conductance G of the line. He found that when $L/R = C/G$, all frequencies travel at the same speed and are attenuated equally: the line becomes *distortionless*. Real lines had far too little inductance, so Heaviside proposed adding it. His suggestion was ignored for years, partly because of his combative personality, until Michael Pupin and George Campbell independently patented *loading coils* around 1900. Loaded telephone cables, with coils every 1.8 km, made long-distance telephony practical—and a century later, those same coils had to be removed from the copper loop before DSL could work, because they turn the line into a low-pass filter that blocks everything above the voice band (Chapter ??).

1.4 The telephone and the network

1.4.1 Voice over a wire

Alexander Graham Bell's telephone patent was filed on 14 February 1876, the same day Elisha Gray filed a caveat on a similar idea. Bell's transmitter converted the pressure variations of speech into a proportional current: an *analog* signal. The telegraph had sent discrete symbols; the telephone sent a continuous waveform, and it demanded a new kind of engineering. A telegraph signal only had to be detectable; a telephone signal had to be reproduced faithfully, which meant controlling the frequency response, noise and distortion of every component in the path.

Experiments with listeners established that intelligible speech needs roughly the band from 300 to 3400 Hz. That number—about 3.1 kHz of useful bandwidth inside a 4 kHz channel allocation—was fixed by the 1920s and still defines the digital telephone network: the 8 kHz sampling rate of pulse-code modulation (Chapter 5) is twice the 4 kHz channel width.

1.4.2 Switching and the first automation

Early telephones were connected through manual switchboards. The first automatic switch was invented by Almon Strowger, an undertaker in Kansas City who, according to the story, suspected that the local operator was diverting his customers to a competitor. His step-by-step switch of 1891 moved a wiper in response to the pulses of a rotary dial: dialling “3” produced three pulses that stepped the switch three positions. The telephone network became the largest machine on Earth, and its growth drove the development of switching theory, traffic theory (A. K. Erlang's queueing formulas, which still size cellular networks; Chapter ??) and ultimately the digital computer.

1.4.3 Amplification and the long-distance problem

Loading coils extended the reach of a telephone call to perhaps 1500 km. Beyond that the signal simply became too weak. The solution was the *repeater*, an amplifier inserted into the line, and the device that made it possible was the vacuum tube. Lee de Forest's Audion triode (1906) could amplify, but only erratically; engineers at AT&T led by Harold Arnold turned it into a reliable high-vacuum amplifier, and on 25 January 1915 the first transcontinental telephone call connected New York and San Francisco through a chain of tube repeaters.

Amplifiers created a new problem. Every repeater added a little distortion, and a transcontinental circuit had dozens of them. In August 1927 Harold Black, a young Bell Labs engineer, sketched on a page of the *New York Times* during his ferry commute the idea of the **negative feedback amplifier**: sacrifice most of the amplifier's gain by feeding an inverted fraction of the output back to the input, and in exchange the overall gain depends almost entirely on the stable passive feedback network rather than on the drifting tube. Negative feedback is now so universal that it is hard to imagine electronics without it, and the stability analysis it required—by Harry Nyquist in 1932 and Hendrik Bode—founded control theory.

1.4.4 Carrier telephony: frequency-division multiplexing

Once a wire could carry frequencies far above the voice band, it was natural to put many conversations on it at once, each shifted to its own frequency slot. This is **frequency-division multiplexing** (FDM). Its key enabler was *single-sideband* modulation, patented by John Carson of AT&T in 1915: by transmitting only one of the two sidebands of an amplitude-modulated carrier, each voice channel occupied only its own 4 kHz. The first carrier system went into service in 1918. By the 1940s the Bell System's L-carrier on coaxial cable stacked 600 voice channels, and by the 1970s the L5 system carried 10 800 voice channels on each coaxial tube, organised into a hierarchy of groups (12 channels), supergroups (60), mastergroups (600) and beyond. Chapter 4 builds this hierarchy from first principles.

1.5 Wireless

1.5.1 From Maxwell to Marconi

In 1865 James Clerk Maxwell published the equations that unified electricity, magnetism and light and predicted electromagnetic waves travelling at the speed of light. Heinrich Hertz generated and detected such waves in his laboratory between 1886 and 1888, using a spark gap as a transmitter and a small loop with a gap as a receiver. He thought the discovery had no practical use.

Guglielmo Marconi thought otherwise. Beginning in 1895, he improved the range of spark-gap systems by raising the antennas and grounding them, and on 12 December 1901 he reported receiving the letter S in Morse code at Signal Hill, Newfoundland, sent from Poldhu in Cornwall—across 3500 km of ocean and, unknown to him, bent around the Earth's curvature by the ionosphere. Wireless telegraphy became essential at sea. The sinking of the *Titanic* in 1912, when a nearby ship's radio operator had gone to bed, led directly to international rules requiring continuous radio watch and to the first licensing of radio stations.

Spark transmitters were extremely wasteful. A spark produces a damped oscillation whose spectrum spreads over a wide band, so only a few stations could operate in one area. The next generation used continuous waves from high-frequency alternators and arc converters, and eventually from vacuum-tube oscillators. A continuous-wave transmitter concentrates its energy at one frequency, which allows receivers to be selective and many stations to coexist: the start of spectrum management.

1.5.2 Voice by radio, and Armstrong's three inventions

Reginald Fessenden, a Canadian engineer, understood that a continuous wave could be *modulated* by voice, and he reportedly broadcast speech and music on Christmas Eve 1906 from Brant Rock, Massachusetts. Amplitude modulation (AM) became the basis of broadcasting; the first scheduled commercial broadcast is usually credited to KDKA in Pittsburgh, which reported the results of the Harding–Cox presidential election on 2 November 1920.

The receivers of the day were poor, and three inventions by Edwin Howard Armstrong fixed them. The first, the *regenerative receiver* (1912), fed part of a tube amplifier's output back to its input with positive feedback, giving enormous gain from one tube. The second, the **superheterodyne receiver** (1918), mixed every incoming signal down to a fixed *intermediate frequency* (IF), where selective, high-gain amplifiers could be built once and optimised. Nearly every radio built since—including the software-defined radio in your lab, which is a superheterodyne whose IF happens to be zero—follows this architecture. The third, wideband *frequency modulation* (FM, patented 1933), we return to shortly.

KEY IDEA The superheterodyne principle

It is hard to build a filter that is both very selective and tunable over a wide range. It is easy to build a very selective filter at one fixed frequency. The superhet therefore moves the signal to the filter rather than moving the filter to the signal, using a mixer and a tunable local oscillator. Chapter 4 derives the image-frequency problem that this creates; Chapter 7 shows how modern zero-IF and low-IF receivers solve it digitally.

1.5.3 FM: the answer to a different question

In 1922 John Carson analysed frequency modulation as a way of saving bandwidth and showed, correctly, that it could not: an FM signal always occupies at least as much bandwidth as the equivalent AM signal. He concluded that FM “inherently distorts without any compensating advantages.” Armstrong asked a different question: what if we deliberately use *more* bandwidth? His 1936 paper showed that wideband FM trades bandwidth for signal-to-noise ratio: a broadcast FM signal that uses about 180 kHz to carry 15 kHz of audio is dramatically quieter than AM. This was the first practical demonstration of a principle that Shannon would make exact twelve years later: bandwidth and signal power are interchangeable currencies. FM broadcasting, the audio of analog television, two-way police and military radio, and the first cellular systems were all built on it.

1.6 The theorists

While engineers built networks, a handful of theorists at Bell Labs and elsewhere worked out what the networks could possibly do. Their results are the foundation of this book.

Harry Nyquist showed in 1924 and 1928 that a channel of bandwidth B can carry at most $2B$ independent pulses per second without intersymbol interference, and derived the condition a pulse must satisfy to achieve it—the *Nyquist criterion* of Chapter ???. With John B. Johnson he also explained thermal noise in 1928: every resistor at temperature T generates noise with power kTB , the floor under every radio receiver (Chapter 3).

Ralph Hartley proposed in 1928 that information be measured as the logarithm of the number of distinguishable messages, so that a channel sending $2B$ symbols per second from an alphabet of M levels carries $2B \log_2 M$ bits per second.

Vladimir Kotelnikov in the Soviet Union (1933), following E. T. Whittaker (1915), stated the sampling theorem: a signal band-limited to B is completely determined by samples taken $2B$ times per second.

Norbert Wiener and Andrey Kolmogorov developed, during the Second World War, the theory of optimal linear filtering of noisy signals, the ancestor of the MMSE equalizer.

Stephen O. Rice of Bell Labs published in 1944–45 the mathematical analysis of random noise that gave the Rayleigh and Rician distributions their place in fading channel theory (Chapter ??).

1.6.1 Claude Shannon

In 1948 Claude Shannon, then 32 and working at Bell Labs, published “A Mathematical Theory of Communication.” It is one of the rare papers that created a field fully formed. Shannon modelled communication with the block diagram of Figure 1.3, defined information as a measurable quantity—the entropy of the source, in bits—and proved two theorems that define the limits of the field. The *source coding theorem* says that a source can be compressed to its entropy and no further. The *channel coding theorem* says that every channel has a capacity C : at any rate below C it is possible to communicate with arbitrarily small error probability, and at any rate above it it is not. For the band-limited Gaussian channel,

$$C = B \log_2 \left(1 + \frac{S}{N} \right) \quad \text{bits per second.} \quad (1.3)$$

Shannon’s proof was not constructive: it showed that good codes exist by averaging over randomly chosen codes, and it said nothing about how to build or decode one. The rest of the history of coding theory, from Richard Hamming’s single-error-correcting codes of 1950 to polar codes in 2009, is the story of the search for practical codes that approach Shannon’s limit. Chapters ??–?? tell it.

KEY IDEA Shannon’s lesson for the working engineer

Before designing anything, compute the capacity of the channel you have. It tells you

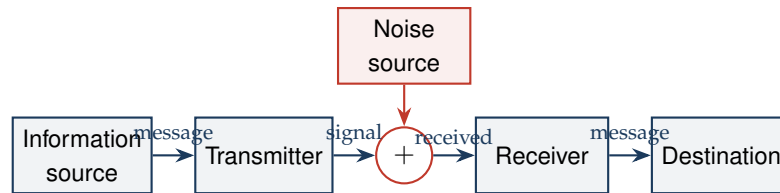


Figure 1.3: Shannon’s model of a communication system (redrawn after his 1948 paper). Every system in this book—telegraph, telephone, radio, satellite, fibre, cellular—fits this diagram.

whether your goal is impossible, easy or somewhere in between, and it tells you whether to spend your effort on more power, more bandwidth, more antennas or a better code. Every link budget in this book ends with that comparison.

1.7 War, secrecy and the birth of digital communication

The Second World War accelerated communication technology as much as any peacetime decade. Three developments stand out.

Pulse-code modulation, invented by Alec Reeves in 1937, represented a voice signal as a sequence of numbers. At first it seemed absurdly wasteful of bandwidth, but it had one decisive advantage: a digital signal can be *regenerated*. An analog signal accumulates noise at every repeater; a digital signal can be detected and re-transmitted perfectly clean, so noise does not accumulate. The first operational use of digital voice was SIGSALY, the secure telephone system that connected Roosevelt and Churchill from 1943, which combined a vocoder, quantized samples and one-time-pad encryption from phonograph records.

Frequency hopping was patented in 1942 by the actress Hedy Lamarr and the composer George Antheil, who proposed synchronising the frequency changes of a torpedo’s radio link using piano-roll mechanisms, so that an enemy could not jam it. The patent was not used during the war, but the principle of spreading a signal over a wide band to make it resistant to jamming and interception—**spread spectrum**—became central to military communications, GPS, Bluetooth and code-division cellular systems (Chapter ??).

Radar and microwaves. The cavity magnetron, brought from Britain to the United States in 1940, made high-power microwave transmitters practical, and the MIT Radiation Laboratory developed the microwave engineering—waveguides, mixers, low-noise receivers, the theory of detection in noise—that post-war microwave relay links and satellite communication were built on. The *matched filter*, which maximises the signal-to-noise ratio of a known pulse in noise, was derived by D. O. North for radar in 1943; it is the optimal receiver front end of Chapter ??.

1.8 Transistors, satellites and fibre

1.8.1 The solid-state revolution

In December 1947 John Bardeen, Walter Brattain and William Shockley at Bell Labs demonstrated the point-contact transistor. The integrated circuit followed in 1958–59 (Jack Kilby at Texas Instruments, Robert Noyce at Fairchild), and with it the exponential scaling that would eventually make it cheaper to process a radio signal digitally than with analog components. The telephone network's first large digital transmission system, the T1 carrier of 1962, sent 24 PCM voice channels over two twisted pairs at 1.544 Mb/s, with regenerative repeaters every 1.8 km—exactly where the loading coils had been.

1.8.2 Communication satellites

In a 1945 article in *Wireless World*, the science-fiction writer and radar engineer Arthur C. Clarke observed that a satellite at an altitude of about 35 786 km above the equator orbits once per sidereal day and therefore appears fixed in the sky; three such satellites could cover almost the whole Earth. Twelve years later the Soviet Union launched Sputnik 1 (4 October 1957), whose radio beacons on 20 and 40 MHz were tracked by amateurs worldwide—and whose Doppler shift, measured at Johns Hopkins, led directly to the Transit satellite navigation system and eventually to GPS.

Progress was then rapid. NASA's Echo 1 (1960) was a passive 30 m metallised balloon that simply reflected signals. AT&T's Telstar 1 (1962) carried the first live transatlantic television, but in a low orbit it was visible to its ground stations for only minutes at a time. Syncom 2 (1963) reached geosynchronous orbit, and Syncom 3 (1964), in true geostationary orbit, relayed television of the Tokyo Olympics to the United States. Intelsat I, "Early Bird", began commercial transatlantic service in 1965 with 240 telephone circuits. Chapter ?? develops orbital mechanics, satellite link budgets and modern systems from DVB-S2 broadcasting to the low-Earth-orbit constellations that, sixty years after Telstar, have returned to Telstar's orbit in their thousands.

1.8.3 Light

In 1966 Charles Kao and George Hockham argued that the high loss of glass fibre was due to impurities rather than to anything fundamental, and that purified silica could reach losses below 20 dB/km, making optical fibre competitive with coaxial cable. Corning achieved that figure in 1970; modern fibre loses about 0.17–0.2 dB/km at 1550 nm. The erbium-doped fibre amplifier (1987) amplified light directly, without converting it to electricity, and allowed many wavelengths to be amplified at once. The first transatlantic fibre cable, TAT-8 (1988), carried 280 Mb/s; a modern submarine cable carries hundreds of terabits per second using coherent detection, the same QAM and DSP you will study in Chapters ??–??, applied to light (Chapter ??). Kao received the Nobel Prize in Physics in 2009.

1.9 Networks of computers

The ARPANET connected its first two nodes in 1969, and Robert Metcalfe's Ethernet (1973) connected computers in a building by letting them share a coaxial cable with a listen-before-talk protocol derived from the University of Hawaii's ALOHAnet radio network. The data

modems that carried computer traffic over the telephone network are a history of physical-layer engineering in miniature: from 300 b/s frequency-shift keying in the 1960s to 9600 b/s with 32-point QAM and trellis-coded modulation (Gottfried Ungerboeck’s 1982 invention) in V.32, to 33.6 kb/s in V.34—within a few decibels of the Shannon capacity of a telephone channel. Figure 1.4 puts these numbers in perspective.

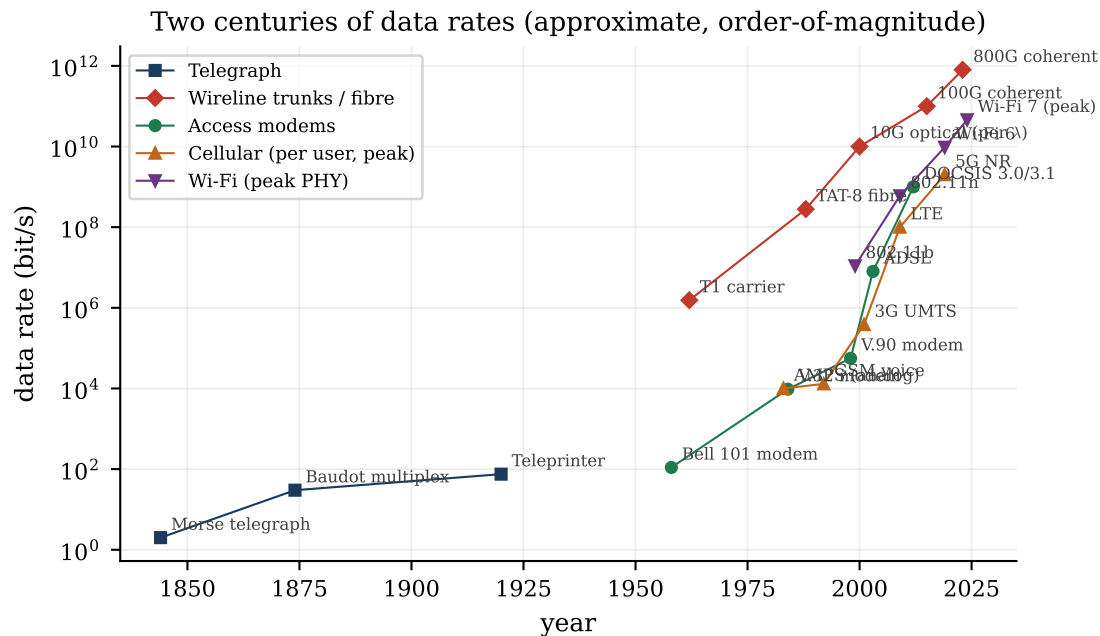


Figure 1.4: Approximate data rates of landmark systems. Each family advanced by roughly a factor of ten every six to ten years once it was digital; the values are order-of-magnitude illustrations of typical or peak rates rather than precise specifications.

1.10 Cellular radio

The cellular concept was described in a 1947 Bell Labs memorandum by Douglas Ring and Rae Young: instead of one powerful transmitter covering a city, use many low-power base stations, each covering a small *cell*, and reuse the same frequencies in cells far enough apart. Capacity then grows with the number of cells rather than being fixed by the spectrum. The idea waited three decades for the technology—microprocessors to manage handoff between cells, and spectrum allocations to use.

The first commercial cellular network opened in Tokyo in 1979; in the United States, the Advanced Mobile Phone System (AMPS) launched in Chicago in October 1983, six and a half years after Martin Cooper of Motorola had made the first handheld cellular call on a New York street in April 1973. Table 1.1 summarises what followed. Each generation brought the newest physical-layer ideas of its time into billions of devices: digital modulation and speech coding in 2G, code-division multiple access and turbo codes in 3G, OFDM and MIMO in 4G, LDPC and polar codes and massive antenna arrays in 5G. Chapter ?? tells the full story, including the “CDMA wars” of the early 1990s, when a small company named Qualcomm—founded by Irwin Jacobs and Andrew Viterbi, the inventor of the Viterbi algorithm—persuaded a

Table 1.1: Five generations of cellular technology.

Generation	Launch	Key systems	Defining physical-layer ideas
1G	1979–83	NTT (Tokyo), AMPS, NMT, TACS	Analog FM voice, 25–30 kHz FDMA channels, handoff
2G	1991–95	GSM, IS-95 (cdmaOne), IS-136	Digital speech codecs, TDMA with GMSK (GSM), CDMA with power control and soft handoff (IS-95), SMS
3G	2001	UMTS/WCDMA, CDMA2000	Wideband CDMA, turbo codes, fast power control, HSPA packet data
4G	2009	LTE, LTE-Advanced	OFDMA downlink, SC-FDMA uplink, MIMO, all-IP, carrier aggregation
5G	2019	NR	Flexible OFDM numerology, LDPC and polar codes, massive MIMO, mmWave beamforming
6G	≈2030	(Release 21 onward)	Evolved OFDM, sensing, AI-native design, non-terrestrial integration

sceptical industry that CDMA could work.

1.11 Unlicensed radio and Wi-Fi

In 1985 the US Federal Communications Commission opened the industrial, scientific and medical (ISM) bands at 902 MHz, 2.4 GHz and 5.8 GHz to unlicensed spread-spectrum communication. Nobody expected much of it: 2.4 GHz is where microwave ovens heat water. The IEEE 802.11 standard followed in 1997 at 1–2 Mb/s, 802.11b reached 11 Mb/s in 1999, and the industry adopted the brand name Wi-Fi. Today more Internet traffic reaches its final destination over Wi-Fi than over any other medium, and the bands that were once a spectral junkyard are among the most valuable in the world. The lesson for the spectrum engineer is that flexible rules and cheap hardware can create more value than exclusive licences. Chapter ?? follows 802.11 from 1 Mb/s to Wi-Fi 7’s 46 Gb/s peak.

1.12 The coding renaissance and the MIMO revolution

Between 1993 and 2009, coding theory went from “several decibels from capacity” to “within a fraction of a decibel”. In 1993 Claude Berrou, Alain Glavieux and Punya Thitimajshima presented *turbo codes*, which used two simple convolutional codes and an iterative decoder to come within 0.7 dB of the Shannon limit; many experts initially assumed the results were wrong. In 1996 David MacKay and Radford Neal rediscovered Robert Gallager’s 1960 *low-density parity-check* codes and showed they were as good. And in 2009 Erdal Arıkan introduced *polar codes*, the first codes proven to achieve capacity with low-complexity encoding and decoding. 5G uses both LDPC and polar codes.

At the same time, Gerard Foschini at Bell Labs (1996) and Emre Telatar (1995) showed that a channel with multiple antennas at both ends has a capacity that grows linearly with the number of antennas—an entirely new resource, obtained from the same bandwidth and

power by exploiting the multipath scattering that radio engineers had always regarded as a nuisance. Siavash Alamouti's two-antenna space-time code (1998) was simple enough to put into every handset. Multiple-input multiple-output (MIMO) transmission is now in every Wi-Fi access point and cellular base station (Chapter ??).

1.13 The software-defined radio

For most of the twentieth century a radio *was* its hardware: its filters, oscillators and detectors fixed what it could do. As digital processing grew cheaper, it became possible to digitise the signal close to the antenna and do everything else in software. Joseph Mitola coined the term *software radio* in 1992. GNU Radio, an open-source framework for building radios in software, began in 2001, and the Universal Software Radio Peripheral (USRP) from Ettus Research gave students and researchers affordable hardware for it. Today the “software radio” idea is simply how radios are built: a 5G base station is largely software running on general-purpose or specialised processors, and the Ettus B200 on your bench is architecturally a scaled-down cousin of a cellular radio unit (Chapter 7).

1.14 Lessons from two centuries

Looking back over this history, a few patterns recur so often that they amount to engineering principles.

1. **Every channel is eventually limited by noise and dispersion.** Kelvin's cable, Nyquist's telegraph and a 5G link differ in scale, not in kind. Intersymbol interference and noise are the two enemies, and every technique in this book fights one or both.
2. **Digital wins because it can be regenerated.** Once a signal is represented by discrete symbols, errors can be prevented from accumulating and corrected by coding. This is why every analog system has been replaced by a digital one.
3. **Bandwidth, power, antennas and complexity are interchangeable currencies.** Armstrong traded bandwidth for SNR; Shannon quantified the exchange rate; turbo and LDPC codes traded computation for power; MIMO traded antennas for bandwidth. Good design is choosing the cheapest currency.
4. **Old ideas return when technology catches up.** Gallager's LDPC codes, the cellular concept, frequency hopping and multicarrier modulation (first proposed in the 1960s) all waited decades for cheap enough electronics.
5. **Compatibility shapes everything.** The 4 kHz voice channel, the 8 kHz sampling rate and the 15 kHz subcarrier spacing persist because new systems had to live with old ones.

SIMULATION LAB Chapter 1 simulations

The figure script `figscripts/ch01_figs.py` reproduces Kelvin's cable model and the key-click spectrum. Try changing the cable length and keying speed until the pulse pattern becomes unreadable, and estimate the maximum “words per minute” of a cable

twice as long. Lab 1 (`lab01_baseband.ipynb`) then moves from telegraph pulses to the complex baseband signals of a modern radio.

Problems

- 1.1 Using Thomson's law of squares, if the 1866 cable (about 3000 km) supported 8 words per minute, what rate would a 1500 km cable of the same construction support? What would halving the capacitance per unit length achieve?
- 1.2 The Chappe telegraph could display 98 positions, of which about 92 carried data, at about one position every 20 s. Compute its information rate in bits per second using Hartley's measure. Compare with a Morse operator sending 20 words per minute (assume 5 letters per word and 4.7 bits per letter of English text).
- 1.3 Morse code assigns E the shortest code and Q a long one. Using the timing rules in the text, compute the average duration in dot units of a letter of English, given the approximate letter frequencies E 12.7%, T 9.1%, A 8.2%, O 7.5%, I 7.0%, N 6.7%, S 6.3%, H 6.1%, R 6.0% (assign the remaining 36.4% an average code of 3 symbols with 2 dashes). Compare with a fixed-length code.
- 1.4 A carrier telephone system stacks 600 single-sideband voice channels, each allotted 4 kHz. What total bandwidth does it require? If double-sideband AM had been used instead, how many channels would fit in the same band?
- 1.5 Use Equation (1.3) to compute the capacity of a 3.1 kHz telephone channel with a 35 dB signal-to-noise ratio. How close did the V.34 modem (33.6 kb/s) come?
- 1.6 Why was the ionosphere essential to Marconi's 1901 experiment? Estimate the maximum line-of-sight distance between two 150 m antenna masts on a spherical Earth of radius 6371 km.
- 1.7 (*Research*) Choose one of the "old ideas" listed in the lessons—LDPC codes, multicarrier modulation, frequency hopping or the cellular concept—and write a one-page account of what technology had to change before it could be deployed.

Further Reading

- T. Standage, *The Victorian Internet*, Walker, 1998. A lively history of the telegraph and its social impact.
- J. Gleick, *The Information: A History, a Theory, a Flood*, Pantheon, 2011. From talking drums to Shannon, beautifully written.
- J. Gertner, *The Idea Factory: Bell Labs and the Great Age of American Innovation*, Penguin, 2012.
- J. Soni and R. Goodman, *A Mind at Play: How Claude Shannon Invented the Information Age*, Simon & Schuster, 2017.

- C. E. Shannon, "A Mathematical Theory of Communication," *Bell System Technical Journal*, 27, 1948. Still readable, and still the best introduction to its subject.
- P. J. Nahin, *Oliver Heaviside: The Life, Work, and Times of an Electrical Genius of the Victorian Age*, Johns Hopkins, 2002.
- A. C. Clarke, "Extra-Terrestrial Relays," *Wireless World*, October 1945.

Signals, Spectra and Systems

A communication engineer thinks in two domains at once: the time domain, where a signal is a waveform on an oscilloscope, and the frequency domain, where it is a distribution of energy across a spectrum analyzer screen. This chapter builds fluency in both. We start from phasors and Fourier series, develop the Fourier transform and its properties, study linear time-invariant systems and the distortion they introduce, define bandwidth in the several ways engineers actually use, move to discrete time with the DFT and FFT, and finish with the complex-envelope representation of bandpass signals that the rest of the book is written in.

2.1 Signals as the language of communication

A **signal** is any quantity that varies with time (or space) and carries information: the voltage from a microphone, the current in a telegraph line, the electric field at an antenna, the light intensity in a fibre. Communication engineering is the art of shaping signals so that the information they carry survives a journey through a physical medium. The medium always does three things to a signal: it *attenuates* it, it *distorts* it (different frequencies are delayed and attenuated differently), and it *adds noise*. The tools of this chapter describe the first two exactly; Chapter 3 describes the third.

We use a handful of standard signal classes. *Deterministic* signals are known functions of time, like a carrier $\cos(2\pi f_c t)$ or a pulse shape $p(t)$. *Random* signals, like noise or a stream of data symbols, are described statistically. *Energy signals* have finite energy $E = \int |x(t)|^2 dt$ (a pulse); *power signals* have finite average power $P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$ (a sinusoid, or a data stream that goes on indefinitely). Real signals are measured by real instruments, but we will quickly find that complex-valued signals are the natural mathematical description of real radio waves.

2.2 Sinusoids, phasors and negative frequency

The sinusoid $x(t) = A \cos(2\pi f_0 t + \phi)$ is the atom of signal analysis, for a simple reason: linear time-invariant systems (Section 2.5) map sinusoids to sinusoids of the same frequency, changing only amplitude and phase. Euler's formula,

$$e^{j\theta} = \cos \theta + j \sin \theta, \quad (2.1)$$

lets us write the sinusoid as the real part of a rotating complex number, the **phasor**:

$$A \cos(2\pi f_0 t + \phi) = \operatorname{Re}\{A e^{j\phi} e^{j2\pi f_0 t}\} = \frac{A}{2} e^{j\phi} e^{j2\pi f_0 t} + \frac{A}{2} e^{-j\phi} e^{-j2\pi f_0 t}. \quad (2.2)$$

The second form, shown in Figure 2.1, is the source of the *negative frequencies* that appear in every spectrum in this book. A real cosine is the sum of two counter-rotating phasors, one at $+f_0$ and one at $-f_0$; their imaginary parts cancel and their real parts add. Negative frequency is not a mathematical fiction: a complex signal $e^{-j2\pi f_0 t}$ is a phasor rotating clockwise, and a radio receiver with I and Q outputs can tell it apart from one rotating counter-clockwise. For a *real* signal, however, the spectrum at $-f$ is always the complex conjugate of the spectrum at $+f$, so the negative-frequency half carries no new information.

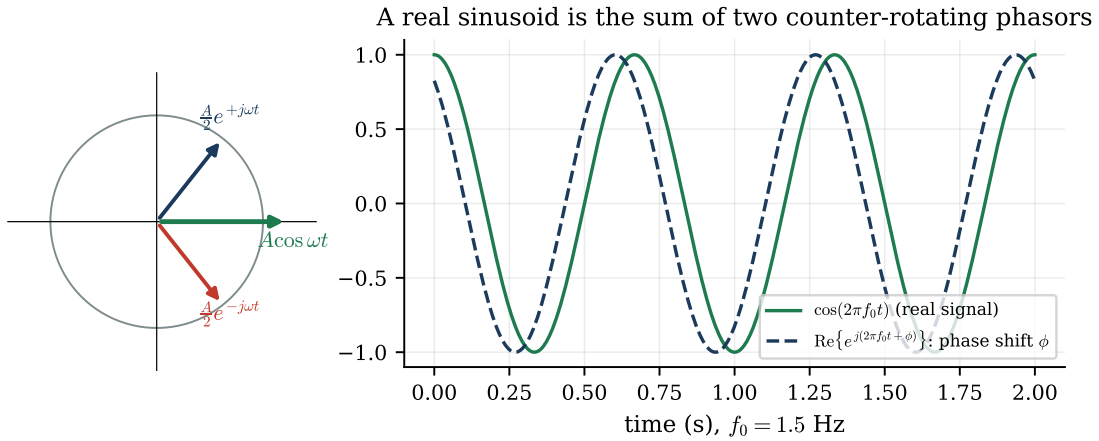


Figure 2.1: Left: a real cosine is the sum of two phasors of half amplitude rotating in opposite directions. Right: a phase offset ϕ rotates the phasor and shifts the waveform in time. Carrier phase recovery (Chapter ??) is the problem of estimating ϕ .

KEY IDEA Think in phasors

Multiplying a signal by $e^{j2\pi\Delta f t}$ shifts its whole spectrum by Δf ; multiplying by $e^{j\phi}$ rotates it by ϕ . Almost every operation in a radio—mixing, tuning, carrier recovery, Doppler correction, OFDM—is one of these two multiplications.

2.3 Fourier series

In 1807 Joseph Fourier claimed, to the scepticism of Lagrange, that any periodic function could be written as a sum of sinusoids. For a signal with period T_0 and fundamental frequency $f_0 = 1/T_0$,

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{j2\pi k f_0 t}, \quad c_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-j2\pi k f_0 t} dt. \quad (2.3)$$

The coefficients c_k are the amplitudes of the harmonics. For a square wave of unit amplitude, only odd harmonics appear and their amplitudes fall as $4/(\pi k)$, as Figure 2.2 shows. Partial sums converge to the square wave everywhere except at the jumps, where they overshoot by about 9% no matter how many terms are used: the **Gibbs phenomenon**. It reappears whenever a spectrum is truncated abruptly—in the ringing of a sharp filter’s step response and in the sidelobes of a rectangular window (Section 2.7.2).

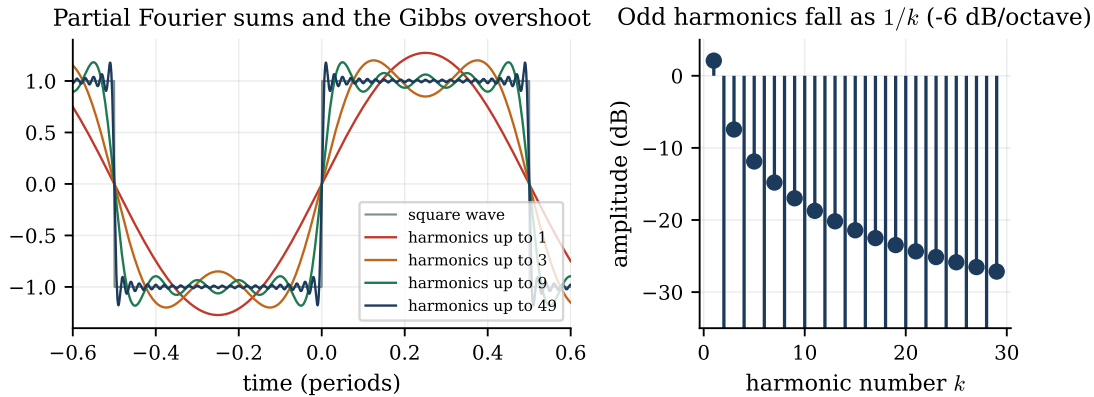


Figure 2.2: Fourier synthesis of a square wave. The overshoot at the discontinuity does not shrink as harmonics are added; it only narrows. The spectrum falls at only 6 dB per octave because the waveform has jumps.

IN PRACTICE Harmonics are everywhere

A transmitter's power amplifier driven into saturation turns a sinusoid into something closer to a square wave, generating harmonics at $2f_c$, $3f_c$ and above. Regulators set strict limits on these *spurious emissions*—the FCC's general limit for many licensed services is 43 dB plus $10 \log_{10}$ of the transmit power in watts below the carrier—so every transmitter ends with a harmonic filter. The rate at which harmonics decay is set by the smoothness of the waveform: a jump gives $1/k$ decay, a jump in slope gives $1/k^2$, and so on. This is the same smoothness–bandwidth trade that governs pulse shaping in Chapter ??.

2.4 The Fourier transform

Letting the period grow without bound turns the Fourier series into the Fourier transform, which describes non-periodic signals:

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt, \quad x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi ft} df. \quad (2.4)$$

We use ordinary frequency f in hertz rather than angular frequency ω because it removes all the stray factors of 2π and because engineers read spectra in hertz. $X(f)$ is complex: its magnitude says how much of each frequency is present, and its phase says how the components are aligned in time.

2.4.1 Properties worth knowing by heart

Table 2.1 lists the properties used most often. Each has a physical meaning. *Time shift* multiplies the spectrum by a linear phase: delay is phase slope. *Frequency shift* (modulation) moves the spectrum: this is how every transmitter places a signal at its carrier. *Scaling* says that compressing a signal in time stretches its spectrum: short pulses need wide bandwidth. *Convolution* in time is multiplication in frequency: this is why filters are described by frequency responses. *Parseval's theorem* says energy can be computed in either domain.

Table 2.1: Fourier transform properties ($x(t) \leftrightarrow X(f)$).

Property	Time domain	Frequency domain
Linearity	$a x(t) + b y(t)$	$a X(f) + b Y(f)$
Time shift	$x(t - \tau)$	$e^{-j2\pi f\tau} X(f)$
Frequency shift	$x(t) e^{j2\pi f_0 t}$	$X(f - f_0)$
Modulation	$x(t) \cos(2\pi f_0 t)$	$\frac{1}{2}[X(f - f_0) + X(f + f_0)]$
Scaling	$x(at)$	$\frac{1}{ a } X(f/a)$
Conjugation	$x^*(t)$	$X^*(-f)$
Convolution	$x(t) * y(t)$	$X(f) Y(f)$
Multiplication	$x(t) y(t)$	$X(f) * Y(f)$
Differentiation	$\frac{d}{dt} x(t)$	$j2\pi f X(f)$
Duality	$X(t)$	$x(-f)$
Parseval	$\int x(t) ^2 dt$	$\int X(f) ^2 df$

2.4.2 Transform pairs

A few pairs appear over and over (Figure 2.3). The rectangular pulse of width T transforms to a sinc function,

$$\text{rect}(t/T) \leftrightarrow T \text{sinc}(fT), \quad \text{sinc}(x) = \frac{\sin \pi x}{\pi x}, \quad (2.5)$$

whose first nulls are at $\pm 1/T$ and whose sidelobes decay slowly, as $1/f$. By duality, an ideal “brick-wall” low-pass spectrum corresponds to a sinc pulse in time—the pulse Nyquist proposed for signalling without intersymbol interference. The triangle, the convolution of two rectangles, transforms to sinc^2 with faster-decaying sidelobes. The Gaussian is its own transform, and among all pulses it minimises the product of time duration and bandwidth (measured as standard deviations): this is the time–bandwidth *uncertainty principle*, $\sigma_t \sigma_f \geq 1/(4\pi)$. GSM’s GMSK modulation uses Gaussian-filtered pulses for exactly this reason (Chapter ??). Two idealised pairs complete the toolkit: the impulse $\delta(t) \leftrightarrow 1$, and the impulse train $\sum_n \delta(t - nT) \leftrightarrow \frac{1}{T} \sum_k \delta(f - k/T)$, which is the mathematics of sampling.

2.4.3 Energy and power spectral density

For an energy signal, $|X(f)|^2$ is the **energy spectral density**: it tells how the energy is distributed in frequency. For power signals—which include every random data stream—we need the **power spectral density** (PSD) $S_x(f)$, defined through the autocorrelation function by the Wiener–Khinchin theorem (Chapter 3):

$$S_x(f) = \int_{-\infty}^{\infty} R_x(\tau) e^{-j2\pi f\tau} d\tau, \quad R_x(\tau) = \mathbb{E}[x(t + \tau)x^*(t)]. \quad (2.6)$$

The PSD is what a spectrum analyzer displays, what a regulator’s emission mask constrains, and what we compute for every modulation format in this book.

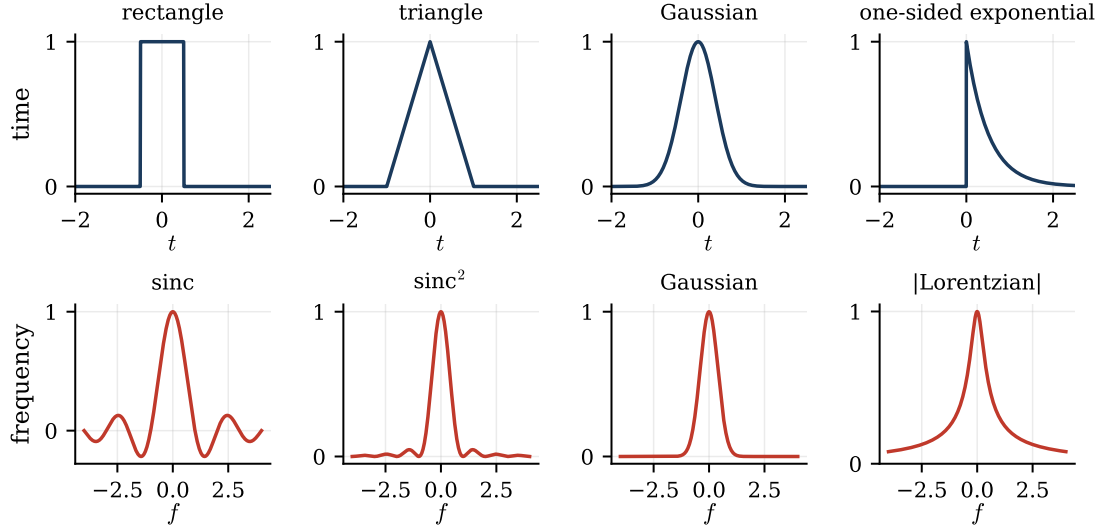


Figure 2.3: Four Fourier transform pairs. Smoother time functions have spectra that decay faster; narrower time functions have wider spectra.

2.5 Linear time-invariant systems

A system is *linear* if the response to a sum of inputs is the sum of the responses, and *time-invariant* if delaying the input merely delays the output. Wires, cables, filters, amplifiers operated in their linear range, and radio channels over short enough intervals are all well modelled as LTI systems. An LTI system is completely described by its **impulse response** $h(t)$, and its output is the convolution

$$y(t) = \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau \iff Y(f) = H(f) X(f). \quad (2.7)$$

The **frequency response** $H(f)$ is the transform of $h(t)$. A sinusoid at frequency f passes through with its amplitude multiplied by $|H(f)|$ and its phase advanced by $\angle H(f)$. The two numbers that summarise an LTI channel are therefore the gain and the phase at every frequency—and a channel is “good” when both are well behaved across the signal’s band.

2.5.1 Distortionless transmission

A channel that only scales and delays, $y(t) = a x(t - \tau)$, has $H(f) = a e^{-j2\pi f\tau}$: constant magnitude and *linear* phase. Any departure from either causes distortion. Amplitude distortion changes the relative sizes of frequency components; phase distortion misaligns them in time. It is convenient to describe phase behaviour by the **group delay**,

$$\tau_g(f) = -\frac{1}{2\pi} \frac{d\angle H(f)}{df}, \quad (2.8)$$

the delay experienced by a narrow packet of energy near frequency f . A distortionless channel has constant group delay; Heaviside’s distortionless line of Chapter 1 is a physical example.

2.5.2 Filters: the classic trade-offs

Figure 2.4 compares four classic fifth-order analog low-pass filter families, all designed for a 1 kHz cutoff. The *Butterworth* filter is maximally flat in the passband. The *Chebyshev* filter allows passband ripple in exchange for a steeper transition. The *elliptic* (Cauer) filter allows ripple in both bands and gives the steepest transition of all for a given order. The *Bessel* filter gives up selectivity for nearly constant group delay. The time-domain panel reveals the price of selectivity: the sharp filters have large group-delay peaks near the band edge, and a rectangular pulse passing through them rings. For data signals, where the pulse shape is the information, this ringing is intersymbol interference; that is why data receivers often prefer linear-phase digital FIR filters (Chapter 6).

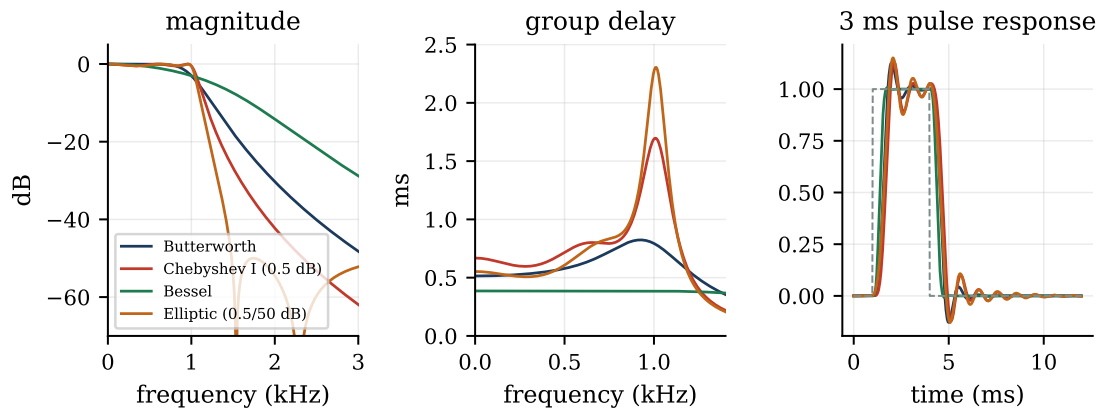


Figure 2.4: Four fifth-order low-pass filters with a 1 kHz cutoff. Left: magnitude. Centre: group delay. Right: response to a 3 ms rectangular pulse. Steeper magnitude responses bring larger group-delay variation and more ringing.

WORKED EXAMPLE Filter selection for an audio channel

A voice channel must pass 300–3400 Hz and reject a 4 kHz pilot tone by 40 dB. The transition band is only $4000/3400 = 1.18$ octaves wide. Standard design charts give the required orders as roughly 15 (Butterworth), 7 (Chebyshev, 0.5 dB ripple) or 5 (elliptic). The elliptic design is the cheapest in components but has the worst group delay; for voice, where the ear is insensitive to moderate phase distortion, it is the usual choice. For a data modem using the same channel, the group-delay variation would have to be equalized (Chapter ??).

2.6 Bandwidth

“Bandwidth” sounds like a single number, but there are several useful definitions, and confusing them causes real errors in system design. Figure 2.5 illustrates them for the spectrum of a rectangular pulse of duration T .

3 dB bandwidth: the width of the band where the PSD is within 3 dB of its peak; for the rectangular pulse, about $0.89/T$.

Null-to-null bandwidth: the width of the main lobe, $2/T$ here.

Occupied bandwidth: the band containing a stated fraction, usually 99%, of the power. This is the regulator's definition, and for the rectangular pulse it is over $20/T$ because the sidelobes decay so slowly.

Noise-equivalent bandwidth: the width B_N of an ideal rectangular filter that passes the same white-noise power as the actual filter, $B_N = \int |H(f)|^2 df / |H(0)|^2$. It is the right number to use in noise calculations (Chapter 3).

Nyquist bandwidth: the minimum bandwidth, $R_s/2$ at baseband, in which symbols at rate R_s can be sent without intersymbol interference (Chapter ??).

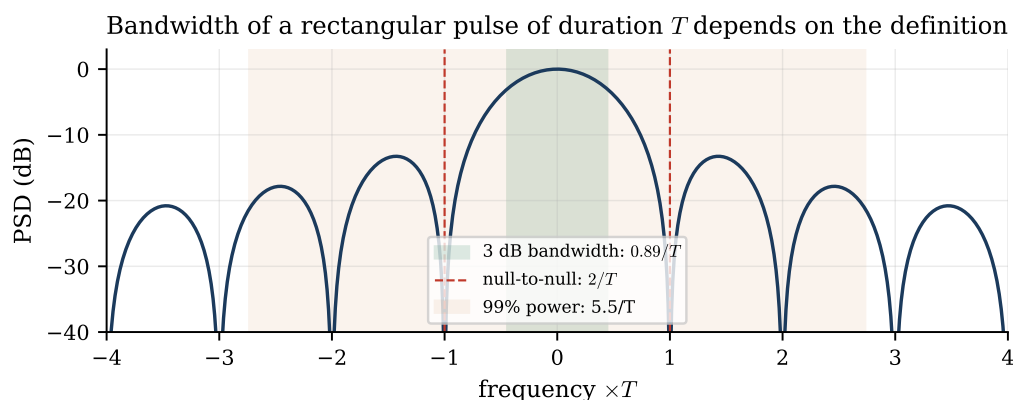


Figure 2.5: Three bandwidth definitions applied to the spectrum of a rectangular pulse. Because the sidelobes decay slowly, the 99% occupied bandwidth is more than ten times the null-to-null bandwidth.

PITFALL Comparing bandwidths across definitions

A datasheet may quote a filter's 3 dB bandwidth, a standard may specify an occupied bandwidth, and a noise calculation needs a noise-equivalent bandwidth. For a root-raised-cosine spectrum with roll-off 0.35 at 1 Msymbol/s, the three are roughly 1.0, 1.3 and 1.0 MHz; for a Gaussian filter they differ more. Always state which definition you are using.

2.7 Discrete time: from DTFT to FFT

Modern radios process sampled signals. If $x[n] = x(nT_s)$ is sampled at rate $f_s = 1/T_s$, its spectrum is the *discrete-time Fourier transform*

$$X(e^{j2\pi f/f_s}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j2\pi f n/f_s}, \quad (2.9)$$

which is periodic in f with period f_s . The periodicity is the frequency-domain face of sampling: a sampled signal cannot distinguish f from $f + kf_s$. Chapter 5 develops the sampling theorem

and aliasing in full; for now it is enough to remember that all the information of a correctly sampled signal lies in one period, $-f_s/2 \leq f < f_s/2$.

2.7.1 The discrete Fourier transform

A computer can only handle a finite block of N samples and a finite set of frequencies. The **discrete Fourier transform** (DFT) evaluates the spectrum at N equally spaced frequencies $f_k = kf_s/N$:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}, \quad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N}. \quad (2.10)$$

The DFT treats the block as one period of a periodic signal. That implicit periodicity has two consequences that dominate practical spectrum analysis: *leakage* and *resolution*.

2.7.2 Windows and leakage

If a sinusoid does not complete an integer number of cycles in the block, the periodic extension has a jump at the block boundary, and energy “leaks” from the true frequency into all other bins, following the sidelobes of the sinc-shaped transform of the rectangular block. Multiplying the block by a smooth **window** that tapers to zero at both ends removes the jump. Figure 2.6 shows the trade: the rectangular window has the narrowest main lobe but sidelobes only 13 dB down, so a weak tone 60 dB below a strong one 30 Hz away is completely hidden. The Hann window lowers sidelobes to -31 dB with a faster roll-off, the Blackman–Harris window to -92 dB, and the flat-top window sacrifices resolution for amplitude accuracy. Choosing a window is choosing between resolution, dynamic range and amplitude accuracy.

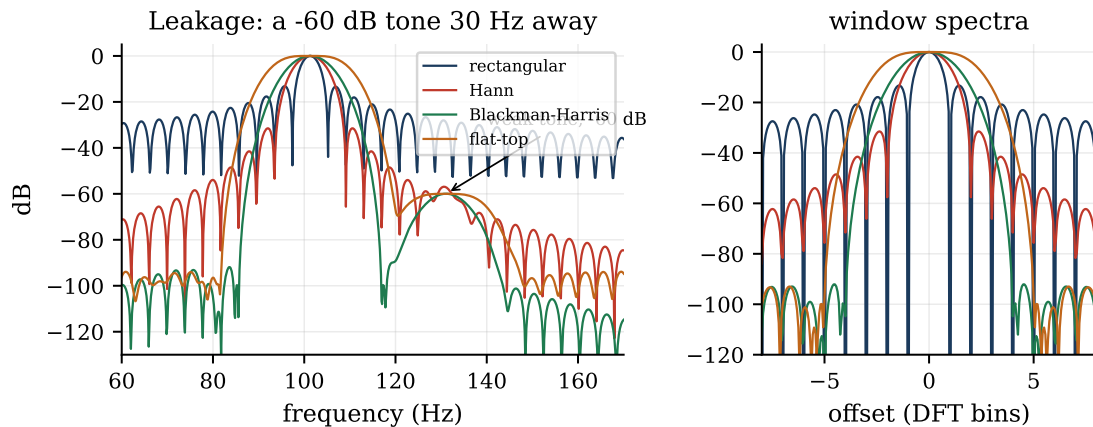


Figure 2.6: Left: DFT of a strong tone and a tone 60 dB weaker, 30 Hz away, with four windows. Only low-sidelobe windows reveal the weak tone. Right: the spectra of the windows themselves, in units of DFT bins.

2.7.3 Resolution and zero padding

Two tones closer together than about f_s/N Hz—the reciprocal of the observation time NT_s —cannot be resolved, no matter how the data are processed. Padding the block with zeros

computes the spectrum at more closely spaced frequencies, which makes plots smoother and helps locate an isolated peak precisely, but it does not add resolution (Figure 2.7). Resolution requires observation time.

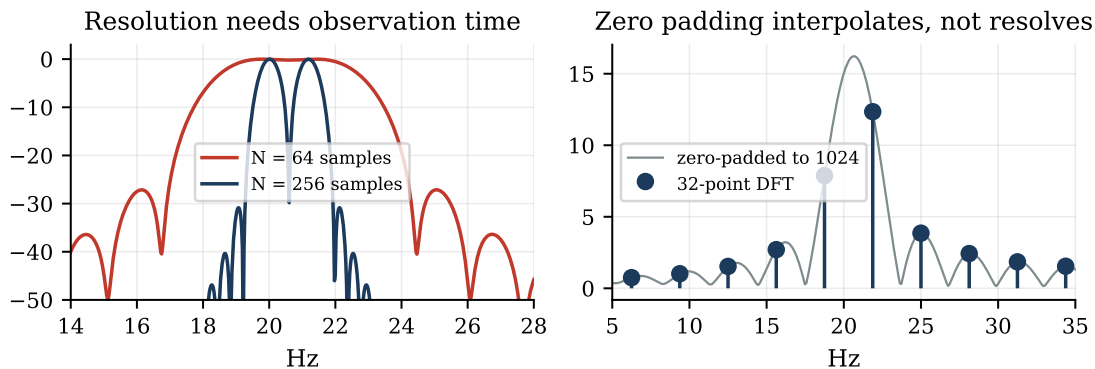


Figure 2.7: Left: two tones 1.2 Hz apart are merged with 64 samples at 100 Hz and resolved with 256. Right: zero padding a 32-point DFT interpolates the same underlying spectrum more finely.

2.7.4 The fast Fourier transform

A direct DFT takes N^2 complex multiplications. In 1965 James Cooley and John Tukey published an algorithm that computes it in about $\frac{N}{2} \log_2 N$ by recursively splitting the sum into even- and odd-indexed samples. (Carl Friedrich Gauss had used the same trick in 1805 to interpolate the orbits of asteroids, but the idea went unnoticed.) For $N = 4096$, the FFT is about 700 times faster. The FFT made digital spectrum analysis routine, and forty years later it made OFDM practical: a Wi-Fi 7 receiver computes a 4096-point FFT every $12.8 \mu\text{s}$ (Chapter ??).

IN PRACTICE Swept and FFT spectrum analyzers

A classic swept spectrum analyzer is a superheterodyne receiver whose local oscillator sweeps across the band; its *resolution bandwidth* (RBW) is the width of its IF filter, and the sweep must be slow enough for that filter to settle, roughly sweep time $\approx k \cdot \text{span} / \text{RBW}^2$. A real-time or FFT analyzer instead digitises a band and computes overlapping windowed FFTs, capturing every event in the band. The displayed noise floor rises by $10 \log_{10}$ of the RBW: halving RBW lowers the noise floor by 3 dB and reveals weaker signals, at the cost of time. GNU Radio's QT GUI Frequency Sink in the gro1 hardware example is an FFT analyzer, and its FFT size sets the RBW.

2.8 Bandpass signals and the complex envelope

Radio signals occupy a narrow band around a carrier frequency f_c that is much larger than their bandwidth: a 20 MHz Wi-Fi channel at 5.5 GHz, a 200 kHz FM station at 100 MHz. Such **bandpass signals** can be described completely by a low-frequency complex signal, the **complex envelope** or *baseband equivalent*.

2.8.1 The analytic signal and the Hilbert transform

Because a real signal's spectrum has Hermitian symmetry, its negative-frequency half is redundant. The **analytic signal** removes it and doubles the positive half:

$$x_+(t) = x(t) + j\hat{x}(t), \quad X_+(f) = 2X(f)u(f), \quad (2.11)$$

where $u(f)$ is the unit step and $\hat{x}(t)$ is the **Hilbert transform** of $x(t)$, obtained by a filter with $H(f) = -j \operatorname{sgn}(f)$ that shifts every positive frequency by -90° . The Hilbert transform of $\cos$ is $\sin$. Shifting the analytic signal down by f_c gives the complex envelope,

$$\tilde{x}(t) = x_+(t) e^{-j2\pi f_c t} = x_I(t) + jx_Q(t), \quad x(t) = \operatorname{Re}\{\tilde{x}(t) e^{j2\pi f_c t}\}. \quad (2.12)$$

Figure 2.8 shows the three spectra. The complex envelope is centred on zero frequency and has the same bandwidth as the original signal's positive half, so it can be sampled at a rate set by the signal bandwidth rather than the carrier frequency. Expanding the real part in (2.12) gives the I/Q form every radio engineer knows,

$$x(t) = x_I(t) \cos(2\pi f_c t) - x_Q(t) \sin(2\pi f_c t). \quad (2.13)$$

The in-phase component x_I modulates a cosine carrier and the quadrature component x_Q modulates a sine carrier; because the two carriers are orthogonal, the receiver can separate them.

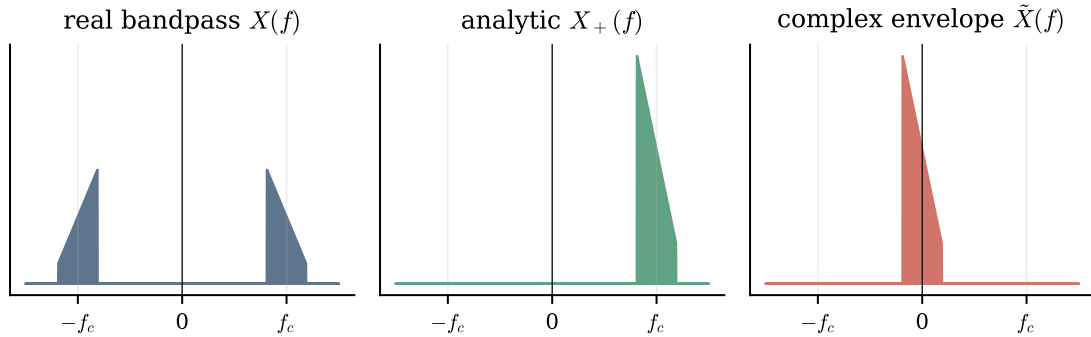


Figure 2.8: From a real bandpass spectrum (left) to the analytic signal (centre) to the complex envelope (right). The complex envelope need not be symmetric about zero.

2.8.2 Baseband-equivalent systems

If a bandpass signal passes through a bandpass system with impulse response $h(t) = \operatorname{Re}\{\tilde{h}(t)e^{j2\pi f_c t}\}$, the complex envelope of the output is simply $\tilde{y}(t) = \tilde{h}(t) * \tilde{x}(t)$. Every bandpass calculation can therefore be done at baseband, with complex numbers and a sample rate set by the bandwidth. One consequence deserves emphasis: a pure delay τ in the passband becomes, at baseband, a delay τ and a phase rotation $e^{-j2\pi f_c \tau}$. At a carrier of 2.4 GHz, a path-length change of 12.5 cm rotates the phase by a full cycle. That single fact explains carrier recovery, fading and beamforming, the subjects of Chapters ??, ?? and ??.

KEY IDEA Why receivers produce I and Q

A quadrature downconverter multiplies the received signal by $\cos(2\pi f_c t)$ and by $-\sin(2\pi f_c t)$ and low-pass filters both products. The two outputs are exactly $x_I(t)$ and $x_Q(t)$. The B200's AD9364 transceiver does this in analog circuitry before its ADCs; Chapter 7 examines what happens when the two paths are not perfectly matched.

SIMULATION LAB Signals and spectra

Lab 1 (`lab01_baseband.ipynb`) performs up- and down-conversion of a two-tone signal with an asymmetric baseband spectrum and demonstrates the image created by IQ imbalance. The chapter's figure script, `figscripts/ch02_figs.py`, contains the window, resolution and filter comparisons; modify the tone spacing and window length to find the point where each window fails to separate the tones.

Problems

- 2.1 Show from (2.3) that the Fourier coefficients of a real signal satisfy $c_{-k} = c_k^*$. What does this imply about the spectrum at negative frequencies?
- 2.2 A rectangular pulse of duration $1\ \mu\text{s}$ is transmitted. Compute its null-to-null bandwidth and estimate its 99% occupied bandwidth numerically. Repeat for a triangular pulse of the same energy.
- 2.3 Prove Parseval's theorem using the convolution property. Use it to show that the energy of $\text{sinc}(t)$ is 1.
- 2.4 A channel has $H(f) = 1 + \alpha e^{-j2\pi f\tau}$ (a direct path plus one echo). Sketch $|H(f)|$ and the group delay for $\alpha = 0.5$ and $\tau = 1\ \mu\text{s}$. At which frequencies are the notches, and what is their depth?
- 2.5 A 1024-point FFT is computed on data sampled at 1.024 MHz with a Hann window. What is the bin spacing? What is the noise-equivalent bandwidth of each bin (the Hann window's equivalent noise bandwidth is 1.5 bins)? By how much does the noise floor per bin drop if the FFT length is doubled?
- 2.6 Show that the Hilbert transform of $\cos(2\pi f_0 t)$ is $\sin(2\pi f_0 t)$, and find the analytic signal and complex envelope (with respect to f_c) of $x(t) = \cos(2\pi f_0 t)$ when $f_0 = f_c + \Delta$.
- 2.7 (*Simulation*) Using the figure script, find the minimum tone separation that each of the four windows can resolve when the tones are equal in amplitude, and when one is 40 dB weaker. Explain the results in terms of main-lobe width and sidelobe level.

Further Reading

- A. V. Oppenheim and A. S. Willsky, *Signals and Systems*, 2nd ed., Prentice Hall, 1997. The standard text.

- R. N. Bracewell, *The Fourier Transform and Its Applications*, 3rd ed., McGraw-Hill, 2000. Physical intuition for every transform pair.
- F. J. Harris, "On the use of windows for harmonic analysis with the discrete Fourier transform," *Proceedings of the IEEE*, 66(1), 1978. The definitive window reference.
- R. G. Lyons, *Understanding Digital Signal Processing*, 3rd ed., Prentice Hall, 2010. Exceptionally clear on the DFT, windows and complex signals.

Random Signals and Noise

If there were no noise, a single sample of a voltage could carry an infinite number of bits, and communication engineering would be trivial. Noise is what makes the subject hard and interesting. This chapter gives the probability and random-process tools the rest of the book needs, explains where noise comes from physically, shows how engineers characterise receivers with noise figure and noise temperature, and ends with the link budget—the single calculation that decides whether a communication system will work at all, from a Wi-Fi access point to a spacecraft at the edge of the solar system.

3.1 The floor under everything

In 1928 John B. Johnson, measuring amplifiers at Bell Labs, found that every resistor produced a small fluctuating voltage whose power was proportional to its temperature and to the measurement bandwidth, and independent of what the resistor was made of. Harry Nyquist explained the result the same year from thermodynamics. Thermal noise is not a defect of any particular device; it is the random motion of charge carriers in any conductor above absolute zero, and it sets a floor below which no receiver can hear. Every other impairment—distortion, interference, fading—can in principle be designed away. Noise can only be outwitted: by using more power, more bandwidth, more antennas, longer observation, or cleverer codes. To reason about it we need probability.

3.2 Random variables in communication

3.2.1 The Gaussian distribution and the central limit theorem

A random variable X is described by its probability density function (pdf) $p_X(x)$, its mean $\mu = \mathbb{E}[X]$ and variance $\sigma^2 = \mathbb{E}[(X - \mu)^2]$. The most important distribution in communications is the Gaussian (normal),

$$p_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}. \quad (3.1)$$

Its importance comes from the **central limit theorem**: the sum of many independent random contributions of comparable size tends to a Gaussian, whatever the individual distributions. Thermal noise is the sum of the effects of an astronomical number of electrons; the interference from many distant transmitters, the error of a quantizer with many levels and the sum of many multipath components all tend to Gaussian for the same reason. Figure 3.1 shows how quickly the convergence happens: the sum of just four uniform random variables

is already hard to tell from a Gaussian by eye, though the tails—which matter for error probability—converge more slowly.

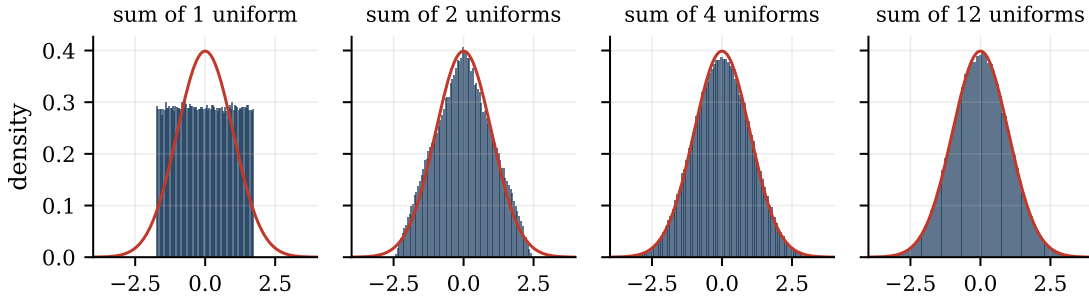


Figure 3.1: The central limit theorem in action: normalised sums of n independent uniform random variables (histograms) approach the Gaussian density (red curves).

3.2.2 The Q function

Nearly every error probability in this book is a Gaussian tail probability. We define the **Q function** as the probability that a zero-mean, unit-variance Gaussian exceeds x ,

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-u^2/2} du = \frac{1}{2} \operatorname{erfc}\left(x/\sqrt{2}\right). \quad (3.2)$$

If a receiver decides between $+A$ and $-A$ corrupted by Gaussian noise of standard deviation σ , the error probability is $Q(A/\sigma)$. The function falls extremely fast (Figure 3.2): $Q(3.09) = 10^{-3}$, $Q(4.75) = 10^{-6}$, $Q(6.0) = 10^{-9}$. Increasing the argument by about 50%—a 3.5 dB increase in signal-to-noise ratio—reduces the error rate by a factor of a thousand. This steepness is why a working digital link has a “cliff”: a few decibels separate perfect reception from none. The bounds $Q(x) \leq \frac{1}{2}e^{-x^2/2}$ and $Q(x) < e^{-x^2/2}/(x\sqrt{2\pi})$ are useful for quick estimates and for proofs.

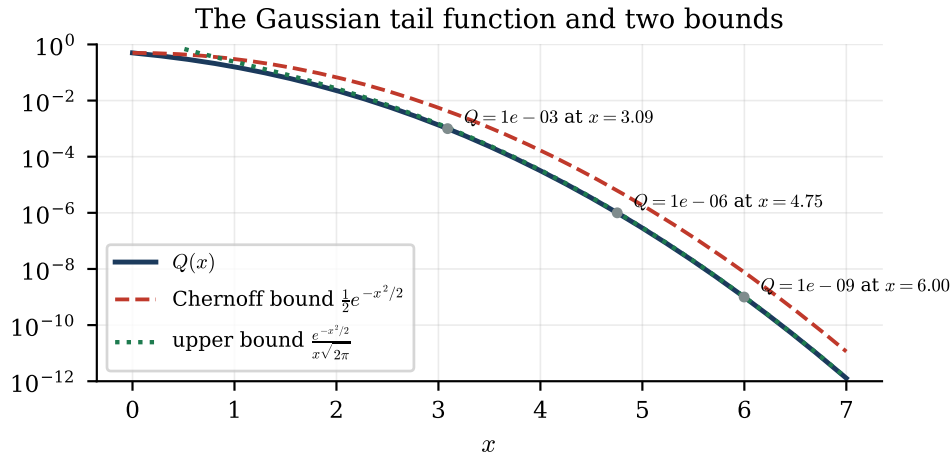


Figure 3.2: The Q function (Gaussian tail probability) on a logarithmic scale, with two classical bounds. Values at three reference error rates are marked.

3.2.3 Complex Gaussian noise and its envelope

Because receivers work with complex baseband signals (Section 2.8), noise is complex: $n = n_I + jn_Q$ with independent zero-mean Gaussian parts of equal variance $\sigma^2 = N_0/2$. Such noise is *circularly symmetric*: its distribution does not change under rotation, so its phase is uniform on $[-\pi, \pi)$ and its total variance is $\mathbb{E}|n|^2 = N_0$. The magnitude $R = |n|$ has the **Rayleigh distribution**

$$p_R(r) = \frac{r}{\sigma^2} e^{-r^2/2\sigma^2}, \quad r \geq 0, \quad (3.3)$$

and the power $|n|^2$ is exponentially distributed. If a constant component s is added—a sinusoid plus noise, or a line-of-sight path plus scattered paths—the envelope follows the **Rice distribution**,

$$p_R(r) = \frac{r}{\sigma^2} \exp\left(-\frac{r^2 + s^2}{2\sigma^2}\right) I_0\left(\frac{rs}{\sigma^2}\right), \quad (3.4)$$

characterised by the *K-factor* $K = s^2/2\sigma^2$, the ratio of the steady power to the random power (Figure 3.3). Stephen Rice derived these distributions in 1944 to describe radio noise; they returned as the models of multipath fading in Chapter ??.

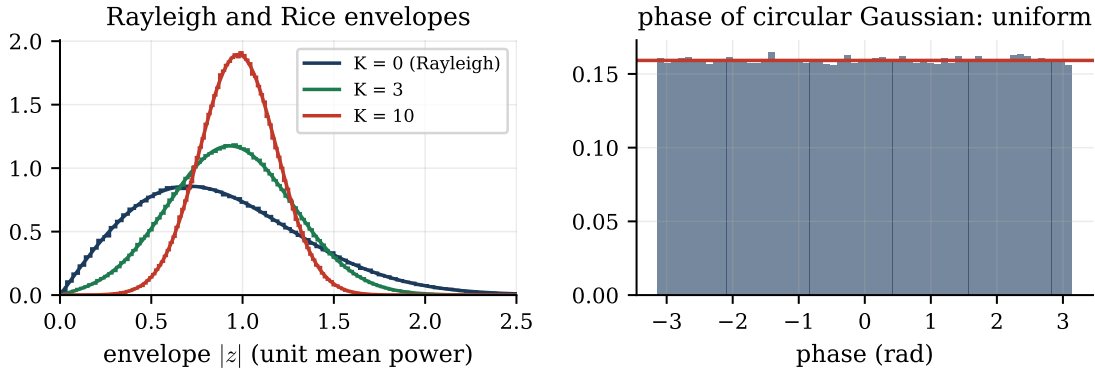


Figure 3.3: Left: envelope distributions of a constant plus circular Gaussian noise, for three Rice K -factors (histograms from simulation, curves from theory). Right: the phase of circularly symmetric Gaussian noise is uniformly distributed.

3.3 Random processes

3.3.1 Stationarity, correlation and spectrum

A **random process** $x(t)$ is a random variable at every instant. We describe it by its mean and its autocorrelation function $R_x(t_1, t_2) = \mathbb{E}[x(t_1)x^*(t_2)]$. If the mean is constant and the autocorrelation depends only on the difference $\tau = t_1 - t_2$, the process is *wide-sense stationary* (WSS), and its power spectral density is the Fourier transform of the autocorrelation—the **Wiener–Khinchin theorem**:

$$S_x(f) = \int R_x(\tau) e^{-j2\pi f\tau} d\tau, \quad \mathbb{E}|x(t)|^2 = R_x(0) = \int S_x(f) df. \quad (3.5)$$

The autocorrelation measures how quickly a process “forgets” its past; the PSD says how its power is spread over frequency. A process that changes quickly has a narrow autocorrelation and a wide spectrum, and vice versa—the same time–bandwidth reciprocity as for deterministic signals.

Many communication signals are not quite stationary: a data stream sent at symbol rate R_s has statistics that repeat with period $1/R_s$. Such *cyclostationary* processes are handled by averaging the autocorrelation over one period, which yields the PSD formulas of Chapter ?? . Cyclostationarity is also a resource: it lets a receiver detect and classify signals buried in noise, because noise has no such periodic structure.

3.3.2 Filtering random processes

When a WSS process passes through an LTI filter $H(f)$, the output is WSS with

$$S_y(f) = |H(f)|^2 S_x(f). \quad (3.6)$$

This single equation carries most of the noise analysis in this book. It says, for example, that noise power at a receiver output is the input noise PSD integrated over the receiver’s *power* response, which is why the noise-equivalent bandwidth of Section 2.6 is the right bandwidth to use.

3.3.3 White noise

Thermal noise has a flat PSD from DC to far beyond any radio frequency (up to about 6 THz at room temperature, where quantum effects set in). We idealise it as **white noise** with two-sided PSD $N_0/2$ W/Hz, whose autocorrelation is an impulse: $R_n(\tau) = \frac{N_0}{2}\delta(\tau)$. White noise has infinite power and does not exist, but once filtered it becomes perfectly well behaved: through a filter of noise bandwidth B , the output power is N_0B (counting both positive and negative frequencies of a real filter). Figure 3.4 shows white noise and low-pass filtered noise: filtering smooths the sample paths, narrows the spectrum and widens the autocorrelation.

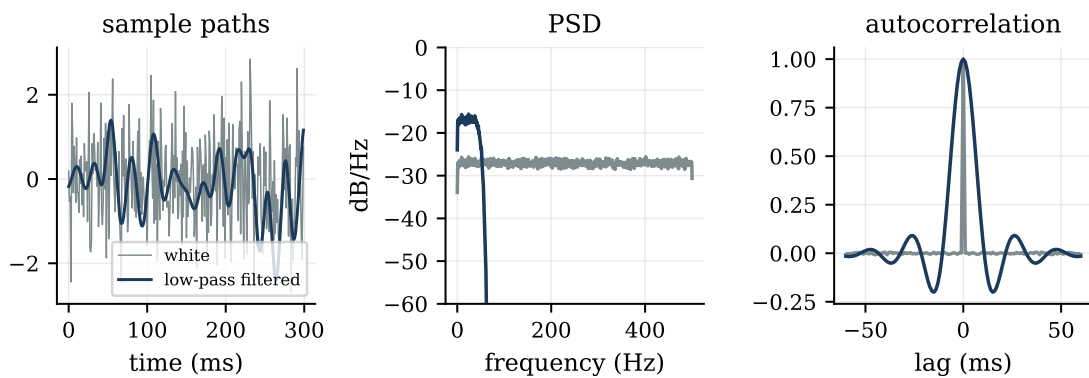


Figure 3.4: White Gaussian noise before and after a 50 Hz low-pass filter: sample paths, power spectral densities and autocorrelation functions.

3.3.4 Bandpass noise

At the output of a radio's IF filter, noise occupies a band around the carrier. Its complex envelope, like any bandpass signal's, can be written $\tilde{n}(t) = n_I(t) + jn_Q(t)$. If the filter is symmetric about f_c , the in-phase and quadrature noise components are independent, each with PSD N_0 over the baseband bandwidth $B/2$, and the complex baseband noise has PSD N_0 across $|f| < B/2$. This is the noise model used in every simulation in the labs: independent Gaussian I and Q samples, each of variance $N_0/2$.

PITFALL Factors of two

Noise PSDs are quoted one-sided (N_0 W/Hz over $f > 0$) or two-sided ($N_0/2$ over all f), and complex baseband noise has variance N_0 split equally between I and Q. Most errors of exactly 3 dB in simulations come from mixing these conventions. The labs use one convention throughout: $\mathbb{E}|n|^2 = N_0$ per complex sample and unit-energy symbols, so that E_s/N_0 is the ratio of symbol energy to per-sample noise variance at the matched-filter output.

3.4 Physical noise sources

3.4.1 Thermal noise

A resistor R at absolute temperature T can deliver to a matched load an available noise power

$$P_n = kTB, \quad k = 1.380649 \times 10^{-23} \text{ J/K}, \quad (3.7)$$

independent of R . At the reference temperature $T_0 = 290$ K, $kT_0 = 4.00 \times 10^{-21}$ W/Hz, or -174 dBm/Hz. This number deserves memorising, together with its common scalings: -144 dBm in 1 kHz, -114 dBm in 1 MHz, -101 dBm in 20 MHz. A GPS signal arrives at about -130 dBm spread over 2 MHz, some 16 dB *below* the thermal noise in that band—yet receivers track it routinely, by despreading (Chapter ??).

3.4.2 Other noise mechanisms

Shot noise arises because current consists of discrete charges crossing a barrier at random times; its PSD is $2qI$ (one-sided, in A^2/Hz), and it dominates in photodiodes and therefore in optical receivers (Chapter ??). *Flicker* or $1/f$ noise rises at low frequencies in all active devices; it is why zero-IF receivers, which move the signal to DC, must fight excess low-frequency noise, and why oscillators have close-in phase noise that rises steeply near the carrier. At very high frequencies and low temperatures the classical kT gives way to the quantum limit hf ; at 300 GHz and 290 K the two are still within a factor of 20, so quantum noise is relevant only in optical and deep-cryogenic systems.

3.5 Noise figure and noise temperature

3.5.1 Definitions

Every real amplifier adds noise. We characterise a two-port device by its **noise factor** F , the ratio of the input SNR to the output SNR when the input noise is thermal at $T_0 = 290$ K; its **noise figure** is $\text{NF} = 10 \log_{10} F$ dB. Equivalently, the device can be modelled as noiseless with an extra noise source at its input of **equivalent noise temperature**

$$T_e = (F - 1) T_0. \quad (3.8)$$

A 1 dB noise figure corresponds to $T_e = 75$ K; 3 dB to 290 K; 0.5 dB to 35 K. Noise temperature is the natural unit when the input noise is not at 290 K—for example, an antenna looking at the cold sky—and it is the unit satellite and radio-astronomy engineers prefer. A passive lossy component (cable, filter, attenuator) with loss L at physical temperature T_0 has noise factor $F = L$: a 2 dB cable in front of a receiver adds exactly 2 dB to the noise figure.

3.5.2 Cascades: the Friis formula

For a chain of stages with gains G_i and noise factors F_i , Harald Friis showed in 1944 that

$$F = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \dots, \quad T_e = T_{e1} + \frac{T_{e2}}{G_1} + \frac{T_{e3}}{G_1 G_2} + \dots \quad (3.9)$$

The noise of each stage is divided by the total gain before it. The first stage dominates, provided it has enough gain; this is the reason every sensitive receiver starts with a low-noise amplifier (LNA), and why the LNA should be as close to the antenna as possible. Figure 3.5 makes the point numerically: moving a 1 dB LNA from after a 2 dB cable and a 1.5 dB filter to directly at the antenna lowers the system noise figure from about 5.1 dB to 2.2 dB, the equivalent of doubling the transmitter power.

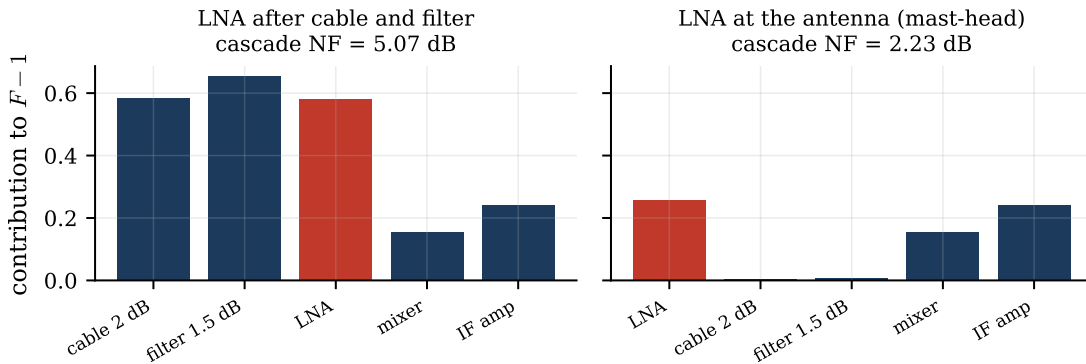


Figure 3.5: Contribution of each stage to the cascade noise factor for two orderings of the same components. Losses in front of the LNA add directly; everything after a high-gain LNA barely matters.

IN PRACTICE Mast-head amplifiers and satellite LNBs

Cellular base stations often place a tower-mounted amplifier at the top of the mast, before tens of metres of lossy coaxial cable. A satellite TV dish goes further: its low-noise

block downconverter (LNB) at the focus amplifies the 10.7–12.75 GHz signal with a noise temperature under 100 K and immediately converts it to 950–2150 MHz, where cable losses are tolerable. On the Ettus B200, the AD9364’s internal LNA gives a noise figure under 8 dB—fine for bench work, but an external LNA with 1 dB NF will improve weak-signal reception by several decibels if placed at the antenna.

3.5.3 Antenna and system noise temperature

An antenna collects noise from whatever it sees. Pointed at the cold sky at microwave frequencies, it may see only 10–50 K; pointed at the ground, about 290 K; at HF, galactic and man-made noise can reach millions of kelvin (Figure 3.6). The *system noise temperature* referred to the antenna terminals is $T_{\text{sys}} = T_{\text{ant}} + T_e$, and the receiver’s noise power is $kT_{\text{sys}}B$. The figure explains two facts of radio engineering. At HF (3–30 MHz), external noise is so strong that receiver noise figure hardly matters: a 10 dB NF receiver is as good as a perfect one. In the “microwave window” from about 1 to 10 GHz, the sky is quiet and receiver noise dominates, which is why satellite, deep-space and radio-astronomy systems live there and use cryogenically cooled amplifiers.

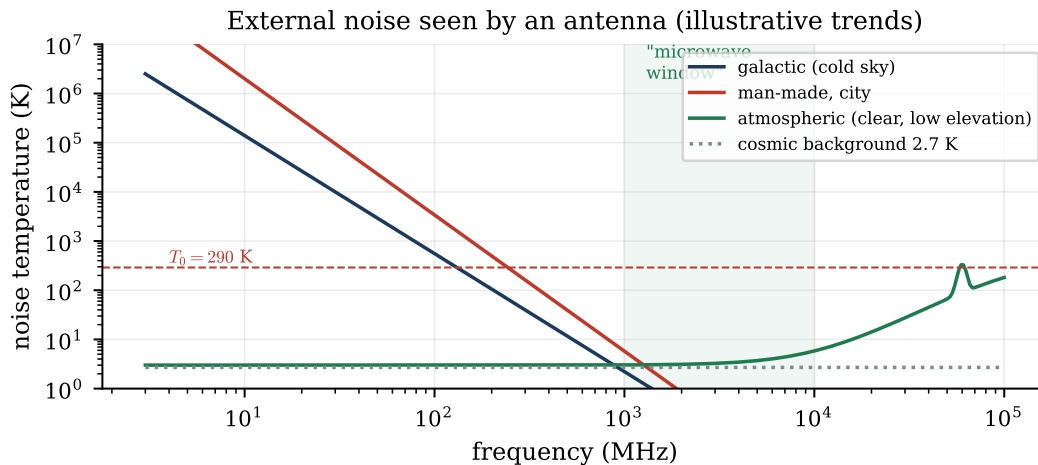


Figure 3.6: Illustrative external noise temperatures versus frequency. Below about 30 MHz, galactic and man-made noise dwarf receiver noise; between roughly 1 and 10 GHz the sky is cold and receiver noise dominates; above 20 GHz atmospheric absorption adds noise. (Trends after ITU-R P.372; values are order-of-magnitude.)

3.6 SNR, E_b/N_0 and C/N_0

Three ratios describe how strong a signal is relative to noise, and translating between them is daily work.

- The **signal-to-noise ratio** $\text{SNR} = P/(N_0B)$ depends on the bandwidth B in which it is measured.
- The **carrier-to-noise-density ratio** $C/N_0 = P/N_0$, in dB-Hz, is independent of bandwidth. It is the natural figure of merit for a link budget, because it can be computed

before the modulation is chosen. GPS receivers report it for each satellite; 45 dB-Hz is a strong GPS signal.

- The **energy per bit to noise density** $E_b/N_0 = P/(R_b N_0)$ normalises by the bit rate R_b . It is the fair basis for comparing modulation and coding schemes, because it charges each scheme for the energy it spends per bit of information.

They are related by

$$\frac{E_b}{N_0} = \frac{C}{N_0} \cdot \frac{1}{R_b} = \text{SNR} \cdot \frac{B}{R_b}, \quad \frac{E_s}{N_0} = \frac{E_b}{N_0} \cdot \log_2 M \cdot R_c, \quad (3.10)$$

where E_s is the energy per symbol of an M -ary scheme with code rate R_c .

WORKED EXAMPLE From C/N_0 to bit rate

A receiver measures $C/N_0 = 60$ dB-Hz. A QPSK link with a rate-1/2 code needs $E_b/N_0 = 2$ dB at the target error rate. The maximum bit rate is $R_b = (C/N_0)/(E_b/N_0) = 10^{(60-2)/10} \approx 630$ kb/s. With 3 dB of implementation margin, plan for 315 kb/s.

3.7 The link budget

3.7.1 The Friis transmission equation

In free space, the power received by an antenna of gain G_r at distance d from a transmitter of power P_t and antenna gain G_t is

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d} \right)^2, \quad L_{fs} = \left(\frac{4\pi d}{\lambda} \right)^2. \quad (3.11)$$

The free-space path loss L_{fs} appears to grow with frequency, but that is an artefact of using antennas of fixed gain: an antenna of fixed *area* A has gain $4\pi A/\lambda^2$, so between two dishes of fixed size the received power actually *rises* with frequency. This is why point-to-point microwave links and satellites use high frequencies and large dishes. In decibels, the link budget is a sum:

$$P_r [\text{dBm}] = P_t + G_t + G_r - L_{fs} - L_{\text{other}}, \quad L_{fs} [\text{dB}] = 32.45 + 20 \log_{10} f_{\text{MHz}} + 20 \log_{10} d_{\text{km}}. \quad (3.12)$$

3.7.2 Margins

A real link budget adds losses for cables, polarisation mismatch, rain, walls, body blockage and, above all, *fading*. Multipath fading can drop the received power 20–30 dB below its average for a small fraction of the time (Chapter ??), so the budget must include a *fade margin* chosen for the required availability. The final comparison is between the received E_b/N_0 or SNR and the value the modem needs, plus margin. Figure 3.7 draws an indoor Wi-Fi budget as a waterfall: 20 dBm from the access point, antenna gains of 3 dBi, about 102 dB of path loss at 50 m with a path-loss exponent of 3 at 5.5 GHz, and a wall, leaves about –84 dBm—14 dB above the noise floor of a 20 MHz receiver with 7 dB noise figure. That supports 16-QAM with moderate coding (Chapter ??).

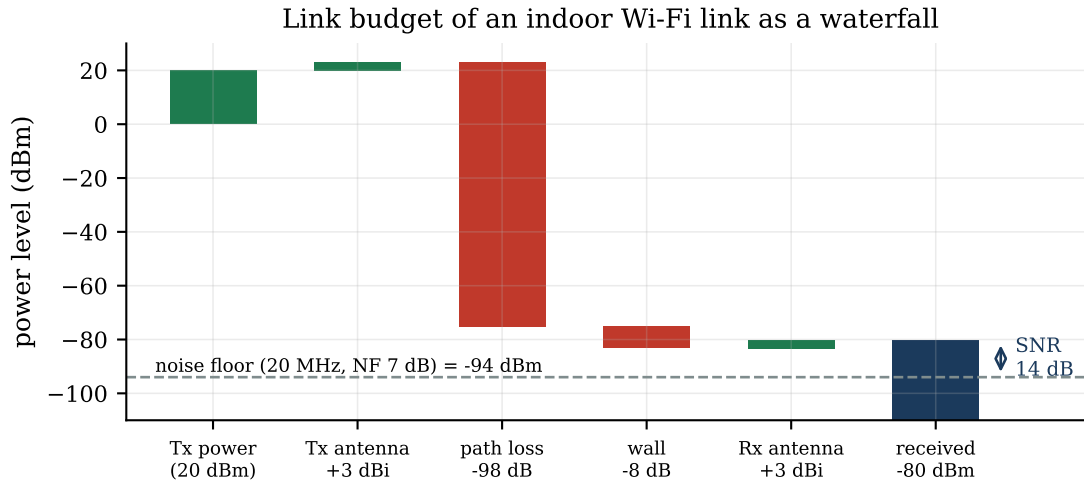


Figure 3.7: A link budget drawn as a waterfall. Gains lift the level, losses lower it; the gap between the received power and the noise floor is the SNR available to the modem.

HISTORY Talking to Voyager

Voyager 1, launched in 1977, is now more than 160 times as far from the Sun as the Earth is. It transmits roughly 20 W at 8.4 GHz from a 3.7 m dish, and its signal is received by the 70 m antennas of NASA's Deep Space Network, whose cryogenic receivers have system noise temperatures of a few tens of kelvin. At more than 2.4×10^{10} km, the free-space path loss exceeds 300 dB, and the received power is below 10^{-19} W. The link still carries 160 b/s of science data, thanks to high antenna gains, very low noise temperatures, a narrow bandwidth and powerful concatenated error-correcting codes—a convolutional code and a Reed–Solomon code, the same pair of codes that protected digital television for a generation (Chapter ??). Every term of Equation (3.11) has been pushed to its limit.

SIMULATION LAB Noise and link budgets

Lab 1 contains an interactive sensitivity calculator: vary bandwidth, noise figure and required SNR and watch the free-space range change. The figure script `figscripts/ch03_figs.py` contains the cascade-noise-figure calculator used for Figure 3.5; rearrange the stages of your own receiver and find the best placement for the filter. Chapter ??'s lab will convert these SNRs into bit error rates.

Problems

- 3.1 Show that if X and Y are independent $\mathcal{N}(0, \sigma^2)$, then $R = \sqrt{X^2 + Y^2}$ is Rayleigh distributed and R^2 is exponential. What fraction of the time is a Rayleigh-fading signal more than 10 dB below its mean power? 20 dB?
- 3.2 Compute the thermal noise power in dBm for bandwidths of 12.5 kHz (narrowband FM), 200 kHz (GSM), 20 MHz (Wi-Fi) and 400 MHz (5G mmWave), all at 290 K.

- 3.3 A receiver has an antenna with $T_{\text{ant}} = 50 \text{ K}$, a feed line with 0.5 dB loss at 290 K, an LNA with NF 0.8 dB and 30 dB gain, and a second stage with NF 10 dB. Compute the system noise temperature. How much does moving the LNA before the feed line improve it?
- 3.4 Derive the Friis formula (3.9) for two stages from the definition of noise temperature.
- 3.5 A geostationary satellite 36 000 km away transmits 50 dBW EIRP at 12 GHz. A 60 cm dish with 65% efficiency and a 60 K system noise temperature receives it. Compute C/N_0 . What bit rate can a DVB-S2 QPSK rate-3/4 carrier (needing about 4 dB E_s/N_0) support?
- 3.6 A Bluetooth Low Energy link transmits 0 dBm at 2.4 GHz with 0 dBi antennas. The receiver sensitivity is -95 dBm . What is the free-space range? Repeat with a path-loss exponent of 3 beyond 1 m.
- 3.7 (*Simulation*) Generate complex Gaussian noise in Python and verify numerically that (a) its phase is uniform, (b) its magnitude is Rayleigh, and (c) after a filter $H(f)$ its PSD is $|H(f)|^2 N_0$. Then measure how many samples are needed to estimate a 10^{-6} tail probability to within 10%.

Further Reading

- A. Papoulis and S. U. Pillai, *Probability, Random Variables and Stochastic Processes*, 4th ed., McGraw-Hill, 2002.
- S. O. Rice, “Mathematical analysis of random noise,” *Bell System Technical Journal*, 1944–45. The classic.
- H. T. Friis, “Noise figures of radio receivers,” *Proc. IRE*, 32, 1944.
- Recommendation ITU-R P.372, *Radio noise*. Measured and modelled external noise levels from 0.1 Hz to 100 GHz.
- J. H. Yuen (ed.), *Deep Space Telecommunications Systems Engineering*, JPL, 1982 (free online). The engineering of links like Voyager’s.

Analog Modulation and the Classic Radio

For most of the twentieth century, radio meant analog modulation: a voice or a picture impressed directly onto the amplitude, phase or frequency of a carrier. Analog systems are still on the air—AM and FM broadcasting, aviation voice, marine and land-mobile radio, amateur single sideband—and their ideas are everywhere inside digital radios. The superheterodyne receiver, the phase-locked loop, the I/Q modulator, pre-emphasis, the capture effect and the threshold effect were all discovered by engineers building analog equipment. This chapter treats the classic systems the way their designers did: as engineering trade-offs between power, bandwidth, receiver complexity and fidelity.

4.1 Why modulate at all?

A microphone produces a baseband signal $m(t)$ with energy from roughly 100 Hz to a few kilohertz. Radiating it directly is hopeless for three reasons. First, antennas are efficient only when their size is comparable to a wavelength; at 1 kHz the wavelength is 300 km. Second, every station would occupy the same spectrum, so no two could operate in the same region. Third, the propagation of radio waves depends strongly on frequency, and the designer wants to choose the band that suits the service: medium waves that follow the ground at night, VHF that stays local, microwaves that go straight to a satellite. **Modulation** solves all three by translating the message to a **carrier** frequency f_c chosen by the engineer and allocated by the regulator.

A sinusoidal carrier $A_c \cos(2\pi f_c t + \phi)$ has only two things that can be varied: its amplitude and its angle. Every analog scheme in this chapter varies one or both, and every digital scheme in Part III does the same with a discrete alphabet. In the language of Chapter 2, a modulated signal is

$$s(t) = \text{Re}\{\tilde{s}(t) e^{j2\pi f_c t}\}, \quad (4.1)$$

and each modulation is nothing more than a rule for building the complex envelope $\tilde{s}(t)$ from $m(t)$:

Scheme	Complex envelope $\tilde{s}(t)$	Bandwidth
AM (DSB-LC)	$A_c[1 + \mu m(t)]$	$2W$
DSB-SC	$A_c m(t)$	$2W$
SSB (upper)	$A_c[m(t) + j\hat{m}(t)]$	W
PM	$A_c e^{jk_p m(t)}$	$\approx 2(\Delta f + W)$
FM	$A_c e^{j2\pi k_f \int m(\tau) d\tau}$	$\approx 2(\Delta f + W)$

Here W is the message bandwidth, $\hat{m}(t)$ the Hilbert transform of $m(t)$ and Δf the peak frequency deviation. The table is worth a moment's study, because it is the whole chapter in five lines: amplitude modulations have *real* or *analytic* envelopes and occupy a bandwidth tied to W ; angle modulations have constant-magnitude envelopes and trade bandwidth for noise immunity.

4.2 Amplitude modulation

4.2.1 The AM signal

Conventional AM, known formally as double sideband with large carrier (DSB-LC), was the first broadcast modulation and is still used on medium and short waves and for aviation voice around 118–137 MHz. With the message normalised so that $|m(t)| \leq 1$,

$$s_{AM}(t) = A_c [1 + \mu m(t)] \cos(2\pi f_c t), \quad (4.2)$$

where $0 < \mu \leq 1$ is the **modulation index**. The envelope of the RF waveform traces $A_c[1 + \mu m(t)]$, which is always positive as long as $\mu \leq 1$, so a diode, a capacitor and a resistor can recover the message. Figure 4.1 shows a tone-modulated carrier at three indices. At $\mu = 1.4$ the envelope would need to go negative; it cannot, so the detector output folds and the audio is badly distorted. Broadcast transmitters limit negative peaks to prevent this **over-modulation**, while allowing positive peaks somewhat above 100% because those do not break the envelope.

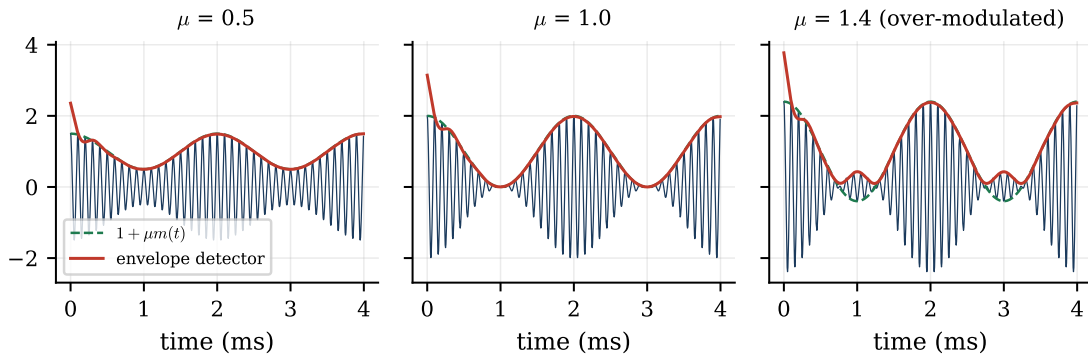


Figure 4.1: AM with a 500 Hz tone on a 10 kHz carrier for three modulation indices. The dashed green curve is the ideal envelope $1 + \mu m(t)$; the red curve is the output of a simulated full-wave envelope detector. Over-modulation ($\mu = 1.4$) makes the envelope fold, and the detected audio is distorted.

Taking the Fourier transform,

$$S_{\text{AM}}(f) = \frac{A_c}{2} [\delta(f - f_c) + \delta(f + f_c)] + \frac{A_c \mu}{2} [M(f - f_c) + M(f + f_c)]. \quad (4.3)$$

The spectrum is the message spectrum copied to either side of the carrier: an **upper sideband** and a **lower sideband**, each a complete copy of the information, plus an unmodulated carrier line. The occupied bandwidth is $2W$; AM broadcast channels are 10 kHz wide in the Americas and 9 kHz elsewhere, which limits the audio to about 4.5 to 5 kHz.

4.2.2 Power efficiency

With a tone message $m(t) = \cos 2\pi f_m t$, the carrier power is $A_c^2/2$ and each sideband carries $A_c^2 \mu^2/8$. The fraction of the transmitted power that carries information is

$$\eta_{\text{AM}} = \frac{\mu^2 P_m}{1 + \mu^2 P_m} \implies \eta = \frac{1}{3} \text{ for a tone at } \mu = 1, \quad (4.4)$$

where P_m is the normalised message power. Speech has a high peak-to-average ratio, so with peaks held at 100% modulation the average P_m is often below 0.1 and η falls to a few percent. A 50 kW clear-channel station may radiate only a kilowatt or two of sidebands.

Why accept such waste? Because in 1920 the only practical receiver was an envelope detector, and a large carrier makes envelope detection possible. The transmitter is expensive but there is one of it; the receivers are cheap and there are millions. That asymmetry—put complexity where there are few units—recurs throughout the history of communications, from broadcast television to GPS.

IN PRACTICE title=Modern AM transmitters

A modern high-power AM transmitter does not use a linear amplifier. It uses *pulse step* or *digital* modulation: dozens of solid-state modules, each switched fully on or off, whose outputs are combined so that the number of active modules follows the audio. Switching amplifiers are 90% efficient or better. The same idea, envelope tracking, is used in every 4G and 5G handset to power the PA efficiently while it amplifies an OFDM signal (Chapter 7).

4.2.3 The envelope detector

The classic detector is a diode followed by a parallel RC network. On each positive half-cycle the diode charges the capacitor to the peak; between peaks the capacitor discharges through R . Two constraints fix the time constant:

$$\frac{1}{f_c} \ll RC \ll \frac{1}{W}. \quad (4.5)$$

If RC is too short the output ripples at the carrier frequency. If it is too long the capacitor cannot follow the falling envelope, and the output shows *diagonal clipping*. A tighter analysis for a tone at index μ gives $RC \leq \sqrt{1 - \mu^2} / (2\pi f_m \mu)$. For broadcast AM with $f_c \approx 1$ MHz and $W = 5$ kHz there is a comfortable factor of 200 between the limits.

WORKED EXAMPLE title=Designing an envelope detector

A 455 kHz IF carries AM with audio up to 5 kHz at up to 90% modulation. The diagonal clipping limit at 5 kHz and $\mu = 0.9$ is $RC \leq \sqrt{1 - 0.81} / (2\pi \cdot 5000 \cdot 0.9) = 15.4 \mu\text{s}$. The ripple condition asks for $RC \gg 1/455 \text{ kHz} = 2.2 \mu\text{s}$. Choosing $R = 10 \text{ k}\Omega$ and $C = 1 \text{ nF}$ gives $RC = 10 \mu\text{s}$, which satisfies both. The remaining 455 kHz ripple is removed by a simple audio low-pass filter.

4.2.4 Double sideband with suppressed carrier

If the receiver can regenerate the carrier, it does not need to be transmitted. Removing it gives DSB-SC,

$$s_{\text{DSB}}(t) = A_c m(t) \cos(2\pi f_c t), \quad (4.6)$$

generated by a **balanced modulator**—a product of two signals, in practice a Gilbert cell or a diode ring. All transmitted power is now in the sidebands, but the envelope is $|m(t)|$, so an envelope detector no longer works. The receiver must multiply by a *coherent* local carrier $\cos(2\pi f_c t + \theta)$ and low-pass filter:

$$\text{LPF}\{s_{\text{DSB}}(t) \cos(2\pi f_c t + \theta)\} = \frac{A_c}{2} m(t) \cos \theta. \quad (4.7)$$

A phase error θ scales the output by $\cos \theta$; at 90° the output vanishes. A frequency error Δf makes the output beat as $m(t) \cos(2\pi \Delta f t)$. Coherent detection therefore needs a carrier recovery loop. John Costas's 1956 answer—two product detectors in quadrature whose outputs multiply to give a phase error—is the **Costas loop** that Chapter ?? uses to recover the carrier of every BPSK and QPSK signal.

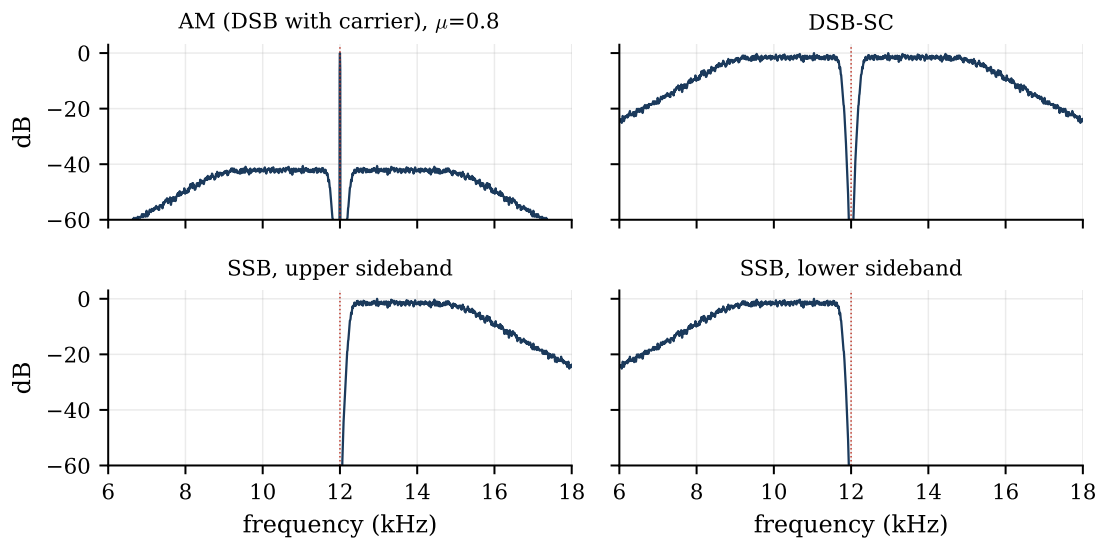


Figure 4.2: Spectra of the four amplitude modulations with a voice-like message on a 12 kHz carrier: conventional AM carries a strong carrier line; DSB-SC removes it; SSB keeps only one sideband; VSB keeps one sideband plus a gradual vestige of the other.

4.3 Single sideband

4.3.1 Only half the spectrum is needed

For a real message, the lower sideband is the mirror image of the upper; transmitting both wastes half the bandwidth. **Single sideband** (SSB) sends one. Its complex envelope is the analytic signal of the message,

$$\tilde{s}_{\text{USB}}(t) = m(t) + j\hat{m}(t), \quad s_{\text{USB}}(t) = m(t) \cos 2\pi f_c t - \hat{m}(t) \sin 2\pi f_c t, \quad (4.8)$$

where $\hat{m}$ is the Hilbert transform (Chapter 2). Changing the sign of the second term gives the lower sideband. SSB occupies bandwidth W , puts all its power into information, and fades more gracefully than AM because there is no carrier to suffer selective fading. It has been the standard for long-haul HF voice since the 1950s: military, maritime, aeronautical and amateur. Its analog-telephony cousin, SSB frequency-division multiplexing, carried the long-distance telephone network (Chapter 1) with 12-channel groups stacked into supergroups and mastergroups on coaxial cables and microwave relays.

4.3.2 Three ways to generate SSB

Filter method. Generate DSB-SC at a low intermediate frequency and remove one sideband with a sharp crystal or mechanical filter. The filter must pass 300 Hz and reject -300 Hz, a transition band of 600 Hz at, say, 9 MHz—hence the need for crystal lattices and the reason voice SSB traditionally cuts off below 300 Hz.

Phasing method. Implement (4.8) directly: two balanced modulators driven in quadrature, one fed with $m(t)$ and the other with a 90° -shifted copy. The unwanted sideband cancels only as well as the two paths match. Figure 4.3 shows how quickly suppression degrades: a 2° phase error limits it to about 35 dB. This is precisely the *image rejection* problem of every I/Q transmitter and direct-conversion receiver, including the B200 of Chapter 7.

Weaver method. Shift the audio band to be centred on 0 Hz with a quadrature oscillator at the band centre, low-pass filter each branch, then up-convert in quadrature. It avoids both the sharp filter and the wideband 90° network, and it is the method most DSP-based SSB implementations use.

4.3.3 Receiving SSB

An SSB receiver is a coherent product detector with a local beat-frequency oscillator. A frequency error Δf does not destroy the signal as it does for DSB-SC; it shifts every audio component by Δf . Speech stays intelligible up to about 50 Hz of error, which is why SSB could be used with free-running oscillators, and why a mistuned SSB voice sounds like “Donald Duck”. Music, whose harmonics lose their integer ratios, cannot tolerate even a few hertz, which is one reason SSB was never adopted for broadcasting.

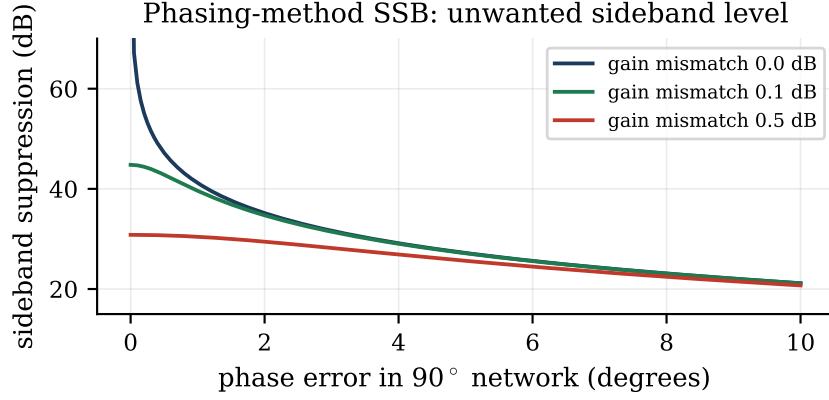


Figure 4.3: Unwanted-sideband suppression of the phasing method as a function of the phase error of the 90° network, for three gain mismatches. The same curves give the image rejection of any I/Q modulator or zero-IF receiver.

4.4 Vestigial sideband and analog television

Television video extends down to DC. A clean SSB filter with a transition at DC is impossible, and DSB wastes spectrum a TV channel cannot afford. **Vestigial sideband** (VSB) keeps one sideband and a gradually rolled-off *vestige* of the other. If the filter's response is antisymmetric about f_c — $H(f_c + f) + H(f_c - f) = \text{constant}$ over the message band—coherent or envelope detection recovers the message without distortion. Readers who have met the Nyquist criterion for zero intersymbol interference (Chapter ??) will recognise the same symmetry condition, discovered twice.

NTSC television, used in North America from 1941 to 2009, put a VSB video carrier 1.25 MHz above the lower edge of a 6 MHz channel, with 4.2 MHz of luminance bandwidth, an FM audio carrier 4.5 MHz above the video carrier, and colour information on a 3.58 MHz subcarrier using *quadrature* amplitude modulation of two colour-difference signals. The colour subcarrier frequency was chosen as an odd multiple of half the line rate so that its energy interleaved between the harmonics of the luminance spectrum—a frequency-domain multiplexing trick of real elegance. The digital ATSC standard that replaced NTSC kept VSB, now as 8-VSB carrying 19.39 Mb/s in the same 6 MHz.

4.5 Angle modulation

4.5.1 Phase and frequency modulation

Angle modulation keeps the amplitude constant and varies the phase:

$$s(t) = A_c \cos(2\pi f_c t + \phi(t)). \quad (4.9)$$

In **phase modulation** (PM), $\phi(t) = k_p m(t)$. In **frequency modulation** (FM), the instantaneous frequency $f_i(t) = f_c + \frac{1}{2\pi} \dot{\phi}(t)$ follows the message,

$$f_i(t) = f_c + k_f m(t), \quad \phi(t) = 2\pi k_f \int_{-\infty}^t m(\tau) d\tau. \quad (4.10)$$

FM of m is PM of the integral of m ; the two differ only by a filter in front of the modulator. The **peak frequency deviation** is $\Delta f = k_f \max |m(t)|$, and for a tone of frequency f_m the **modulation index** is $\beta = \Delta f / f_m$, the peak phase excursion in radians.

4.5.2 The spectrum of FM: Bessel functions

FM is nonlinear: the spectrum of an FM signal is not a shifted copy of the message spectrum. For a tone,

$$s(t) = A_c \cos(2\pi f_c t + \beta \sin 2\pi f_m t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta) \cos(2\pi(f_c + n f_m)t), \quad (4.11)$$

where J_n is the Bessel function of the first kind. A single tone produces an infinite set of sidebands spaced f_m apart, with amplitudes $J_n(\beta)$ (Figure 4.4). Three facts follow.

- The total power is constant, $\sum J_n^2(\beta) = 1$: modulation redistributes power from the carrier to the sidebands.
- The carrier amplitude $J_0(\beta)$ vanishes at $\beta = 2.405, 5.520, \dots$. Engineers use these *Bessel nulls* to calibrate deviation: raise the audio level until the carrier disappears on a spectrum analyser, and the index is exactly 2.405.
- For small β , $J_1(\beta) \approx \beta/2$ and higher orders are negligible: **narrowband FM** looks like AM with the carrier in quadrature, and its bandwidth is $2f_m$.

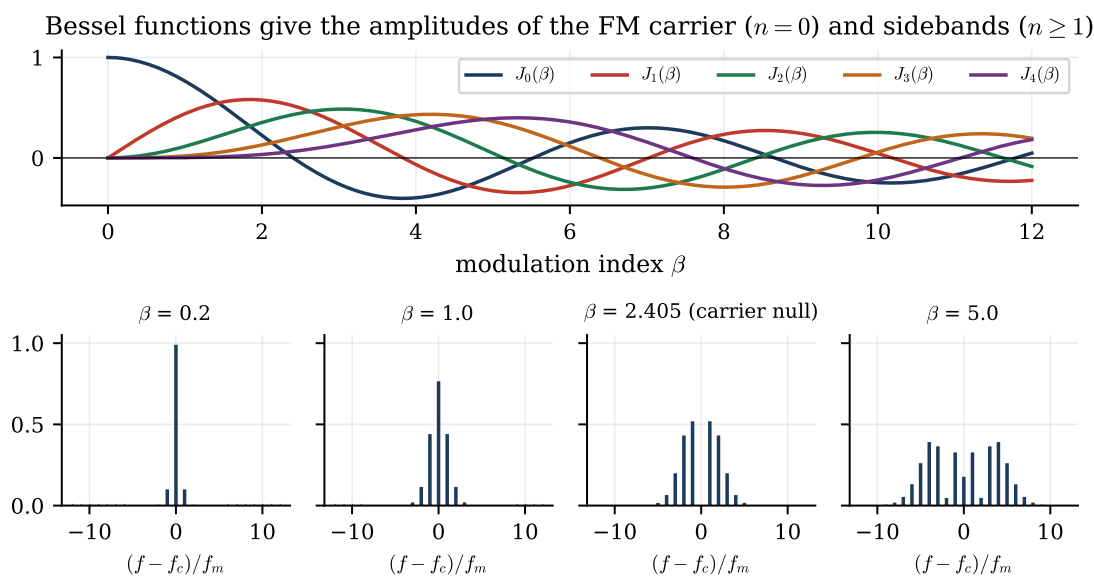


Figure 4.4: Top: Bessel functions J_0 to J_4 versus modulation index. Bottom: line spectra of tone-modulated FM for four indices, including the first carrier null at $\beta = 2.405$.

4.5.3 Carson's rule

In theory FM bandwidth is infinite; in practice sideband amplitudes fall rapidly once $|n| > \beta + 1$. John Carson's 1922 rule of thumb—ironically from a paper arguing that FM could not

reduce bandwidth, which is true—estimates the band containing about 98% of the power:

$$B_T \approx 2(\Delta f + W) = 2(\beta + 1)W. \quad (4.12)$$

Figure 4.5 compares it with the exact 98% bandwidth. FM broadcasting uses $\Delta f = 75$ kHz and $W = 15$ kHz ($\beta = 5$), so $B_T = 180$ kHz, which fits the 200 kHz channel raster. Two-way land-mobile radio uses $\Delta f = 2.5$ or 5 kHz and $W = 3$ kHz, fitting 12.5 or 25 kHz channels.

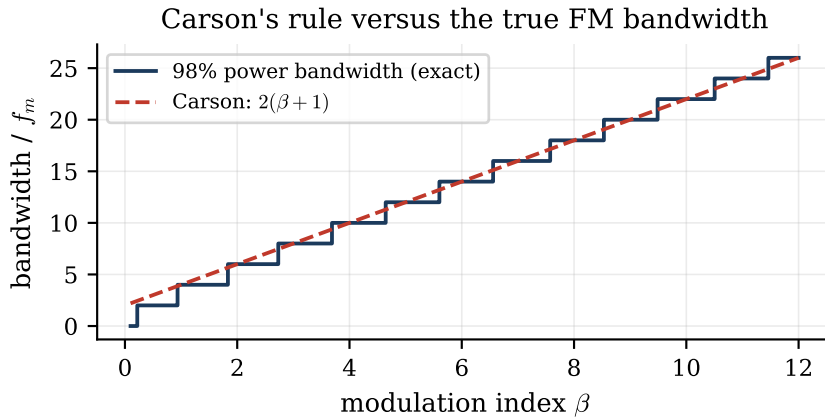


Figure 4.5: Carson's rule (dashed) against the exact bandwidth containing 98% of the power of tone-modulated FM (steps). The rule is accurate across the range of practical modulation indices.

4.5.4 Generating FM

The obvious FM modulator is a voltage-controlled oscillator (VCO) whose frequency follows the message, typically an LC oscillator tuned by a varactor diode. Its centre frequency drifts, so the VCO is enclosed in a phase-locked loop whose bandwidth is below the lowest audio frequency: the loop holds the average frequency while the modulation passes through. Armstrong's original *indirect* method generated narrowband PM of the integrated audio with a crystal-stable oscillator and then multiplied the frequency many times, which multiplied the deviation by the same factor. Today most FM is generated digitally: an NCO in an FPGA or DSP accumulates $2\pi k_f m[n] / f_s$ each sample and an I/Q modulator translates the result to RF. A software-defined radio produces broadcast-quality stereo FM in a few lines of code.

4.5.5 Demodulating FM

A demodulator must output a voltage proportional to instantaneous frequency. The historical sequence of solutions is a tour of analog circuit ingenuity:

Slope detector Tune a resonant circuit so the carrier sits on its skirt; frequency variations become amplitude variations, which an envelope detector recovers.

Foster–Seeley discriminator and ratio detector Balanced versions using a centre-tapped transformer; the ratio detector adds built-in amplitude limiting. These dominated radios from the 1930s to the 1970s.

Quadrature detector Multiply the signal by a copy delayed through a phase-shift network whose phase varies linearly with frequency; used in countless FM IF chips.

PLL demodulator Lock a VCO to the incoming signal; the VCO's control voltage is the demodulated audio. PLL demodulators extend the threshold (below).

Digital With complex baseband samples $z[n]$, the phase difference between successive samples is the instantaneous frequency:

$$\hat{f}[n] = \frac{f_s}{2\pi} \arg(z[n] z^*[n-1]). \quad (4.13)$$

This single line of code is the FM demodulator in GNU Radio's `Quadrature Demod` block and in almost every SDR.

4.6 Noise in analog systems

4.6.1 The figure of merit

To compare modulations fairly, define the received carrier power P_R , the noise density N_0 , and the message bandwidth W . The reference is baseband transmission, with output SNR $\gamma = P_R/(N_0W)$. For each scheme we compute the output SNR relative to γ . The results for a tone message are the classic table:

System	Output SNR	Transmission bandwidth
Baseband	γ	W
DSB-SC (coherent)	γ	$2W$
SSB (coherent)	γ	W
AM, envelope detector (above threshold)	$\frac{\mu^2 P_m}{1 + \mu^2 P_m} \gamma \leq \gamma/3$	$2W$
FM (above threshold)	$3\beta^2 P_m \gamma$	$2(\beta + 1)W$

DSB-SC and SSB tie with baseband: coherent linear modulation neither gains nor loses. AM loses at least 4.8 dB to the carrier. FM is different in kind: its output SNR grows as β^2 , so every doubling of deviation (and roughly of bandwidth) buys 6 dB. This is the first appearance in this book of a principle that information theory (Chapter ??) makes exact: *bandwidth can be exchanged for signal-to-noise ratio*. For broadcast FM with $\beta = 5$ and $P_m = 1/2$, the improvement is 37.5, or 15.7 dB, before pre-emphasis.

4.6.2 Why FM noise rises with frequency

In the demodulator, additive noise perturbs the phase by an amount inversely proportional to the carrier amplitude, and the discriminator differentiates phase. Differentiation multiplies the noise spectrum by f^2 , so FM's output noise PSD is *parabolic*: low audio frequencies are quiet and high frequencies are noisy. Both broadcast FM and analog TV sound exploit this with **pre-emphasis**: the transmitter boosts high audio frequencies with a first-order high-pass shelf of time constant $75 \mu\text{s}$ (Americas, Korea) or $50 \mu\text{s}$ (Europe and most of the world), and

the receiver applies the matching de-emphasis. Because speech and music have little energy at high frequencies, the boost hardly increases the deviation, while de-emphasis cuts the parabolic noise by a further 10 to 13 dB. Dolby noise reduction in cassette recorders used the same idea.

4.6.3 The threshold effect

The FM advantage holds only above a carrier-to-noise ratio of about 10 dB in the IF bandwidth. Below it, the noise occasionally drives the received phasor around the origin, adding a full 2π to the phase. The discriminator turns each of these **clicks** into an impulse, and the output SNR collapses. Figure 4.6 shows a Monte Carlo simulation: above threshold, the curves follow the $3\beta^2 P_m \gamma$ theory and higher deviation wins; below it, the curves bend sharply down, and the wider-band signal crosses threshold first because it collects more noise. The threshold is the analog ancestor of the “cliff effect” of digital systems: a good FM link sounds perfect until it suddenly does not. PLL and feedback demodulators lower the threshold by a few decibels by effectively narrowing the noise bandwidth; they were developed for the space programme, where every decibel of a deep-space link mattered.

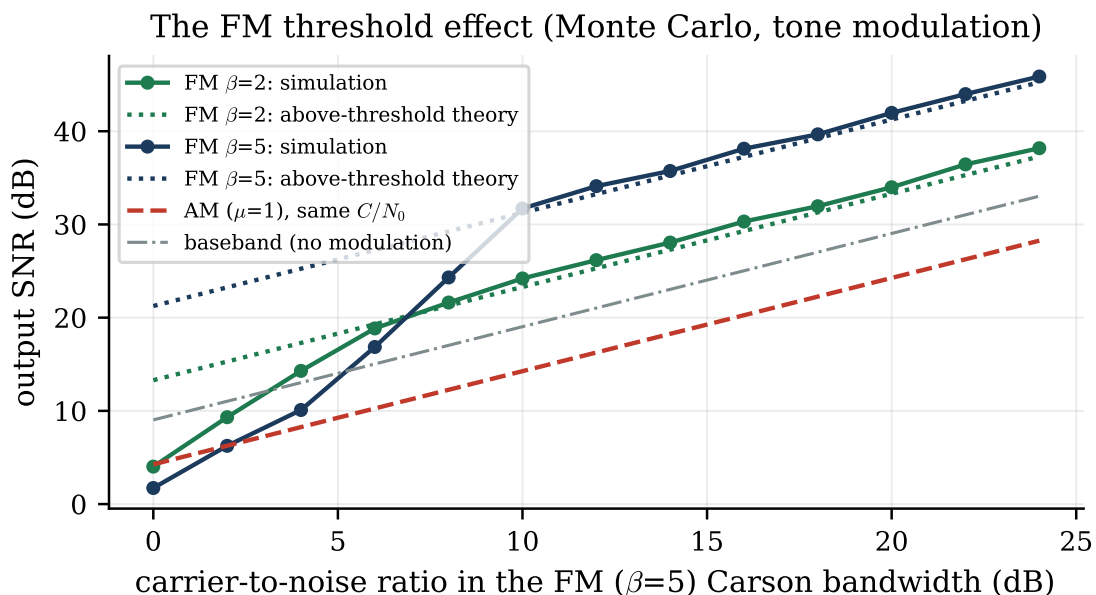


Figure 4.6: Output SNR versus carrier-to-noise ratio for FM with $\beta = 2$ and $\beta = 5$ (Monte Carlo with a tone), compared with AM and baseband. Above threshold FM beats AM by up to 20 dB; below about 10 dB CNR the “clicks” take over.

4.6.4 The capture effect

When two FM signals arrive on the same channel, the demodulator output is dominated by the stronger one, and the weaker is suppressed almost completely once the ratio exceeds a few decibels (the *capture ratio*, 1 to 2 dB in a good receiver). AM has no such effect: an interferer 20 dB weaker is still clearly heard. Capture is why FM made VHF mobile radio practical: co-channel stations could be reused much closer together. It is also why aircraft still

use AM on VHF: two pilots transmitting simultaneously produce an audible heterodyne on AM, whereas FM would silently hide the weaker transmission.

4.7 FM broadcasting in detail

4.7.1 Stereo multiplex

FM stereo, standardised in the USA in 1961, had to remain compatible with millions of mono receivers. The solution (Figure 4.7) builds a **multiplex** (MPX) baseband that frequency-modulates the carrier:

- the sum signal $L + R$ from 30 Hz to 15 kHz, which a mono receiver plays as usual;
- a 19 kHz **pilot** tone at 8 to 10% of deviation;
- the difference $L - R$ as DSB-SC on a suppressed 38 kHz subcarrier, occupying 23 to 53 kHz.

The stereo decoder doubles the pilot to regenerate the 38 kHz carrier coherently, recovers $L - R$, and forms $L = \frac{1}{2}[(L + R) + (L - R)]$ and $R = \frac{1}{2}[(L + R) - (L - R)]$. Because the $L - R$ channel sits high in the MPX spectrum where FM noise is parabolic, stereo reception is about 20 dB noisier than mono, which is why receivers blend to mono on weak signals.

Above the stereo band, the **Radio Data System** (RDS, or RBDS in North America) carries station names, programme type and traffic information as differentially encoded BPSK at 1187.5 b/s on a 57 kHz subcarrier, the third harmonic of the pilot. Older subsidiary carriers at 67 and 92 kHz carried background music and reading services. The MPX spectrum is a compact lesson in frequency-division multiplexing, coherent carrier regeneration, and backward compatibility.

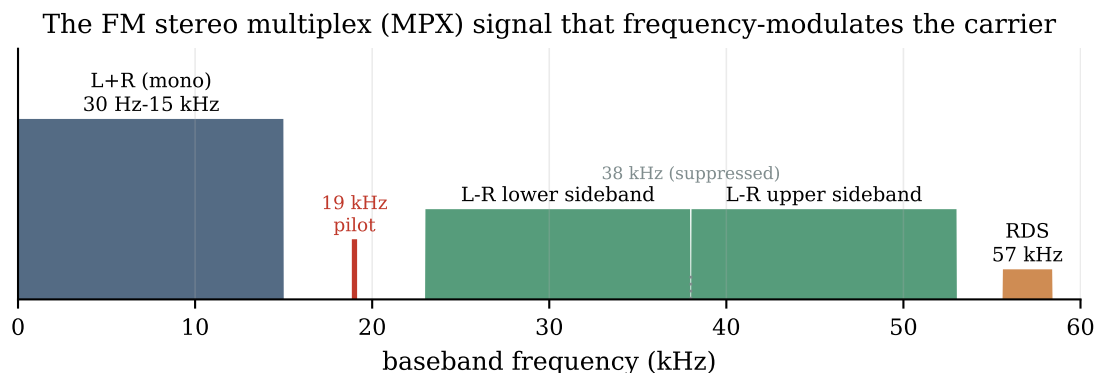


Figure 4.7: The FM stereo multiplex baseband: mono-compatible $L + R$, the 19 kHz pilot, the $L - R$ DSB-SC signal around a suppressed 38 kHz subcarrier, and RDS at 57 kHz.

4.7.2 HD Radio and DAB

Digital radio arrived in two philosophies. Europe's DAB (1995) and its successor DAB+ use OFDM in new VHF spectrum, multiplexing a dozen services in a 1.5 MHz ensemble with single-frequency networks. The US HD Radio system instead places low-power OFDM

sidebands on either side of existing AM and FM analog carriers (“in-band on-channel”), allowing a gradual transition without new spectrum. Both use the OFDM and coding machinery of Part V.

4.8 The superheterodyne receiver

4.8.1 Armstrong’s architecture

Early receivers amplified directly at the signal frequency in several tuned stages (the tuned radio-frequency, or TRF, receiver). Tracking several tuned circuits across a band was hard, and selectivity and gain varied with tuning. Edwin Armstrong’s **superheterodyne**, conceived in France in 1918, mixes every incoming station down to a single fixed **intermediate frequency** (IF), where all the selectivity and most of the gain can be designed once and for all. Nearly every radio receiver built since 1930 is a superheterodyne or a direct descendant of one.

A mixer multiplies the input with a local oscillator (LO). An input at f_{RF} and an LO at f_{LO} produce $|f_{RF} \pm f_{LO}|$; the IF filter selects $f_{IF} = |f_{RF} - f_{LO}|$. With *high-side injection*, $f_{LO} = f_{RF} + f_{IF}$, so tuning the receiver means tuning the LO.

4.8.2 The image problem

The mixer cannot tell whether the input is above or below the LO: an input at $f_{\text{image}} = f_{LO} + f_{IF}$, two IFs away from the wanted signal, produces the same IF. Figure 4.8 illustrates an FM receiver tuned to 98.1 MHz with a 10.7 MHz IF: the image is at 119.5 MHz, which falls in the aeronautical band. A tuned **preselector** ahead of the mixer must attenuate the image before mixing; once mixed, the image is indistinguishable from the wanted station. This sets the first rule of IF selection: *the IF must be high enough that the preselector can reject the image*. The second rule is its opposite: *the IF must be low enough that the IF filter can provide adjacent-channel selectivity*. At 455 kHz, a ceramic filter easily makes a 10 kHz AM channel; at 10.7 MHz, a ceramic filter makes a 200 kHz FM channel.

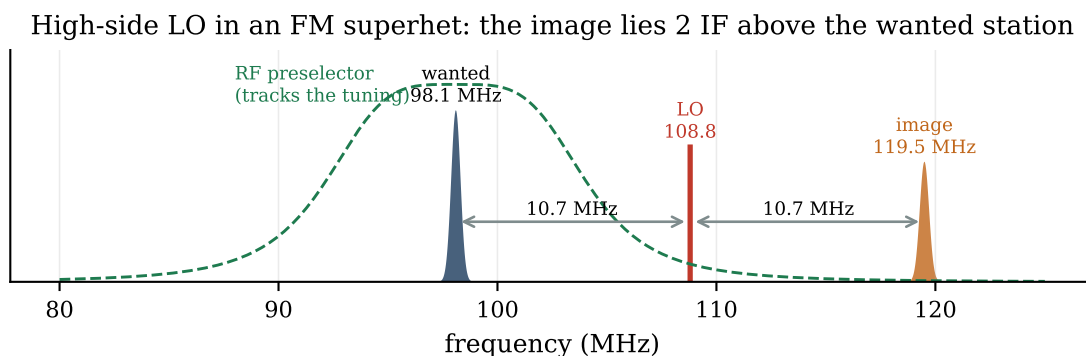


Figure 4.8: Frequency plan of an FM superheterodyne with a 10.7 MHz IF and high-side LO. The preselector must suppress the image two IFs above the wanted station before the mixer.

4.8.3 Double conversion and the path to zero IF

When one IF cannot satisfy both rules, use two. Communications receivers convert first to a high IF (often above the tuning range, so the image is far away and a fixed low-pass filter rejects it) and then to a low IF where the channel filtering is done. Cellular receivers of the 1990s followed this pattern. The logical end point is to convert directly to an IF of zero: the **direct-conversion** or zero-IF receiver, in which the “image” is the signal’s own negative-frequency half and is rejected not by a filter but by I/Q balance. That architecture, with its DC-offset and image-rejection problems, is the subject of Chapter 7.

4.8.4 Automatic gain control

Received signal strengths span more than 100 dB, from a station across town to one at the edge of coverage, and fading adds rapid swings of tens of decibels. **Automatic gain control** (AGC) measures the IF level and adjusts the gain of the RF and IF stages to keep the detector input constant. AGC time constants encode a compromise: fast enough to follow fading, slow enough not to flatten the speech envelope of AM. In digital receivers the same loop protects the ADC from clipping and is the first block in every receive chain.

4.9 Analog systems that survive

Analog modulation is not only history. In 2026 it remains in service where simplicity, backward compatibility or graceful degradation matter:

- AM broadcasting on long, medium and short waves, and aviation VHF voice (AM), for the capture-effect reason above and for decades of installed equipment.
- FM broadcasting worldwide, although some countries (Norway in 2017, Switzerland by 2026) have switched national FM networks off in favour of DAB+.
- Narrowband FM in marine VHF and amateur radio, although professional land-mobile radio has largely moved to digital systems such as P25, TETRA and DMR.
- SSB on HF for maritime, military and amateur long-distance links, increasingly alongside digital HF modems.
- Analog FM links inside satellite transponder tests, in some wireless microphones and in radio astronomy calibration.

KEY IDEA title=What analog radio teaches the digital engineer

- **Everything is a complex envelope.** AM, SSB, PM and FM differ only in how $\tilde{s}(t)$ is built. So do QPSK, QAM and OFDM.
- **Coherent detection needs a carrier reference.** The Costas loop was invented for DSB-SC voice.
- **Bandwidth buys noise immunity**, but only above a threshold. Shannon made both statements exact.

- **Image rejection and I/Q balance are the same problem**, from the phasing SSB exciter of 1950 to the zero-IF RFIC of today.

SIMULATION LAB title=Lab: analog modulation in Python

The notebook `labs/ch04_analog_modulation.ipynb` builds AM, DSB-SC, SSB (phasing and Weaver) and FM modulators and demodulators on complex baseband samples; lets you vary modulation index and noise with sliders while listening-free metrics (SINAD, THD) update; reproduces the FM threshold curve; and decodes a synthetic FM stereo MPX signal with pilot-based 38 kHz regeneration. With the `B200, gnuradio/gr01_spectrum_iq_capture.py` records a real broadcast FM station that the notebook can demodulate.

Problems

- 4.1 An AM transmitter radiates 10 kW of carrier. With tone modulation at $\mu = 0.8$, what is the total power and the power in each sideband? What is the efficiency?
- 4.2 Show that the envelope detector's diagonal-clipping condition for a tone is $RC \leq \sqrt{1 - \mu^2} / (2\pi f_m \mu)$.
- 4.3 A DSB-SC signal is demodulated with a local carrier having a phase error that drifts linearly at 1 Hz. Describe the output. Repeat for SSB.
- 4.4 Derive the Bessel expansion (4.11) from the generating function $e^{j\beta \sin \theta} = \sum_n J_n(\beta) e^{jn\theta}$.
- 4.5 Using Carson's rule, how many 12.5 kHz land-mobile FM channels ($\Delta f = 2.5$ kHz, $W = 3$ kHz) fit in the bandwidth of one broadcast FM channel? Compare the output SNR advantage over AM of each system.
- 4.6 An FM receiver has IF 10.7 MHz and high-side LO. It is tuned to 88.1 MHz. Find the image frequency. What preselector attenuation is needed if an image signal is 40 dB stronger than the wanted one and the demodulator needs 20 dB of protection?
- 4.7 Show that the digital FM demodulator (4.13) is unbiased for any carrier amplitude and explain what happens when $|\hat{f}| > f_s/2$.
- 4.8 Explain why FM stereo is about 20 dB noisier than mono at the same carrier-to-noise ratio. (*Hint*: integrate the parabolic noise PSD over 23–53 kHz and over 0–15 kHz.)
- 4.9 (*Simulation*) Generate an FM stereo MPX signal with different tones in L and R , frequency-modulate it with 75 kHz deviation at 400 kS/s, add noise, and write a decoder. Plot left-right separation versus pilot phase error.

Further Reading

- B. P. Lathi and Z. Ding, *Modern Digital and Analog Communication Systems*, 5th ed., Oxford, 2019, Ch. 4–5. A clear textbook treatment of AM and FM with noise analysis.

- S. Haykin and M. Moher, *Communication Systems*, 5th ed., Wiley, 2009.
- E. H. Armstrong, "A method of reducing disturbances in radio signaling by a system of frequency modulation," *Proc. IRE*, 24(5), 1936. The paper that launched FM.
- J. R. Carson, "Notes on the theory of modulation," *Proc. IRE*, 10, 1922.
- U. L. Rohde, J. C. Whitaker and H. Zahnd, *Communications Receivers: Principles and Design*, 4th ed., McGraw-Hill, 2017. The practitioner's reference on superheterodynes.
- *The ARRL Handbook for Radio Communications* (annual). Practical SSB, FM and receiver circuits, updated every year.

Sampling, Quantization and Digital Telephony

Every digital communication system begins where the analog world ends: at a sampler and a quantizer. This chapter explains what is lost and what is kept when a continuous waveform becomes a sequence of numbers. It develops the sampling theorem and its practical cousins (bandpass sampling, oversampling, reconstruction), the mathematics of quantization and companding, the clever one-bit converters that live in every phone and audio chip, and the performance limits of real data converters. It then tells the story of the first great digital network—the pulse-code-modulated telephone system of 64 kb/s channels, T1 and E1 frames and multiplexing hierarchies—whose design choices still shape the Internet’s backbone.

5.1 From waveforms to numbers

In 1937 Alec Reeves, an engineer at International Telephone and Telegraph’s Paris laboratory, filed a patent on a strange idea: instead of transmitting a voice waveform, transmit a sequence of numbers describing it. Each number would be sent as a short train of on–off pulses, and because a receiver only had to decide whether a pulse was present, noise could be removed completely at every repeater. The idea was called **pulse-code modulation** (PCM). It needed thousands of vacuum tubes per channel and was far ahead of its time; the first commercial PCM system, Bell’s T1 carrier, did not enter service until 1962, once transistors made it cheap. Today PCM, in one form or another, is how every sound, image and radio waveform enters a computer.

PCM has three steps: *sampling* (discretising time), *quantization* (discretising amplitude) and *coding* (writing each quantized value as bits). Sampling, done correctly, loses nothing. Quantization always loses something, and the art is to lose only what does not matter.

5.2 The sampling theorem

5.2.1 Statement and proof sketch

KEY IDEA title=The sampling theorem

A signal $x(t)$ with no energy at frequencies $|f| \geq W$ is completely determined by its

samples $x(nT_s)$ taken at rate $f_s = 1/T_s > 2W$, and can be reconstructed exactly by

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \operatorname{sinc}(f_s(t - nT_s)). \quad (5.1)$$

The rate $2W$ is the **Nyquist rate**. The theorem has many fathers: E. T. Whittaker published the interpolation series in 1915, Harry Nyquist identified $2W$ as the maximum pulse rate of a telegraph channel in 1928, Vladimir Kotelnikov proved the theorem for communications in 1933, and Claude Shannon stated it in its modern form in 1949.

The proof is a picture. Multiplying $x(t)$ by an impulse train of period T_s in time convolves its spectrum with an impulse train of spacing f_s in frequency:

$$X_s(f) = f_s \sum_{k=-\infty}^{\infty} X(f - kf_s). \quad (5.2)$$

The sampled spectrum is the original spectrum plus copies—*images* or *replicas*—centred on every multiple of f_s (Figure 5.1). If $f_s > 2W$ the replicas do not overlap, and an ideal low-pass filter with cutoff $f_s/2$ recovers $X(f)$ exactly; its impulse response is the sinc of (5.1). If $f_s < 2W$ the replicas overlap and the sum cannot be separated: information has been destroyed, not merely hidden.

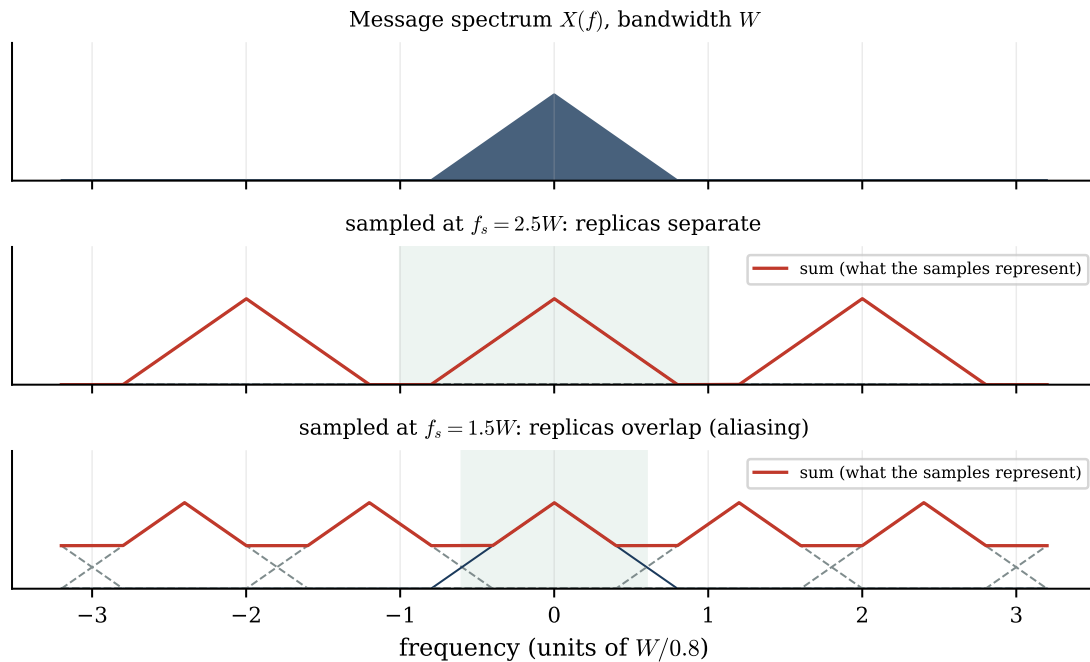


Figure 5.1: Sampling in the frequency domain. Sampling replicates the spectrum at multiples of f_s . Above the Nyquist rate the replicas separate and the original can be recovered by low-pass filtering (green band); below it they overlap and the sum is permanently corrupted.

5.2.2 Aliasing

The overlap has a name, **aliasing**, because a frequency component masquerades as a different one. A sinusoid at f_0 sampled at f_s produces exactly the same samples as one at $f_0 + kf_s$ for any integer k , and (for real signals) as one at $kf_s - f_0$. In Figure 5.2 a 9 Hz cosine sampled at 10 samples per second is indistinguishable from a 1 Hz cosine. The wagon wheels that turn backwards in Western films, the strobe light that freezes a spinning fan, and the moiré patterns on a striped shirt on television are all aliasing.

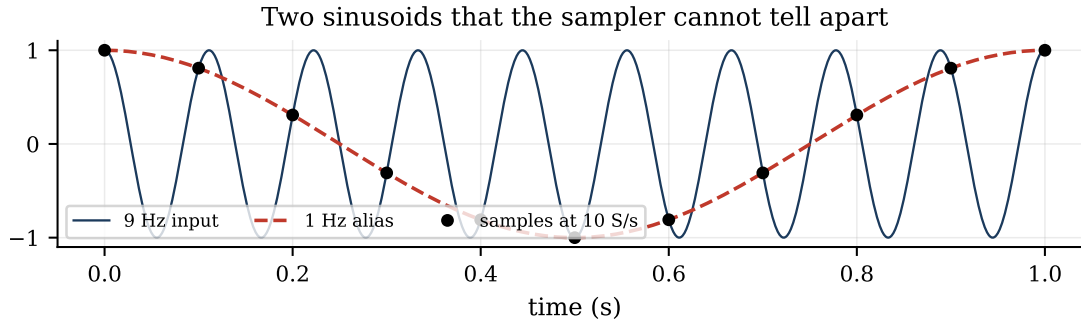


Figure 5.2: Aliasing in time: a 9 Hz cosine (blue) and a 1 Hz cosine (dashed) pass through exactly the same samples at 10 samples per second.

Once aliasing has happened, no processing can undo it. A converter must therefore be preceded by an **anti-aliasing filter** that removes everything above $f_s/2$ before sampling. No real filter has a brick-wall response, so practical systems sample faster than the Nyquist rate and leave a transition band. Telephone speech, band-limited to 300–3400 Hz, is sampled at 8 kHz rather than 6.8 kHz; CD audio, band-limited to 20 kHz, is sampled at 44.1 kHz. The ratio of the sample rate to the Nyquist rate, the **oversampling ratio**, is the designer's lever: more oversampling means an easier analog filter and more work for the digital hardware that follows.

5.2.3 Sampling real and complex signals

A real signal with bandwidth W needs $f_s > 2W$ real samples per second. A complex baseband signal occupying $-W/2 < f < W/2$ needs $f_s > W$ complex samples per second: its two real components, I and Q, each need $f_s > W$, so the total real sample rate is the same $2W$. There is no free lunch, but there is a practical advantage: with complex sampling the spectrum need not be symmetric about zero, so a receiver can see a signal and its mirror image as different things (Chapter 7).

5.2.4 Bandpass sampling

The theorem as stated is for low-pass signals, but a signal that occupies only a band from f_L to $f_H = f_L + B$ carries information at a rate governed by B , not f_H . If f_s is chosen so that the replicas of the band interleave without overlapping, the band aliases cleanly down to a lower frequency. The condition is

$$\frac{2f_H}{k} \leq f_s \leq \frac{2f_L}{k-1}, \quad k = 1, 2, \dots, \lfloor f_H/B \rfloor. \quad (5.3)$$

This **bandpass sampling** (or *undersampling*) lets an ADC digitise an IF directly: a 70 MHz IF with 20 MHz bandwidth can be sampled at about 61 MS/s. The ADC's track-and-hold must still have analog bandwidth above f_H , its aperture jitter must suit the high input frequency (Section 5.7), and the anti-aliasing filter becomes a bandpass filter that must reject all the other bands that would alias onto the same place.

5.2.5 Reconstruction and the zero-order hold

The sinc interpolator of (5.1) is not causal. Real digital-to-analog converters hold each sample constant for one period, a **zero-order hold** (ZOH), and follow it with an analog **reconstruction filter** that removes the images. The hold is a convolution with a rectangle of width T_s , so the output spectrum is multiplied by $\text{sinc}(f/f_s)$: a gentle roll-off of 3.9 dB at $f_s/2$ (Figure 5.3). Designs compensate with a digital pre-emphasis filter, oversample so that the band of interest sits where the droop is small, or both. Every transmitter in this book—including the AD9364 in the B200—has a ZOH DAC and interpolating digital filters in front of it.

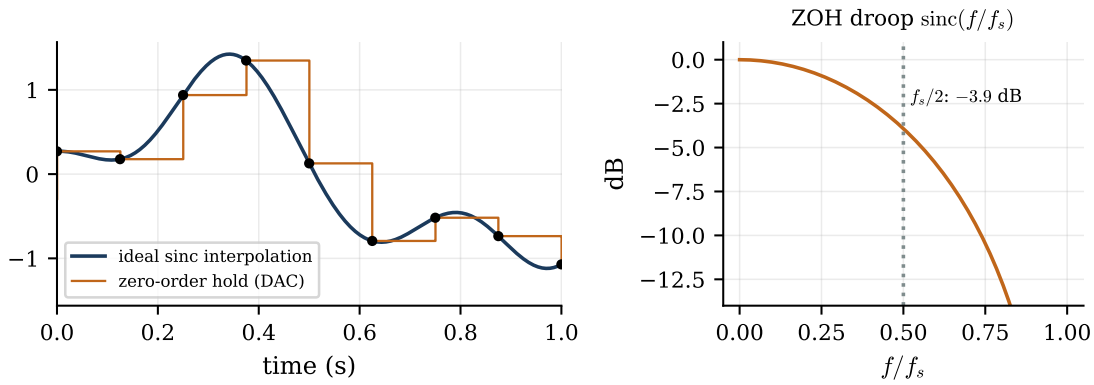


Figure 5.3: Left: ideal sinc interpolation passes smoothly through the samples, while a zero-order hold produces a staircase. Right: the staircase's frequency response $\text{sinc}(f/f_s)$ droops by 3.9 dB at the Nyquist frequency.

5.3 Quantization

5.3.1 The uniform quantizer

A b -bit quantizer maps each input sample to one of $L = 2^b$ levels. The uniform quantizer spaces the levels by a step $\Delta = 2X_{\max}/L$ over a full-scale range $\pm X_{\max}$. The difference between the output and the input is the **quantization error** $e = Q(x) - x$, bounded by $\pm\Delta/2$ as long as the input does not exceed full scale (Figure 5.4). Outside that range the quantizer **clips** and the error grows without bound.

When the input is busy—it spans many levels and is not synchronised to the sampling—the error behaves like a random variable uniformly distributed on $[-\Delta/2, \Delta/2]$, uncorrelated with the signal and white across $[0, f_s/2]$. Its power is

$$\sigma_e^2 = \frac{\Delta^2}{12}. \quad (5.4)$$

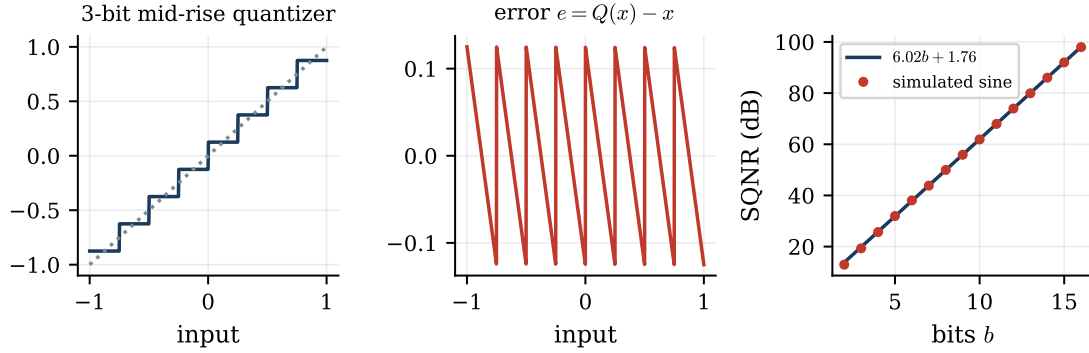


Figure 5.4: A 3-bit mid-rise quantizer, its error sawtooth, and the SQNR of a full-scale sine versus word length. The simulated values match the $6.02b + 1.76$ dB rule to a fraction of a decibel.

For a full-scale sine of amplitude $X_{\max} = 2^{b-1}\Delta$ the **signal-to-quantization-noise ratio** is then

$$\text{SQNR} = \frac{X_{\max}^2/2}{\Delta^2/12} = \frac{3}{2} 2^{2b} \implies \text{SQNR}_{\text{dB}} = 6.02b + 1.76. \quad (5.5)$$

This is the most frequently quoted formula in data conversion: *each bit is worth 6 dB*. A 12-bit converter, such as the B200's, offers 74 dB; a 16-bit audio converter 98 dB; a 24-bit audio converter a theoretical 146 dB, far below its analog noise floor.

Two refinements matter in practice. First, real signals do not sit at full scale. Speech, music and OFDM all have large peak-to-average ratios, so the designer must *back off* the average level to avoid clipping, and every decibel of back-off costs a decibel of SQNR. For a Gaussian-like signal clipped at four standard deviations, the SQNR is about $6.02b - 7.3$ dB. Second, if the signal is narrowband compared with $f_s/2$, digital filtering after the ADC removes the out-of-band quantization noise, giving a **processing gain** of $10 \log_{10}(f_s/2B)$ dB. An SDR that samples at 56 MS/s and filters to 200 kHz gains 21.5 dB, which is why a 12-bit radio can receive a signal 90 dB below full scale.

5.3.2 Dither

When the input is small or slowly varying, the error is no longer noise-like; it is a deterministic function of the signal, heard as harmonic distortion or seen as contour bands in images. Adding a small random **dither** signal before quantization—typically noise of about one LSB—decorrelates the error from the signal. Paradoxically, adding noise improves quality: the distortion becomes a benign noise floor, and sub-LSB signals remain detectable in the averaged output. High-resolution measurement ADCs add dither internally.

5.4 Non-uniform quantization and companding

5.4.1 The problem with speech

Speech has a dynamic range of about 40 dB between loud and soft talkers, and its amplitude distribution is sharply peaked near zero (approximately Laplacian). A uniform quantizer

wastes its fine levels on large amplitudes that rarely occur and gives quiet talkers a poor SQNR. The ideal is a quantizer whose step grows in proportion to the signal amplitude, giving a roughly constant SQNR regardless of level. A logarithmic characteristic does exactly that.

5.4.2 The μ -law and A-law

Companding (compressing then expanding) implements the logarithmic quantizer as a nonlinear compressor followed by a uniform quantizer, with the inverse expander at the receiver. North America and Japan standardised the **μ -law**

$$F(x) = \text{sgn}(x) \frac{\ln(1 + \mu|x|)}{\ln(1 + \mu)}, \quad |x| \leq 1, \mu = 255, \quad (5.6)$$

and the rest of the world the **A-law** with $A = 87.6$, which is linear near zero and logarithmic above $1/A$. Both are specified in ITU-T Recommendation G.711 (1972) as piecewise-linear approximations with eight segments per polarity, which made them easy to implement in the digital logic of the time.

Figure 5.5 shows the result. With a speech-like input, 8-bit μ -law holds an SQNR of 35 to 38 dB over a 40 dB range of input levels, while an 8-bit uniform quantizer degrades 1 dB per decibel of level. Matching μ -law across that range with a uniform quantizer would need about 12 to 13 bits. Eight bits at 8000 samples per second gives the fundamental rate of digital telephony: **64 kb/s**, the **DS0** channel.

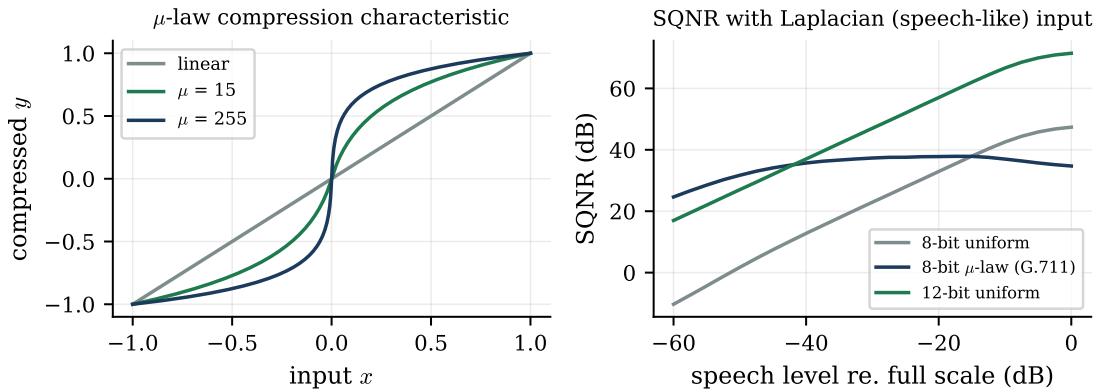


Figure 5.5: Left: the μ -law compressor. Right: SQNR against input level for a speech-like (Laplacian) signal. Eight-bit μ -law keeps the SQNR nearly constant over a 40 dB range, performing like a 12-bit uniform quantizer for quiet talkers.

5.4.3 Optimal quantizers

For a known input distribution, the quantizer that minimises mean-square error satisfies two conditions found by Stuart Lloyd (1957, published 1982) and Joel Max (1960): each decision threshold lies midway between adjacent output levels, and each output level is the centroid (conditional mean) of its decision interval. Iterating the two conditions is the **Lloyd–Max algorithm**. Its multidimensional generalisation, the k -means algorithm of machine learning,

designs vector quantizers for speech and image codecs (Chapter ??). An important asymptotic result: at high resolution, the entropy of a uniform quantizer’s output is within 0.25 bit per sample of the best possible rate for the same distortion, so uniform quantization followed by entropy coding is nearly optimal—the design behind almost every modern codec.

5.5 Differential and predictive coding

Adjacent speech samples at 8 kHz are strongly correlated. Rather than quantize each sample, **differential PCM** (DPCM) predicts the next sample from past reconstructed samples and quantizes only the prediction error, which has much smaller variance. The gain equals the ratio of signal variance to prediction-error variance, typically 6 to 10 dB for speech with a short linear predictor—worth one to two bits per sample. **Adaptive DPCM** (ADPCM) also adapts the quantizer step and predictor to the local statistics. ITU-T G.726 ADPCM at 32 kb/s delivers quality close to 64 kb/s PCM and was used to double the capacity of international circuits and in DECT cordless phones.

Note the structure: the predictor at the encoder uses *reconstructed* samples, exactly what the decoder will have, so that quantization errors do not accumulate. This “closed-loop” prediction is the same principle that makes video codecs and decision-feedback equalizers (Chapter ??) stable.

5.5.1 Delta modulation

Take DPCM to the extreme: sample much faster than Nyquist, predict each sample as the previous reconstructed value, and quantize the error with a single bit. A **delta modulator** transmits 1 if the input is above the staircase approximation and 0 if below, and the staircase steps by $\pm\delta$ accordingly (Figure 5.6). It is trivially simple and robust to bit errors, but it suffers two impairments. *Granular noise* appears when the input is flat and the staircase hunts around it. *Slope overload* appears when the input changes faster than δf_s per second and the staircase cannot keep up. Adaptive delta modulation (CVSD, continuously variable slope delta), which grows the step during runs of identical bits, found long service in military voice and in early Bluetooth headsets.

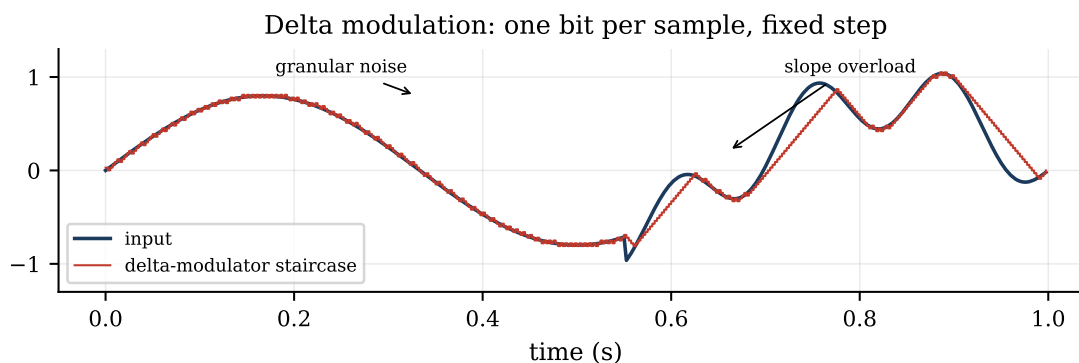


Figure 5.6: Delta modulation: the staircase follows a slow input closely (with granular noise) but cannot follow the steep segment on the right (slope overload).

5.6 Oversampling and sigma-delta conversion

Oversampling spreads the fixed quantization noise power $\Delta^2/12$ over a wider band, so a digital filter that keeps only the signal band gains 3 dB per doubling of the oversampling ratio (OSR)—half a bit per octave. That is too slow to be useful by itself. The **sigma-delta** ($\Sigma\Delta$) modulator changes the game by placing the quantizer inside a feedback loop with an integrator. The loop forces the *average* of the coarse quantizer’s output to track the input, and it filters the quantization error through a high-pass **noise transfer function**:

$$Y(z) = z^{-1}X(z) + (1 - z^{-1})^L E(z) \tag{5.7}$$

for an L th-order modulator. The noise is pushed out of the signal band to high frequencies (Figure 5.7), where a digital decimation filter removes it. The in-band SQNR now improves by $(6L + 3)$ dB per octave of OSR: a second-order modulator with an OSR of 64 and a one-bit quantizer achieves about 85 dB. Because a one-bit quantizer is perfectly linear (two points always define a line), $\Sigma\Delta$ converters avoid the component matching that limits other architectures. They are in every audio codec, most precision measurement ADCs, and—as continuous-time bandpass $\Sigma\Delta$ converters—in many cellular receivers.

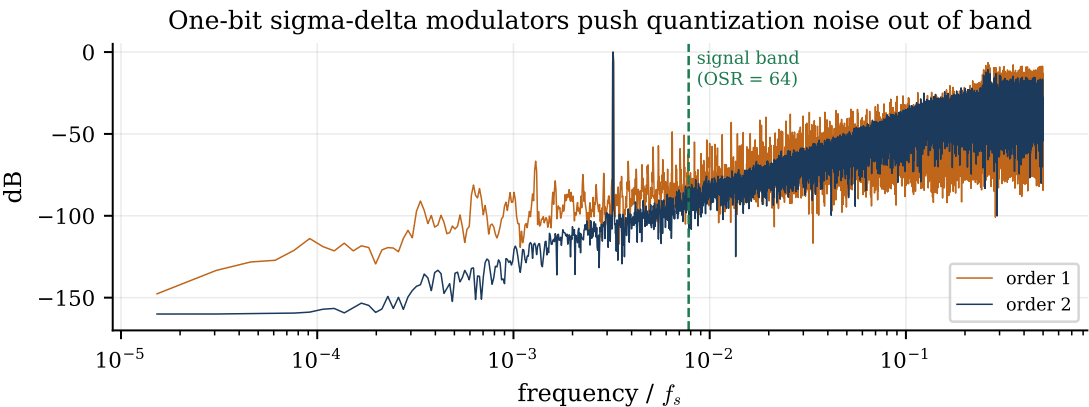


Figure 5.7: Spectra of first- and second-order one-bit sigma-delta modulators with a tone input. Quantization noise is shaped away from low frequencies; the second-order loop leaves far less noise in the signal band.

5.7 Real data converters

5.7.1 Architectures

Architecture	Typical resolution	Typical rate	Where it is used
Flash	4–8 bits	up to tens of GS/s	Optical links, oscilloscopes, sub-100 MHz
Pipeline	10–16 bits	50 MS/s–1 GS/s	Base stations, SDRs, instrumentation
SAR	8–18 bits	1 kS/s–2 GS/s	Low power sensors, time-interleaved
Sigma-delta	16–32 bits	audio to ≈ 100 MHz bandwidth	Audio, precision, cellular RX
Time-interleaved	8–14 bits	up to 100+ GS/s	Direct RF sampling, 5G, radar, 800 MHz

A *flash* converter compares the input with $2^b - 1$ thresholds at once; it is the fastest architecture but its size doubles with every bit. A *successive-approximation* (SAR) converter performs a binary search with one comparator and a capacitor DAC, and has become the most power-efficient design at moderate rates. *Pipeline* converters resolve a few bits per stage and pass the amplified residue on. *Time interleaving* runs M slower converters in rotation, which multiplies the rate but introduces spurs from mismatched gains, offsets and timing that must be calibrated digitally. Modern RF-sampling converters, such as those in AMD's RFSoc devices, sample directly at several gigasamples per second and perform the mixing and decimation in on-chip digital logic, eliminating the analog mixer altogether.

5.7.2 Specifications that matter

The **effective number of bits** (ENOB) inverts equation (5.5): measure the signal-to-noise-and-distortion ratio (SINAD) with a full-scale sine and compute $\text{ENOB} = (\text{SINAD}_{\text{dB}} - 1.76) / 6.02$. A 14-bit RF ADC might deliver 11 effective bits. The **spurious-free dynamic range** (SFDR) is the ratio of the signal to the largest spur, and in a receiver with a strong blocker it often matters more than noise, because a spur can land on a weak wanted signal.

At high input frequencies, **aperture jitter**—random variation of the sampling instant—dominates. A sinusoid of frequency f has maximum slope $2\pi fA$, so a timing error σ_t produces an amplitude error that limits the SNR to

$$\text{SNR}_{\text{jitter}} = -20 \log_{10}(2\pi f \sigma_t) \text{ dB}. \quad (5.8)$$

With 100 fs of jitter, the limit at 1 GHz is 64 dB, barely 10 bits, whatever the converter's resolution (Figure 5.8). This is why clock generation is a serious design discipline in every RF-sampling system, and why the B200 and every base station specify the phase noise of their reference oscillators.

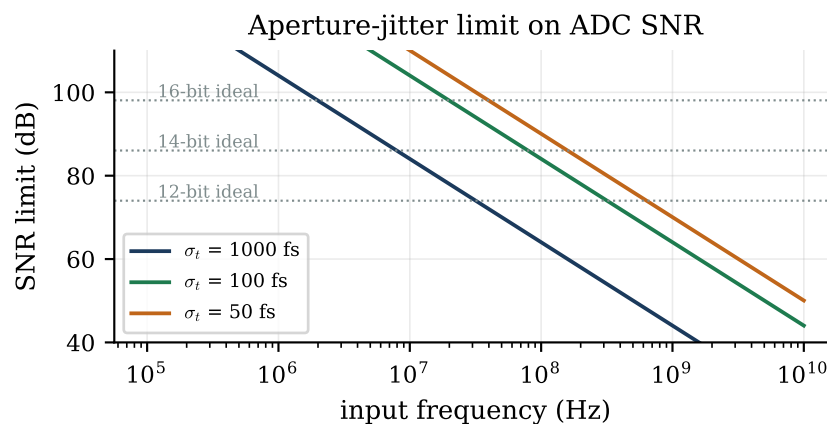


Figure 5.8: The SNR ceiling set by sampling-clock jitter. At gigahertz input frequencies even 50 fs of jitter limits a converter to about 10 to 11 effective bits.

5.8 Digital telephony

5.8.1 The 64 kb/s channel

By the late 1950s, the Bell System faced a crisis in its local networks: the cable ducts between exchanges in big cities were full. Replacing copper pairs was expensive; making each pair carry more calls was not. The answer was T1 carrier, which put 24 PCM voice channels on two twisted pairs, with regenerative repeaters every 6000 feet (the spacing of the existing loading coils, whose manholes were already there). Each repeater received a noisy stream of pulses, decided what each bit was, and retransmitted a clean pulse. For the first time in history, a long-distance voice circuit had the same quality as a short one.

5.8.2 Time-division multiplexing and the T1 and E1 frames

The channels share the line by **time-division multiplexing** (TDM): each channel gets a time slot of eight bits in a repeating frame. The T1 (DS1) frame (Figure 5.9) contains 24 slots plus one framing bit, 193 bits every $125\ \mu\text{s}$, giving $193 \times 8000 = 1.544\text{ Mb/s}$. The framing bits follow a fixed pattern over 12 frames (the superframe) or 24 frames (the extended superframe, which also carries a CRC-6 and a 4 kb/s data link). Signalling (on-hook, off-hook, dial pulses) was originally carried by stealing the least significant bit of each channel in every sixth frame—“robbed-bit” signalling—which is why modem data over digital trunks was limited to 56 kb/s instead of 64. The European E1 frame, standardised by CEPT, uses 32 slots of eight bits: slot 0 for framing and alarms, slot 16 for signalling, and 30 for voice, for 2.048 Mb/s. Its cleaner design carries 64 kb/s “clear channels” without robbed bits.

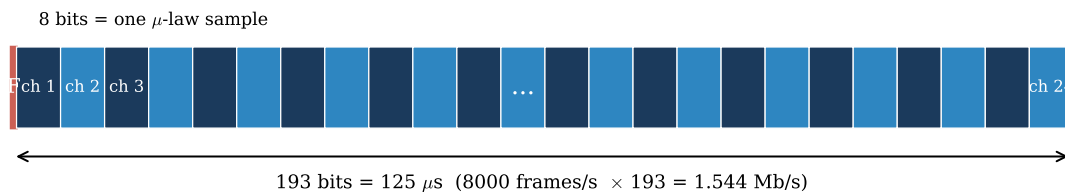


Figure 5.9: The T1 (DS1) frame: one framing bit and 24 eight-bit μ -law samples every $125\ \mu\text{s}$, for 1.544 Mb/s.

5.8.3 The multiplexing hierarchy

Frames are multiplexed again into higher rates. The North American **plesiochronous digital hierarchy** (PDH) combined four DS1s into a DS2 (6.312 Mb/s) and seven DS2s into a DS3 (44.736 Mb/s, 672 voice channels); the European hierarchy went 2.048, 8.448, 34.368 and 139.264 Mb/s. “Plesiochronous” means *nearly* synchronous: each tributary has its own clock, so the multiplexer inserts *justification* (stuffing) bits to absorb small rate differences. Extracting one DS1 from a DS3 required demultiplexing everything.

The **synchronous optical network** (SONET, 1988) and synchronous digital hierarchy (SDH) replaced this with byte-synchronous frames of $125\ \mu\text{s}$ at 51.84 Mb/s (STS-1) and multiples (OC-3 at 155.52 Mb/s, OC-48 at 2.488 Gb/s, OC-192 at 9.953 Gb/s), with pointers that locate each tributary directly. SONET’s $125\ \mu\text{s}$ frame—the sampling period of an 8 kHz voice channel—carried the Internet’s backbone traffic through the 1990s and 2000s, and the

timing distribution networks built for it still synchronise cellular base stations today. The story continues in Chapter ??.

5.8.4 Line coding on the wire

A copper pair cannot carry DC through its transformers, and the receiver must recover the bit clock from the data. T1 therefore uses **alternate mark inversion** (AMI): zeros are sent as no pulse and ones as pulses of alternating polarity, which gives a spectrum with no DC and a built-in error check (two consecutive pulses of the same polarity are a *bipolar violation*). Long runs of zeros would starve the clock recovery, so T1 requires a minimum ones density, later replaced by B8ZS substitution, which replaces eight zeros with a pattern containing deliberate violations that the receiver recognises. E1 uses the similar HDB3. Line codes are treated fully in Chapter ??.

5.8.5 Echo, delay and the end of the circuit

Digital telephony exposed problems analog networks had hidden. The **hybrid** that converts the two-wire subscriber loop to the four-wire transmission network leaks some of the far-end's voice back as echo. On short circuits the echo arrives so quickly that it merges with sidetone, but on satellite circuits (500 ms round trip, Chapter ??) or packet networks it is intolerable. **Adaptive echo cancellers**, LMS filters that learn the echo path and subtract a replica (Chapter ??), were among the first mass applications of adaptive signal processing and are in every telephone network and speakerphone today.

From the 2000s, voice moved from circuits to packets. Voice over IP (VoIP) and voice over LTE (VoLTE) still sample at 8 or 16 kHz, but compress the samples with codecs such as AMR-WB, EVS and Opus at 6 to 64 kb/s (Chapter ??). The 125 μ s frame survives only in legacy interfaces; the 64 kb/s G.711 codec survives as the universal fallback that every VoIP system still supports.

HISTORY title=The 8 kHz heartbeat

The number 8000 appears everywhere in communications because telephone speech was band-limited to 3.4 kHz and the designers of T1 chose a comfortable margin above the 6.8 kHz Nyquist rate. From that choice follow the 125 μ s frame, the 64 kb/s DSo, the 1.544 and 2.048 Mb/s primary rates, SONET's 51.84 Mb/s, the 8 kHz frame clock of ISDN and GSM's 20 ms speech frames. A sampling-rate decision made for a vacuum-tube filter in the 1950s still sets the timing of the global network.

PITFALL title=Clipping is worse than noise

Quantization noise degrades gracefully; clipping does not. A signal that clips even occasionally generates broadband distortion and, in a transmitter, spectral regrowth into adjacent channels. For OFDM signals with 10 dB or more of peak-to-average ratio, the rule of thumb is to keep the RMS level 10 to 12 dB below full scale in the ADC and DAC, and to budget the lost SQNR accordingly. In the B200, setting the digital TX amplitude above about 0.7 is a common way to generate a very clean spectral splatter.

SIMULATION LAB title=Lab: sampling and quantization

The notebook `labs/ch05_sampling_quantization.ipynb` lets you sweep an input frequency through the Nyquist zones and watch it alias; compares sinc, linear and zero-order-hold reconstruction; measures SQNR against bits, back-off and dither; implements μ -law and A-law companding and reproduces Figure 5.5; and builds first- and second-order sigma-delta modulators whose output spectra update as you change the OSR.

Problems

- 5.1 A signal contains components at 3, 7 and 12 kHz and is sampled at 16 kHz without an anti-aliasing filter. At what frequencies do the components appear after ideal reconstruction?
- 5.2 Find all valid bandpass sampling rates for a signal occupying 70–80 MHz. Which is the lowest? What practical margin would you add?
- 5.3 Derive the ZOH frequency response $\text{sinc}(f/f_s)$ and design a three-tap FIR compensation filter that flattens it to within 0.2 dB up to $0.4f_s$.
- 5.4 A 14-bit ADC samples at 125 MS/s. What is the ideal SQNR? If the signal of interest is a 1 MHz channel after digital filtering, what is the in-band SQNR? What jitter would limit it at an input frequency of 200 MHz?
- 5.5 Show that for a uniform quantizer with a Gaussian input of standard deviation σ clipped at $\pm 4\sigma$, $\text{SQNR} \approx 6.02b - 7.3$ dB (neglect the clipping noise).
- 5.6 Implement the Lloyd–Max algorithm for a 3-bit quantizer and a Laplacian input. Compare the MSE with that of the best uniform quantizer.
- 5.7 Compute the bit rates of a T1 superframe’s framing pattern, a DS3 and an OC-12. How many 64 kb/s channels does an OC-192 carry?
- 5.8 Explain why a first-order sigma-delta modulator with a DC input produces periodic patterns (idle tones), and how dither or higher order removes them.
- 5.9 (*Simulation*) Record your voice at 48 kHz (or use any speech file), decimate to 8 kHz, and encode it with 8-bit uniform PCM, 8-bit μ -law and 4-bit ADPCM. Compare SQNR and listen to the results.

Further Reading

- N. S. Jayant and P. Noll, *Digital Coding of Waveforms*, Prentice Hall, 1984. The classic on PCM, DPCM, ADPCM and delta modulation.
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- B. M. Oliver, J. R. Pierce and C. E. Shannon, "The philosophy of PCM," *Proc. IRE*, 36, 1948. A remarkably modern paper.
- J. Bellamy, *Digital Telephony*, 3rd ed., Wiley, 2000. T1, E1, SONET and the engineering of the digital network.
- ITU-T Recommendations G.711 (PCM), G.726 (ADPCM), G.704 (frame structures).

PART II

Digital Signal Processing for Radio

Digital Filters and Multirate Processing

A software-defined radio is, to a first approximation, a collection of digital filters running at different sample rates. Channel selection, pulse shaping, matched filtering, interpolation for timing recovery, decimation after the ADC and interpolation before the DAC are all filtering problems, and the efficiency with which they are solved determines whether a design fits in a phone, an FPGA or a satellite. This chapter is the working engineer's guide to digital filters: how to specify them, design them, implement them in fixed and floating point, and above all how to change sample rates efficiently with polyphase structures, half-band filters, CIC decimators and filter-bank channelizers. It closes with the oscillators and rotation engines—NCOs and CORDIC—that do the mixing in every digital radio.

6.1 Discrete-time systems in one page

A discrete-time linear time-invariant (LTI) system is characterised by its impulse response $h[n]$ and computes the convolution $y[n] = \sum_k h[k] x[n - k]$. Its **z-transform** $H(z) = \sum_n h[n] z^{-n}$ turns convolution into multiplication, and its frequency response is $H(e^{j\omega})$, the z-transform on the unit circle, with $\omega = 2\pi f / f_s$. Most practical systems are described by a difference equation,

$$y[n] = \sum_{k=0}^M b_k x[n - k] - \sum_{k=1}^N a_k y[n - k] \quad \Longleftrightarrow \quad H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}}. \quad (6.1)$$

The roots of the numerator are the **zeros** and those of the denominator the **poles**. A causal system is stable if and only if all its poles lie strictly inside the unit circle. The magnitude response dips near zeros and peaks near poles; with a little practice you can sketch a frequency response from a pole-zero plot, and every experienced DSP engineer does.

If $a_k = 0$ for all k , the filter has only zeros and a finite impulse response: it is an **FIR** filter. Otherwise feedback makes the impulse response infinite: an **IIR** filter. Which to use is the first decision in any filter design.

	FIR	IIR
Phase	Exactly linear if coefficients are symmetric	Nonlinear, worst near band edges
Stability	Always stable	Must be ensured; sensitive to quantization
Order for a sharp filter	High (tens to hundreds of taps)	Low (4 to 10)
Multirate efficiency	Excellent (polyphase)	Limited
Fixed-point behaviour	Benign	Limit cycles, overflow, scaling
Typical use in radios	Pulse shaping, channel filters, decimators	DC blockers, AGC loops, audio

In communications, the need for linear phase (a filter with nonlinear phase smears pulses in time, which is intersymbol interference; Chapter ??) and for efficient rate changes makes FIR filters the default. IIR filters earn their place where phase does not matter or where a very low order is essential.

6.2 Specifying a filter

A low-pass specification has four numbers: the passband edge f_p , the stopband edge f_{st} , the maximum passband ripple δ_p (or A_p in dB) and the minimum stopband attenuation A_s in dB. The transition width $\Delta f = f_{st} - f_p$ and A_s largely determine the cost. A useful estimate of the number of taps of an optimal FIR filter is fred harris's rule of thumb,

$$N \approx \frac{A_s}{22 (\Delta f / f_s)}, \quad (6.2)$$

so a filter with 60 dB of attenuation and a transition width of 1% of the sample rate needs about 270 taps. Two lessons follow. Halving the transition doubles the cost. And the transition width is relative to the sample rate at which the filter runs, which is the key to the multirate savings later in this chapter: *run sharp filters at low rates*.

6.3 FIR filter design

6.3.1 The window method

The ideal low-pass filter has the infinite impulse response $h_d[n] = 2f_c \text{sinc}(2f_c n)$. Truncating it to N taps is the same as multiplying by a rectangular window, which convolves the ideal response with the window's spectrum and produces the *Gibbs* ripple of about 9% (21 dB stopband) no matter how long the filter. Tapering the truncation with a smooth **window**—Hann, Hamming, Blackman, Kaiser—trades a wider transition for lower sidelobes. The **Kaiser window** has an adjustable parameter β and an empirical design formula that gives β and N directly from A_s and Δf . Windowed designs are quick and robust, and adequate for many

tasks, but not optimal: their ripple is largest near the band edge and decays away from it, wasting attenuation where it is not needed.

6.3.2 Optimal designs

The **Parks–McClellan** algorithm (1972), based on the Remez exchange, designs the FIR filter that minimises the maximum weighted error over the specified bands. The result is *equiripple*: the error oscillates with equal amplitude across each band, and for a given length no other filter has lower peak error. *Least-squares* designs instead minimise the integrated squared error; they have slightly higher peaks but less total stopband energy, which can be better when the stopband contains broadband noise rather than a single interferer. Figure 6.1 compares window, least-squares and equiripple designs of the same length. At 41 taps the equiripple filter gains about 15 dB of minimum stopband attenuation over a Hamming window with a similar transition.

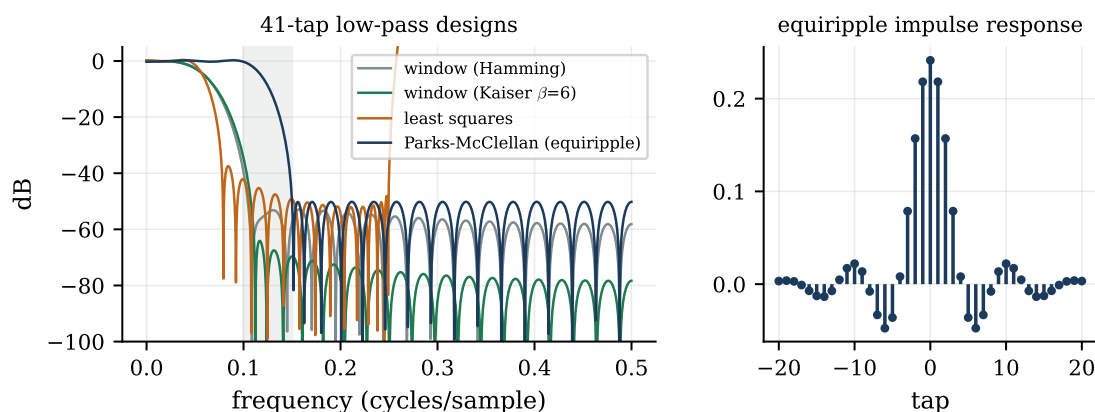


Figure 6.1: Four 41-tap low-pass designs with a transition band from 0.10 to 0.15 cycles/sample (shaded). The Parks–McClellan filter’s stopband is equiripple; the windowed designs have ripples that decay away from the band edge. Right: the equiripple impulse response, symmetric and therefore exactly linear phase.

6.3.3 Linear phase

An FIR filter with symmetric coefficients, $h[n] = h[N - 1 - n]$, has frequency response $H(e^{j\omega}) = A(\omega)e^{-j\omega(N-1)/2}$ with $A(\omega)$ real. Every frequency is delayed by exactly $(N - 1)/2$ samples, so pulse shapes pass undistorted. Symmetry also halves the multiplier count: add $x[n - k]$ and $x[n - N + 1 + k]$ first, then multiply once. The four types of linear-phase FIR (odd or even length, symmetric or antisymmetric) differ in whether they can have zeros at DC and $f_s/2$; an antisymmetric filter is required for a Hilbert transformer or differentiator.

6.4 IIR filter design

6.4.1 From analog prototypes

IIR filters are usually designed by transforming classic analog prototypes. The **bilinear transform**

$$s = \frac{2}{T_s} \frac{1 - z^{-1}}{1 + z^{-1}} \quad (6.3)$$

maps the entire $j\Omega$ axis onto the unit circle once, so there is no aliasing, but it warps frequency as $\Omega = \frac{2}{T_s} \tan(\omega/2)$. The designer *pre-warps* the critical frequencies before designing the analog prototype. The prototypes are the classical families:

- **Butterworth**: maximally flat passband, monotonic, gentlest roll-off.
- **Chebyshev type I**: equiripple passband, monotonic stopband, sharper.
- **Chebyshev type II**: flat passband, equiripple stopband.
- **Elliptic (Cauer)**: equiripple in both bands; the sharpest transition for a given order, with zeros on the unit circle in the stopband.
- **Bessel**: maximally flat group delay; nearly linear phase, very gentle magnitude roll-off (and the bilinear transform spoils its phase property, so digital designs usually approximate it directly).

Figure 6.2 contrasts three sixth-order designs. The elliptic filter reaches 60 dB of attenuation in a small fraction of the transition the Butterworth needs, but its group delay peaks sharply at the band edge—exactly the kind of phase distortion that would cause intersymbol interference in a data receiver.

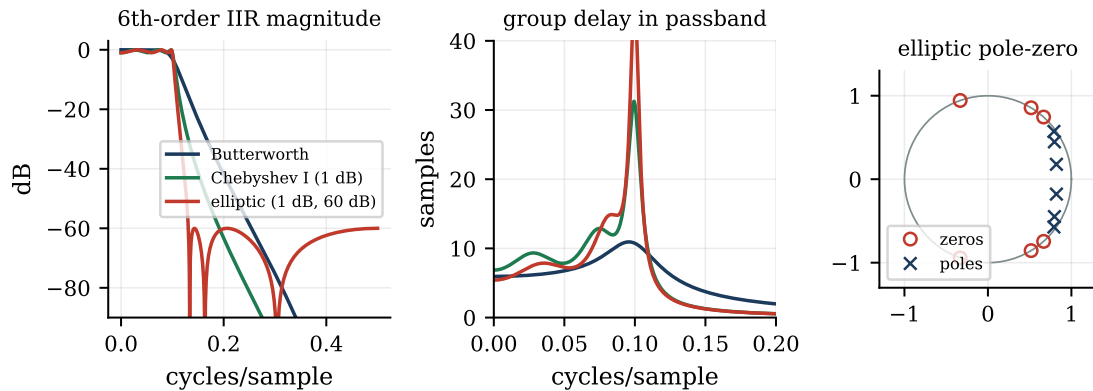


Figure 6.2: Sixth-order IIR low-pass filters with cutoff 0.1 cycles/sample. Left: magnitude. Centre: group delay, which peaks at the band edge, most severely for the sharpest design. Right: the elliptic filter’s poles cluster near the unit circle and its zeros lie on it.

6.4.2 Implementation structures

An N th-order IIR filter can be implemented as one difference equation (*direct form*) or as a cascade of second-order sections (**biquads**). Mathematically they are equivalent. Numerically they are not. The poles of a high-order polynomial are extremely sensitive to its coefficients when they are clustered, as they are in any sharp filter. Figure 6.3 shows an eighth-order

elliptic filter: rounding the direct-form coefficients to 16 or 12 bits moves poles outside the unit circle and the filter becomes unstable, while the same filter as four biquads with 12-bit coefficients is almost indistinguishable from the ideal. *Always implement IIR filters of order above two as cascaded biquads*, with poles paired to their nearest zeros and sections ordered by increasing pole radius to control internal gain.

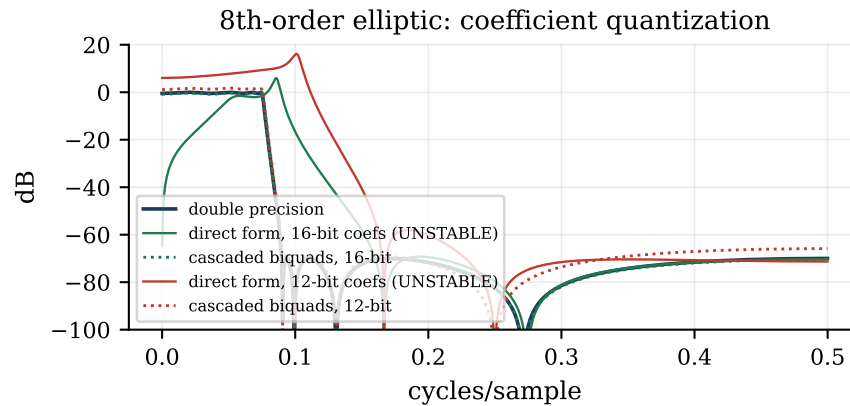


Figure 6.3: Coefficient quantization of an eighth-order elliptic filter. The direct form becomes unstable with 16-bit and 12-bit coefficients; the cascade of biquads tolerates 12 bits with negligible change.

6.5 Fixed-point implementation

Floating point is standard on CPUs and GPUs, but FPGAs and ASICs, where most radio DSP at high sample rates lives, favour fixed point for its cost and power. A number in **Q-format** $Q_{m.n}$ has m integer bits (including sign) and n fractional bits; a 16-bit word in $Q_{1.15}$ represents $[-1, 1)$ with resolution 2^{-15} . Four issues dominate fixed-point design:

Coefficient quantization changes the response, as above. FIR filters degrade gracefully: each bit of coefficient precision buys roughly 6 dB of achievable stopband.

Product rounding adds noise at every multiplier. A $b \times b$ product has $2b$ bits, and keeping only b adds error of variance $2^{-2b}/12$. Accumulating products at full precision and rounding once at the output—as FPGA DSP slices with 48-bit accumulators do—is far better than rounding each product.

Overflow must be prevented by scaling or handled by saturation. Two’s-complement wrap-around is catastrophic in IIR filters (it can sustain large oscillations) but harmless in intermediate FIR sums as long as the final result fits, a useful property.

Limit cycles are small self-sustained oscillations in IIR filters with zero input, caused by rounding in the feedback path. Magnitude truncation or careful structure choice suppresses them.

IN PRACTICE title=Counting the cost

The cost of a filter in hardware is measured in multiply-accumulates per second (MAC/s).

A 100-tap FIR filter at 100 MS/s needs 10^{10} MAC/s: about 25 DSP slices of an FPGA clocked at 400 MHz, or a large fraction of a CPU core using SIMD instructions. GNU Radio's VOLK library selects the best SIMD kernel for each CPU at run time. The multirate techniques below routinely reduce such budgets by factors of 10 to 100.

6.5.1 Fast convolution

For long filters on processors, convolution via the FFT is cheaper. An N -tap filter costs N MACs per sample directly, but only about $2 \log_2(2N)$ complex operations per sample with the FFT, using the *overlap-add* or *overlap-save* method to process a stream in blocks. The crossover is typically between 32 and 128 taps. GNU Radio's `fft_filter` blocks use overlap-save; OFDM receivers are, in a sense, a fast convolution built into the waveform.

6.6 Multirate signal processing

6.6.1 Decimation and interpolation

To **decimate** by an integer M is to keep every M th sample. It divides the sample rate by M and, as Chapter 5 warned, aliases everything between the new Nyquist frequency $f_s/2M$ and the old one into the band. A decimator is therefore a low-pass filter followed by a downsampler (Figure 6.4). To **interpolate** by L is to insert $L - 1$ zeros between samples, which creates $L - 1$ images of the spectrum, followed by a low-pass filter that removes them. A rational rate change by L/M interpolates by L and then decimates by M with a single filter whose cutoff is the smaller of the two Nyquist frequencies.

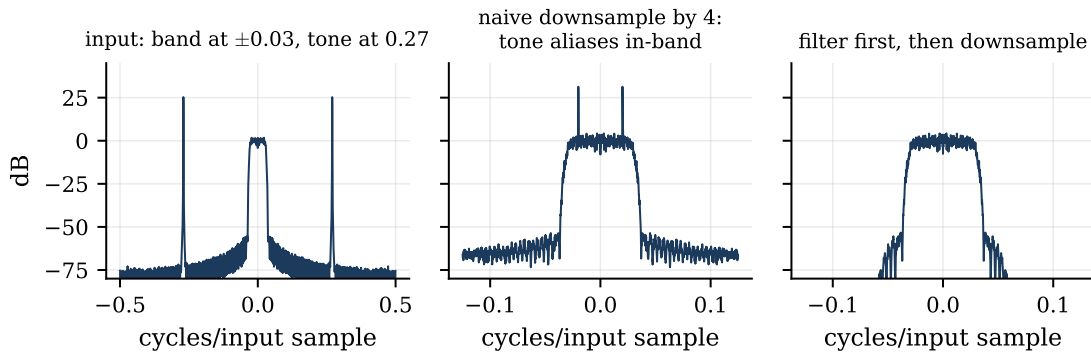


Figure 6.4: Decimation by 4. Downsampling without filtering folds an out-of-band tone at 0.27 cycles/sample into the band of interest; filtering first removes it.

6.6.2 The noble identities and polyphase decomposition

The naive decimator is wasteful: it computes M filter outputs and throws $M - 1$ away. The **noble identities** state that filtering by $H(z^M)$ and then downsampling by M equals downsampling and then filtering by $H(z)$ (and dually for upsampling). Combined with the

polyphase decomposition

$$H(z) = \sum_{p=0}^{M-1} z^{-p} E_p(z^M), \quad e_p[k] = h[kM + p], \quad (6.4)$$

they turn a decimating filter into M short subfilters, each running at the *output* rate and fed by a commutator that deals input samples to them in turn. The computation drops by a factor of M with no change in the result. The same trick applies to interpolation: the L polyphase subfilters produce the L output phases, and the zeros are never multiplied. Polyphase structures are the foundation of efficient DSP, and GNU Radio's `polyphase` (polyphase filter bank) blocks, used in Chapter ?? for symbol timing recovery, are named after them.

6.6.3 Half-band filters

A **half-band filter** has its transition centred on $f_s/4$ with symmetric passband and stopband ripples. Every second coefficient except the centre one is exactly zero, so it costs about a quarter of a general filter of the same length. Cascades of half-band filters, each decimating by 2, are the standard way to decimate by powers of two: the AD9364 in the B200 contains a chain of them (HB1, HB2, HB3) before its programmable FIR, and so does every digital down-converter IC.

6.6.4 CIC filters

When the decimation ratio is large—hundreds or thousands—even polyphase FIR filters become expensive. Eugene Hogenauer's **cascaded integrator-comb** (CIC) filter (1981) needs no multipliers at all. N integrators run at the high input rate, the signal is downsampled by R , and N comb stages $y[n] = x[n] - x[n-1]$ run at the low rate. The overall response is

$$H(e^{j\omega}) = \left[\frac{\sin(\omega R/2)}{R \sin(\omega/2)} \right]^N, \quad (6.5)$$

whose nulls fall exactly at the frequencies that alias onto DC after decimation (Figure 6.5). The price is passband droop, corrected by a short compensation FIR at the low rate, and bit growth of $N \log_2 R$ bits in the integrators, which relies on two's-complement wrap-around being harmless. CIC decimators are the first stage in the B200's FPGA and in almost every wideband digital receiver.

6.6.5 Multistage design

Because FIR cost scales as $A_s/(\Delta f/f_s)$, one large decimation is best done in stages: first a cheap filter with a wide transition (a CIC or half-band) at the high rate, then sharper filters at progressively lower rates, where their relative transition width is larger. Designing a decimate-by-64 channel filter as $2 \times 2 \times 16$ rather than one 64 can cut the computation by an order of magnitude. This is the architecture of the digital down-converter: NCO mixer, CIC, half-band cascade, final channel FIR.

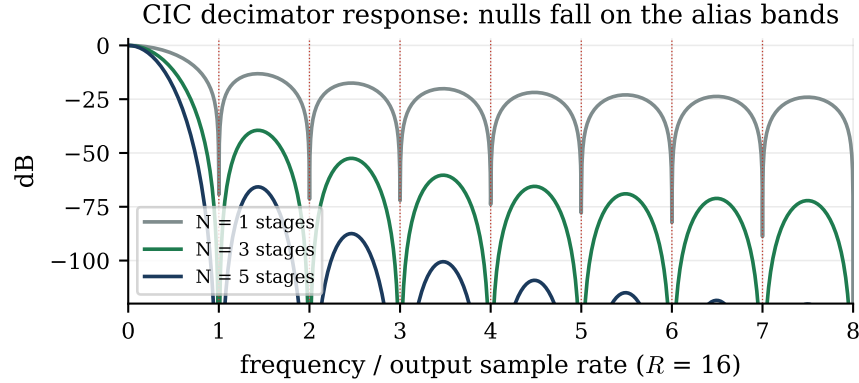


Figure 6.5: CIC decimator magnitude response for decimation $R = 16$ and one, three and five stages, plotted against the output sample rate. The nulls sit on the bands that alias to DC; more stages give deeper nulls and more droop.

6.6.6 Arbitrary and fractional resampling

Real systems often need irrational or awkward ratios: 48 kHz audio to 44.1 kHz ($147/160$), a 61.44 MS/s LTE clock to a 56 MS/s processing rate, or continuous tracking of a clock that drifts by parts per million. Two methods cover these cases. A *polyphase resampler* with many phases (say 32 or 128) picks, for each output sample, the subfilter nearest to the required fractional delay. A **Farrow** structure computes the interpolating filter as a polynomial in the fractional delay μ , so any delay can be synthesised from a few fixed subfilters and a Horner evaluation. The cubic Lagrange interpolator used for timing recovery in Chapter ?? and in `commLib` is a Farrow structure.

6.7 Filter banks and channelizers

A bank of M bandpass filters that splits a wideband input into M adjacent channels would naively cost M times one filter. The **polyphase DFT filter bank** computes all of them for roughly the cost of one polyphase decimator plus an M -point FFT per output sample: the polyphase subfilters of a common low-pass prototype feed an FFT, whose outputs are the M channels (Figure 6.6). Channelizers built this way sit inside satellite payloads that route hundreds of carriers, in spectrum-monitoring receivers, in radio telescopes, and in software base stations that process many narrowband carriers from one wideband ADC. Oversampled and non-maximally decimated variants avoid the aliasing at channel edges; the perfect-reconstruction synthesis banks that recombine channels are the ancestors of the filter-bank multicarrier (FBMC) waveforms studied for 5G.

6.8 Oscillators and rotations

6.8.1 The numerically controlled oscillator

Digital mixing multiplies samples by $e^{-j2\pi f_0 n / f_s}$. A **numerically controlled oscillator** (NCO), or direct digital synthesiser (DDS), keeps a B -bit phase accumulator that adds a frequency

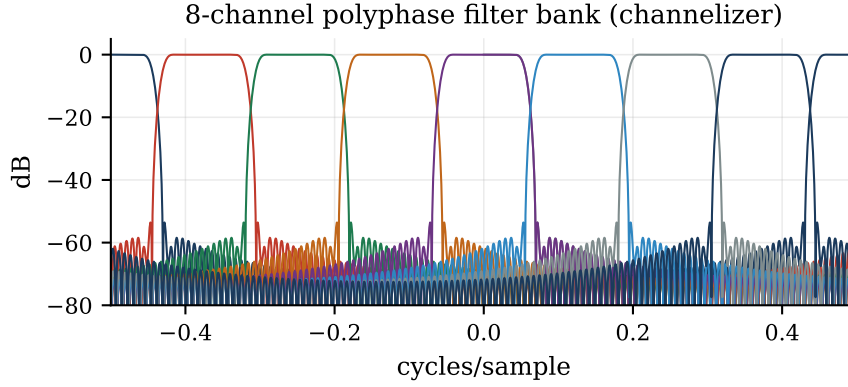


Figure 6.6: An 8-channel uniform filter bank: every channel is a frequency-shifted copy of one prototype low-pass filter, and a polyphase-plus-FFT structure computes them all at once.

word F each clock, giving output frequency $f_0 = Ff_s/2^B$ with resolution $f_s/2^B$ —millihertz at 100 MS/s with a 32-bit accumulator. The top P bits address a sine look-up table. Truncating the phase to P bits creates periodic phase errors and hence spurs, about $6P$ dB below the carrier (Figure 6.7); phase dithering converts the spurs into a noise floor. Amplitude quantization in the table adds its own floor. NCOs tune every digital down-converter and generate the chirps of FMCW radars and the carriers of many transmitters.

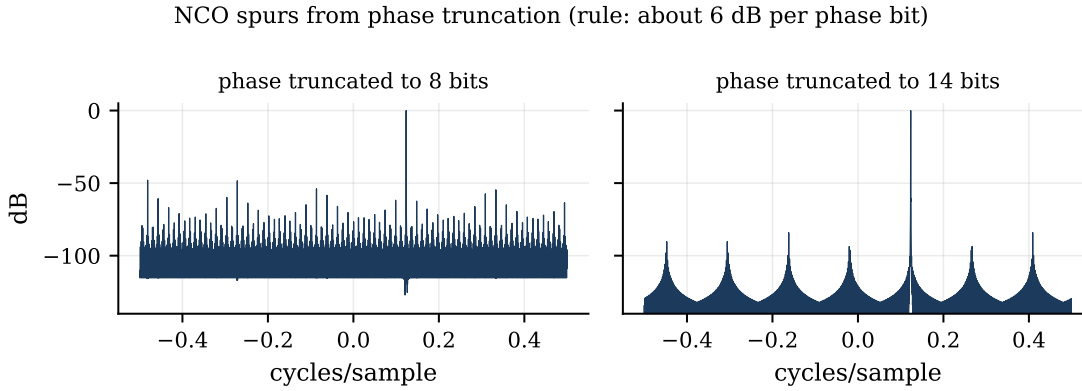


Figure 6.7: Output spectra of an NCO with a 32-bit accumulator when the phase is truncated to 8 and to 14 bits before the sine look-up. Each additional phase bit lowers the spurs by about 6 dB.

6.8.2 CORDIC

Jack Volder’s **CORDIC** algorithm (1959), invented for the B-58 bomber’s navigation computer, rotates a vector by an arbitrary angle using only shifts and adds: at stage i it rotates by $\pm \arctan 2^{-i}$, choosing the sign to drive the residual angle toward zero. After b stages the result is accurate to about b bits, scaled by a known constant $K \approx 1.6468$. Run in *rotation mode* it is a mixer or an NCO; in *vectoring mode* it computes magnitude and phase ($\sqrt{I^2 + Q^2}$ and $\arctan(Q/I)$), which is exactly what AM and FM demodulators, AGC loops and phase

detectors need. CORDIC engines appear in FPGA IP libraries and in the digital front ends of most radio chips.

6.8.3 Two more useful tools

The *Goertzel algorithm* computes a single DFT bin with one second-order recursion per sample; it is the classic detector for dual-tone multi-frequency (DTMF) telephone signalling and is cheaper than an FFT when only a few bins are needed. The *DC blocker* $H(z) = (1 - z^{-1}) / (1 - \alpha z^{-1})$ with α close to 1 is a single-pole high-pass that removes the residual DC offset of a direct-conversion receiver; it is one of the few IIR filters found in almost every SDR data path.

KEY IDEA title=Rules of thumb for radio DSP

- Cost of an FIR $\approx A_s / (22 \Delta f / f_s)$ taps: run sharp filters at low rates.
- Decimate early and in stages: CIC, then half-bands, then the channel filter.
- Use polyphase structures so no multiplier ever computes a discarded sample or multiplies a stuffed zero.
- Implement IIR filters as biquads; accumulate at full precision and round once.
- Budget spurs: every bit of NCO phase is 6 dB, every bit of coefficient 6 dB of stopband.

SIMULATION LAB title=Lab: filters and multirate

`labs/ch06_filters_multirate.ipynb` provides an interactive filter designer (window, least-squares and Parks–McClellan) with live magnitude, phase and group-delay plots and a tap estimate; a biquad-versus-direct-form quantization explorer; a polyphase decimator you can verify sample-for-sample against the naive version; a CIC with compensation filter; a Farrow resampler; and an 8-channel polyphase channelizer that separates a synthetic multi-carrier signal.

Problems

- 6.1 Estimate the number of taps for a low-pass FIR with passband to 20 kHz, stopband from 24 kHz and 80 dB attenuation at a sample rate of 192 kHz. Repeat at 48 kHz. Explain the difference.
- 6.2 Show that an FIR filter with symmetric coefficients has linear phase, and find its group delay.
- 6.3 Design (in Python) a sixth-order elliptic low-pass filter and implement it in direct form and as biquads with Q1.15 coefficients. Find the smallest word length for which each is stable.

- 6.4 Prove the noble identity for decimation. Draw the polyphase structure of a decimate-by-3 FIR filter with 12 taps and count the multiplications per output sample.
- 6.5 Explain why every other coefficient of a half-band filter is zero. How many multiplies per input sample does a 31-tap half-band decimator need, exploiting both symmetry and the zeros?
- 6.6 A CIC decimator has $R = 64$ and $N = 5$ with 14-bit input. How many bits must the integrators have? What is the passband droop at one quarter of the output sample rate?
- 6.7 Design a two-stage decimate-by-40 filter chain for a 200 kHz channel sampled at 20 MS/s with 70 dB of alias protection. Compare the MAC rate with a single-stage design.
- 6.8 An NCO has a 32-bit accumulator clocked at 245.76 MHz. What is its frequency resolution? What phase-word width gives spurs below -90 dBc?
- 6.9 (*Simulation*) Implement CORDIC in vectoring mode in integer arithmetic with 16 stages and measure its magnitude and phase errors over random inputs.

Further Reading

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The Radio Transceiver and the Software-Defined Radio

The mathematics of this book assumes ideal mixers, perfect oscillators and linear amplifiers. Real radios have none of these, and the engineer who understands their imperfections designs systems that work on the first try. This chapter walks through a modern transceiver from the antenna to the samples: receiver architectures, the sensitivity–linearity trade-off and its figures of merit, oscillators and phase noise, power amplifiers and the linearisation techniques that make OFDM affordable, and the digital correction of analog impairments. It ends with the software-defined radio—its history, its architecture as embodied in the USRP B200, and the practical art of streaming samples between a radio and a computer.

7.1 Anatomy of a transceiver

Every radio, from a Bluetooth earbud to a 64-antenna base station, has the same skeleton. On the receive side:

1. An antenna and a *band filter* (a SAW, BAW or cavity filter) that passes only the band of interest and protects the receiver from strong out-of-band signals.
2. A **low-noise amplifier** (LNA) that sets the noise figure of the whole chain (Chapter 3).
3. One or more *mixers* driven by a *local oscillator* (LO) that translate the signal to an intermediate frequency or directly to baseband.
4. Analog filtering and gain (the *baseband* or *IF* section) with automatic gain control.
5. An ADC, followed by digital down-conversion, decimation and channel filtering (Chapter 6).

The transmit side mirrors it: DAC, reconstruction filter, I/Q up-converter, driver, power amplifier (PA), harmonic filter, and antenna, with a switch or duplexer sharing the antenna with the receiver. Each block contributes noise, distortion, spurious signals and frequency errors, and the radio designer’s job is to allocate a budget for each.

7.2 Receiver architectures

7.2.1 Three families

Figure 7.1 sketches the three architectures in use today. The **superheterodyne** of Chapter 4 converts to a fixed IF where high-quality filters select the channel. It remains the choice where

selectivity and dynamic range are paramount: spectrum analysers, military receivers, many base stations, and satellite ground terminals.

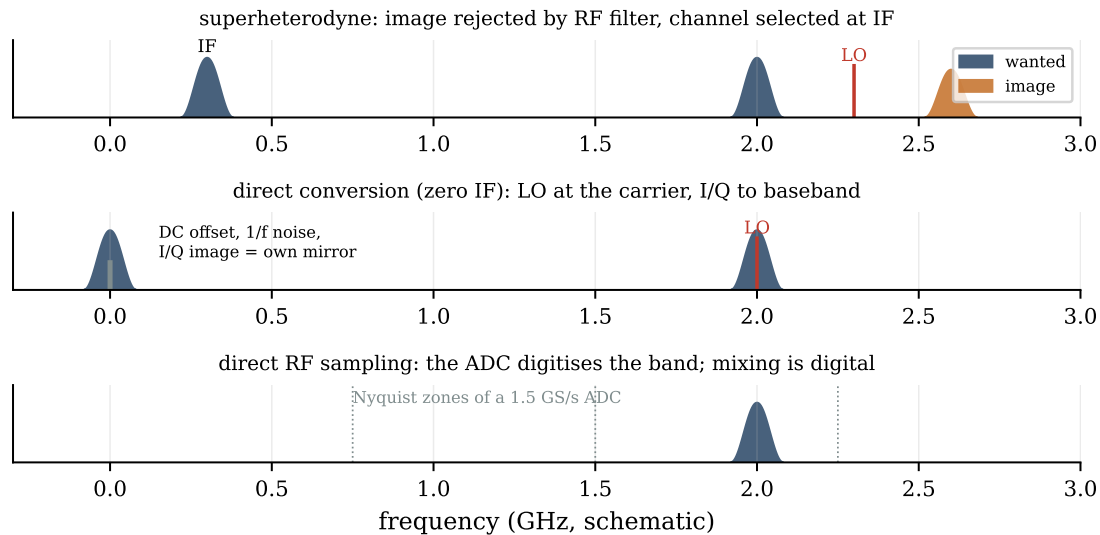


Figure 7.1: Receiver architectures in the frequency domain. Top: the superheterodyne must reject the image before mixing. Middle: the zero-IF receiver’s image is the signal’s own mirror, rejected only by I/Q balance, and its baseband sits on the DC offset. Bottom: a direct RF-sampling receiver digitises the whole band and performs the mixing digitally.

The **direct-conversion** or **zero-IF** receiver sets the LO at the carrier and produces I and Q baseband outputs directly. It needs no image filter and no IF filter—channel selection is a low-pass filter that can be integrated on chip—so it dominates mobile phones and integrated transceivers like the AD9364 in the B200. Its weaknesses are the ones Chapter 1 of the companion labs lets you see: *DC offset* from LO leakage that mixes with itself, *flicker (1/f) noise* in the baseband amplifiers, *second-order intermodulation* that converts strong amplitude-varying blockers directly to baseband, and *I/Q imbalance* that creates a mirror image of the signal onto itself. A *low-IF* receiver places the signal a few hundred kilohertz or a megahertz from DC, avoiding the DC and flicker problems at the cost of needing better image rejection; it is common in Bluetooth, GNSS and FM radio chips.

Direct RF sampling puts the ADC at the antenna, or nearly so. Converters sampling at 3 to 10 GS/s with 12 to 14 bits, integrated with FPGA logic in devices such as AMD’s RFSoc, can digitise everything up to several gigahertz. The mixers, IF filters and channel filters all become DSP. This is the logical end of Mitola’s software radio vision, limited today by power consumption and by the dynamic range needed when strong and weak signals share the band.

7.2.2 The sensitivity–linearity trade-off

A receiver must hear signals near the thermal floor while tolerating others 80 or 100 dB stronger on nearby frequencies. Gain early in the chain improves noise figure (Friis) but pushes later stages toward overload. The two figures of merit are noise figure, for the weak signal, and linearity, for the strong ones.

Linearity is usually characterised by expanding the amplifier's transfer function as a power series, $y = a_1x + a_2x^2 + a_3x^3 + \dots$. Driving it with two tones at f_1 and f_2 produces, among many products, **third-order intermodulation** (IM₃) at $2f_1 - f_2$ and $2f_2 - f_1$. These are the dangerous ones because they fall close to the tones—in the band, where no filter can remove them (Figure 7.2). As the input rises by 1 dB the IM₃ products rise by 3 dB; the extrapolated input level at which they would equal the fundamental is the **third-order intercept point** (IIP₃). A second-order intercept (IIP₂) describes the a_2 products that trouble zero-IF receivers. The 1 dB **compression point** (P_{1dB}), where gain has fallen by 1 dB, sits roughly 10 dB below IIP₃ for a simple amplifier.

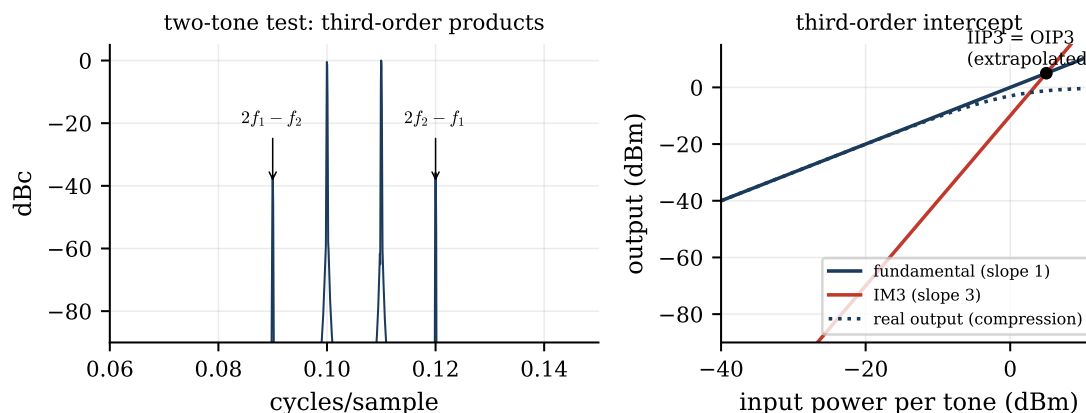


Figure 7.2: Left: a two-tone test through a cubic nonlinearity produces third-order products at $2f_1 - f_2$ and $2f_2 - f_1$. Right: fundamental and IM₃ output against input power; their extrapolated intersection is the third-order intercept.

The **spurious-free dynamic range** combines both ends: the range of input powers over which a signal is above the noise floor while two equal interferers' IM₃ products are still below it,

$$\text{SFDR} = \frac{2}{3}(\text{IIP}_3 - P_{\text{noise}}) \text{ dB}, \quad P_{\text{noise}} = -174 + 10 \log_{10} B + \text{NF dBm}. \quad (7.1)$$

A receiver with NF 6 dB, IIP₃ −10 dBm and a 200 kHz channel has a noise floor of −115 dBm and an SFDR of 70 dB. In cascades, IIP₃ degrades with gain before a stage: $1/\text{IIP}_{3\text{tot}} \approx \sum_k G_{k-1}/\text{IIP}_{3k}$ (linear units, where G_{k-1} is the gain ahead of stage k). Noise wants gain early; linearity wants it late. Every receiver line-up is a negotiation between the two.

7.2.3 Blocking and desensitisation

A strong *blocker* near the wanted channel does harm in three ways even if no intermodulation product lands on the signal: it compresses the LNA or mixer, reducing gain for the weak signal; its presence forces the AGC to reduce gain, raising the effective noise figure; and it mixes with the LO's phase noise to spread noise onto the wanted channel (reciprocal mixing, below). Cellular standards therefore specify blocking tests in which the receiver must maintain its sensitivity with a specified interferer present, and those tests, rather than the plain sensitivity test, usually set the design.

7.3 Oscillators and frequency synthesis

7.3.1 References

Every frequency in a radio derives from a reference oscillator, almost always a quartz crystal. A plain crystal oscillator (XO) is accurate to about ± 20 ppm over temperature; a temperature-compensated crystal oscillator (TCXO), like the B200's, to about ± 2 ppm; an oven-controlled oscillator (OCXO) to 10 ppb; and a GPS-disciplined oscillator (GPSDO) to parts in 10^{12} over a day. A frequency error of ϵ at carrier f_c is a carrier frequency offset of ϵf_c : 2 ppm at 3.5 GHz is 7 kHz, a quarter of a 30 kHz NR subcarrier. Cellular base stations hold ± 50 ppb, and phones lock their oscillators to the base station, which is why Chapter ?? matters so much.

7.3.2 PLL synthesisers

A **phase-locked loop** (PLL) synthesiser multiplies the reference to the desired LO frequency. A voltage-controlled oscillator's output is divided by N and compared in phase with the reference; the loop filter steers the VCO until the divided output tracks the reference, so $f_{\text{VCO}} = N f_{\text{ref}}$. An *integer-N* synthesiser can only step in multiples of f_{ref} ; a *fractional-N* synthesiser alternates the divide ratio with a sigma-delta modulator (Chapter 5) to achieve fine resolution with a high reference frequency. Inside the loop bandwidth, the output phase noise follows the reference multiplied by $20 \log_{10} N$ dB; outside it, the VCO's own noise dominates. The AD9364 contains two such fractional-N synthesisers, one for receive and one for transmit, both locked to the B200's 40 MHz TCXO.

7.3.3 Phase noise

A real oscillator's phase wanders randomly. Its spectrum, instead of a line, has skirts described by the **single-sideband phase noise** $\mathcal{L}(f)$ in dBc/Hz at offset f from the carrier (Figure 7.3). Leeson's model gives the characteristic shape: $1/f^3$ close in (flicker noise upconverted), $1/f^2$ in the middle (white noise integrated by the resonator), and a flat floor far out. Integrating $\mathcal{L}(f)$ over offsets gives the RMS phase error,

$$\sigma_\phi^2 = 2 \int_{f_1}^{f_2} 10^{\mathcal{L}(f)/10} df \quad (\text{rad}^2). \quad (7.2)$$

Phase noise hurts in three ways.

- **Constellation rotation.** Residual phase jitter after carrier tracking rotates dense constellations: 64-QAM tolerates about 1.5° RMS, 1024-QAM less than 0.5° .
- **OFDM impairment.** Slow phase noise produces a common phase error on all subcarriers, which pilots track; fast phase noise produces inter-carrier interference, which they cannot. This is why 5G NR at millimetre-wave uses wider subcarriers and adds phase-tracking reference signals (Chapter ??).
- **Reciprocal mixing.** A strong blocker at offset Δf mixes with the LO's phase noise at that offset and lands on the wanted channel with power $P_{\text{blocker}} + \mathcal{L}(\Delta f) + 10 \log_{10} B$. A -30 dBm blocker 1 MHz away, an LO at -130 dBc/Hz and a 1 MHz channel give -100 dBm of noise—enough to swamp a weak signal.

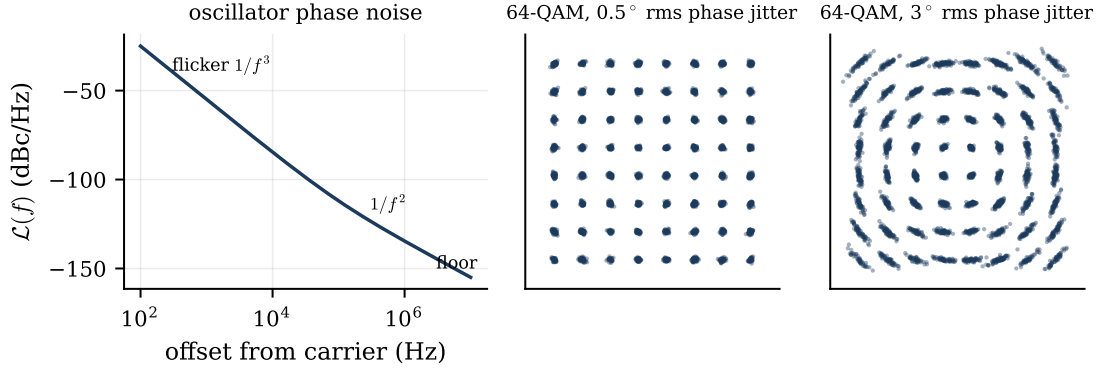


Figure 7.3: Left: typical oscillator phase-noise profile with its $1/f^3$, $1/f^2$ and floor regions. Centre and right: 64-QAM with 0.5° and 3° RMS residual phase jitter; the outer points smear along arcs.

7.4 The transmitter

7.4.1 The I/Q modulator

A modern transmitter synthesises the complex envelope digitally, converts I and Q with two DACs and up-converts them with a quadrature modulator: $s(t) = I(t) \cos 2\pi f_c t - Q(t) \sin 2\pi f_c t$. Its impairments mirror the receiver's: *LO leakage* appears as a carrier spike at the centre of the signal, and I/Q gain and phase mismatch produce a mirror image with the suppression given by Figure 4.3. Both are measured and corrected digitally, and both are part of the transmit EVM budget that standards specify.

7.4.2 Power amplifiers

The PA is the most power-hungry component of most radios and the most nonlinear. Its classes trade linearity for efficiency. *Class A* conducts all the time and is linear but at best 50% efficient; *class AB* and *B* conduct for about half the cycle and reach 60 to 78% at peak power; *class C, D, E* and *F* operate as switches or with harmonic shaping and exceed 80%, but only for constant-envelope signals such as FM, GMSK and FSK. That is why 2G GSM, Bluetooth and most IoT links use constant-envelope modulations: they let a battery-powered device use a saturated, efficient PA.

Linear modulations such as QAM and OFDM have a large peak-to-average power ratio (PAPR), about 10 dB for OFDM at 10^{-3} probability. A PA driven into saturation clips the peaks, and the clipping distortion spreads into adjacent channels as **spectral regrowth** (Figure 7.4). Standards limit it through the **adjacent channel leakage ratio** (ACLR), typically 45 dB for LTE and NR terminals, and limit in-band distortion through EVM. The simplest fix is **back-off**: operating the PA well below saturation. But efficiency falls rapidly with back-off; a class-AB PA backed off by 8 to 10 dB may be only 10 to 15% efficient.

7.4.3 Efficiency and linearisation

Four techniques recover efficiency without giving up linearity:

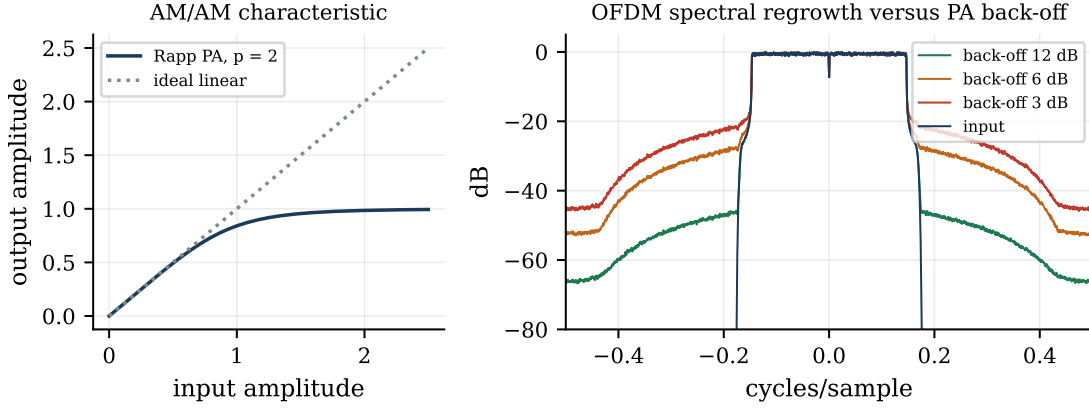


Figure 7.4: Left: AM/AM curve of a solid-state PA (Rapp model). Right: spectrum of an OFDM signal after the PA at 12, 6 and 3 dB of back-off. Each 3 dB less back-off raises the out-of-band emission substantially.

Crest-factor reduction (CFR) clips and filters the digital signal to reduce PAPR by 2 to 4 dB before the PA, trading a small, controlled EVM for a large efficiency gain.

Digital predistortion (DPD) applies the inverse of the PA's nonlinearity in the digital domain, so that the cascade is linear. A small observation receiver samples the PA output; an adaptive algorithm (usually the *indirect learning architecture* with a *memory polynomial* or generalised memory polynomial model) learns the inverse. Figure 7.5 shows a 20 dB reduction in spectral regrowth at 10 dB of back-off. Every macro base station and most small cells use DPD.

Doherty amplifiers combine a carrier amplifier biased in class AB with a peaking amplifier that turns on only for signal peaks. Load modulation keeps efficiency high over about 6 dB of back-off. Doherty PAs with DPD are the standard in cellular base stations.

Envelope tracking (ET) varies the PA's supply voltage to follow the signal envelope, so the transistor is always near saturation. It is used in nearly every 4G and 5G smartphone.

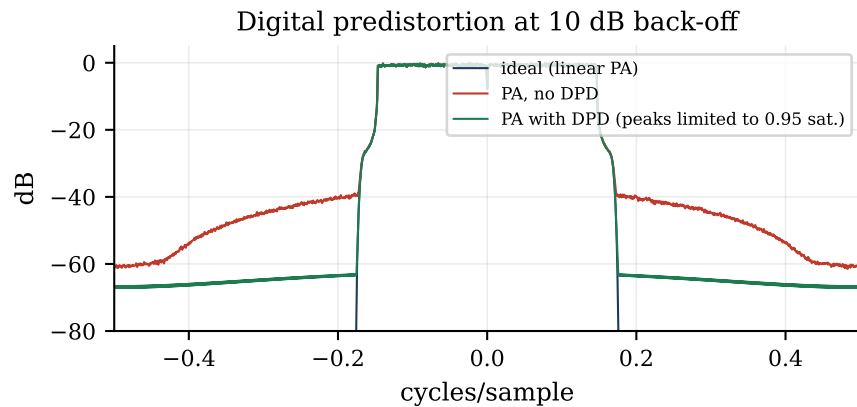


Figure 7.5: Digital predistortion inverts the PA's AM/AM curve. At 10 dB back-off the out-of-band emission drops by about 20 dB, approaching the ideal linear amplifier.

7.5 Correcting analog impairments digitally

The philosophy of the modern transceiver is: build analog circuits that are good enough, then measure and correct the rest in DSP. The standard digital corrections are:

- **DC offset:** a high-pass filter or an estimator that subtracts the running mean, often in hardware (the B200's FPGA does this).
- **I/Q imbalance:** estimate the gain and phase mismatch from the second-order statistics of the received signal (for a proper complex signal, $E[I^2] = E[Q^2]$ and $E[IQ] = 0$), then apply a 2×2 correction. Frequency-dependent imbalance requires a short FIR per branch.
- **Carrier and sampling frequency offset:** estimated and corrected by the synchronisation algorithms of Chapter ??.
- **Transmit LO leakage and image:** calibrated with a loopback path at start-up and tracked in the background.
- **PA nonlinearity:** DPD, as above.

7.6 The software-defined radio

7.6.1 From SPEAKEasy to the USRP

The term *software radio* was coined by Joseph Mitola III in 1991 and popularised in his 1995 paper “The software radio architecture”. The US military's SPEAKEasy programme (1991–1995) demonstrated a radio that could emulate more than a dozen military waveforms by loading different software, motivated by the fact that US forces in the 1983 Grenada invasion could not talk to each other across services. The Joint Tactical Radio System and the Software Communications Architecture followed. In civil communications, the key enabler was the combination of fast ADCs, FPGAs and general-purpose processors. Eric Blossom started GNU Radio in 2001; Matt Ettus built the first Universal Software Radio Peripheral (USRP) in 2004 to give GNU Radio affordable hardware. The RTL-SDR discovery of 2012—that a \$20 DVB-T dongle could stream raw samples—put an SDR receiver in every student's hands. Today, software-defined base stations run on commodity servers (OpenAirInterface, srsRAN) and the Open RAN movement is built on the same idea.

7.6.2 The USRP B200 as a case study

The B200 is a compact example of the architecture:

- **RF:** an Analog Devices AD9364 RFIC covering 70 MHz to 6 GHz, with one transmit and one receive chain, zero-IF architecture, integrated LNAs, mixers, synthesisers, 12-bit ADCs and DACs running up to 61.44 MS/s, half-band and programmable FIR decimators, and hardware DC-offset and quadrature correction. Instantaneous bandwidth is up to 56 MHz.
- **FPGA:** a Xilinx Spartan-6 performs further DSP (CIC and half-band decimation or interpolation, DC-offset correction, timestamping) and packs samples for the host.

- **Host interface:** USB 3.0, carrying 16-bit I/Q samples (or 8-bit for higher rates).
- **Clocking:** a 40 MHz TCXO (± 2 ppm) with inputs for an external 10 MHz reference and a pulse-per-second (PPS) signal; a GPSDO can be fitted internally.

The *master clock rate* (MCR) sets the AD9364's converter rate; the requested host sample rate is MCR divided by an integer decimation in the FPGA. Choosing an MCR that makes the decimation even, or a power of two, gives the best filter performance, and choosing an MCR equal to a multiple of the signal's natural rate (for example 30.72 MHz for LTE and NR) avoids resampling altogether.

7.6.3 Streaming, latency and time

The link between radio and host is a stream of packets. If the host does not consume samples fast enough, the radio's buffer *overflows* and samples are lost (UHD prints $\circ$); if it does not supply transmit samples fast enough, the DAC *underflows* (U). A USB 3.0 link and a modern CPU sustain a few tens of MS/s; beyond that, processing must move into the FPGA. Every packet carries a timestamp from the radio's sample clock, which allows timed transmissions (transmit this burst at time t), coherent operation of several radios sharing a reference and PPS, and measurement of exact delays. Latency through the USB stack and operating system is typically a few milliseconds—fine for most experiments, but far too much for protocols such as Wi-Fi's 16 μ s acknowledgement timing, which is why real-time MAC layers run in the FPGA.

7.6.4 GNU Radio

GNU Radio models a radio as a *flowgraph* of blocks connected by streams of items, each block running in its own thread with ring buffers between them. Blocks can also exchange asynchronous *messages* (for packets and control) and attach *tags* to stream items (for burst lengths, timestamps and frequency changes). GNU Radio Companion (GRC) draws the flowgraph and generates Python. The hardware scripts in this book's companion code are written directly in Python in the same style, so that every block and parameter is visible. GNU Radio 4.0, at release-candidate stage in 2026, rebuilds the runtime around a new scheduler and C++ block model, but the flowgraph concept is unchanged.

Device	Frequency range	Bandwidth	Notes
RTL-SDR (RTL2832U)	≈ 24 MHz–1.7 GHz	≈ 2.4 MHz	8-bit, receive only
ADALM-Pluto	325 MHz–3.8 GHz (extendable)	20 MHz	AD9363, 1×1 , teaching
HackRF One	1 MHz–6 GHz	20 MHz	8-bit, half-duplex
USRP B200 / B210	70 MHz–6 GHz	56 MHz	AD9364 / AD9361, 1×1 / 2×2
bladeRF 2.0	47 MHz–6 GHz	56 MHz	AD9361, 2×2
USRP X410	1 MHz–7.2 GHz	400 MHz	RFSoc, 4×4

PITFALL title=Safe loopback testing

The B200 can transmit more than +10 dBm. Its receiver is damaged by roughly -15 dBm

at the input. Never connect TX to RX without at least 30 dB of attenuation (60 dB is safer), and start with the lowest TX gain. Over the air, transmit only in bands you are licensed for or in ISM bands at low power.

SIMULATION LAB title=Lab: the imperfect transceiver

`labs/ch07_transceiver_impairments.ipynb` models a complete transceiver in Python: a two-tone test with an interactive IIP₃ readout, a receiver line-up calculator (cascaded NF, gain, IIP₃ and SFDR), phase noise synthesis from an $\mathcal{L}(f)$ mask with its effect on 64-QAM and OFDM, a Rapp PA with back-off, ACLR and EVM measurements, a memory-polynomial DPD with indirect learning, and blind I/Q imbalance estimation. With hardware, `gnuradio/gr01_spectrum_iq_capture.py` shows the B200's DC spur and image when its corrections are disabled.

Problems

- 7.1 A receiver has an LNA (NF 1.5 dB, gain 18 dB, IIP₃ +5 dBm), a mixer (NF 9 dB, gain 8 dB, IIP₃ +10 dBm) and an IF amplifier (NF 6 dB, gain 30 dB, IIP₃ +15 dBm). Compute the cascaded NF and IIP₃ and the SFDR in a 1 MHz bandwidth.
- 7.2 Derive the SFDR formula (7.1).
- 7.3 Two interferers at 1001 and 1002 MHz, each −35 dBm, reach a receiver tuned to 1000 MHz with IIP₃ −5 dBm. What is the IM₃ level on the wanted channel?
- 7.4 A 3.5 GHz receiver's LO has $\mathcal{L} = -100$ dBc/Hz at 10 kHz, falling 20 dB/decade. Compute the RMS phase error integrated from 10 kHz to 1 MHz. Is it adequate for 256-QAM?
- 7.5 Explain why a zero-IF receiver is sensitive to IIP₂ but a superheterodyne largely is not.
- 7.6 An OFDM PA delivers 20 W average with 8 dB of back-off and 25% efficiency. Estimate the DC power saved per year if DPD allows 5 dB of back-off at 38% efficiency.
- 7.7 Choose a B200 master clock rate for (a) a 1 MS/s experiment, (b) a 20 MHz LTE carrier sampled at 30.72 MS/s, (c) a 25 MS/s wideband capture. Justify each choice.
- 7.8 (*Simulation*) Implement a memory-polynomial DPD with indirect learning for a PA model with memory (a Wiener–Hammerstein model) and compare its ACLR improvement with a memoryless DPD.

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